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Ministry of Environment and Tourism

Fire Management Strategy for Namibia's Protected Areas



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Ministry of Environment and Tourism

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FOREWORD

Namibia is a relatively dry country with grassland and savanna ecosystems existing in much of its arid and semi-arid areas. Fire is a natural and regular occurrence in these ecosystems and it is important to manage fire in the protected areas occurring within them. In general, fires are frequent in the northeast where annual rainfall up to 700 mm occurs and this frequency diminishes with rainfall in the desert biomes.

Management of fire in Namibia's protected areas is not a new issue and has been dominated by fire exclusion, protection and suppression policies dating back to the 1960s. Over the decades, recognition that fire plays a role in the ecosystem developed in the 1980's with the intentional use of fire to simulate lightning ignited fire regimes, control bush encroachment and to modify grazing pressure.

Implementation has not been widespread and fire prevention and suppression activities have remained the mainstay of fire management in protected areas for the last thirty to forty years. Resources and efforts have been focused on extensive firebreak networks and reactive fire suppression to mitigate uncontrolled wildfires that have impacted negatively on protected area management objectives.

Traditional burning, the use of fire as a land management tool, is prevalent in the neighbouring communal areas of many protected areas, particularly in the northeast. This use includes slash-and-burn agriculture, natural product harvesting, livestock grazing, protection from wildlife, hunting and honey gathering. Insufficient and inconsistent land and fire management legislation, policies and strategies inadequately address the appropriate use of this fire, causing a lack of understanding that results in the uncoordinated use of fire throughout.

Ineffective fire prevention and suppression approaches have resulted in extensive uncontrolled wildfires in the late dry season occurring throughout the majority (up to 80 %) of the northeast Parks every year. Protected areas in other regions are experiencing an increase in uncontrolled wildfires over recent years due to build up of fire fuel loads from years of fire exclusion and a weakening in prevention and suppression capacity.

Over the decades the high intensity fire regime has contributed to the degradation of the PA ecosystems and management objectives by affecting vegetation structure and species composition; reducing ecosystem function and productivity, threatening high value and vulnerable ecosystems, and reducing habitat and biological diversity.

The adherence to this strategy will play a role in ensuring that the Ministry remains committed to the conservation and protection of our wildlife, and that our protected areas are managed properly.

Pohamba Shifeta, MP
MINISTER



PREFACE

In recent years, fire management approaches have been introduced into Namibia's Protected Areas - particularly Etosha National Park and the parks in the north-east - under the premise that fire can be used as an appropriate management tool to actively maintain and rehabilitate all terrestrial habitats within the parks.

Implementation of the practice - in collaboration with neighbouring conservancies and farms in order to share resources and workloads - has been successful in some areas (and in varying degrees) in controlling veld fires, as well as in reducing the area burnt annually and shifting the seasonality of burning to the early dry season in the Protected Areas.

The scale, complexity and urgency of the situation has required the development of an integrated, flexible and comprehensive strategy for dealing with the management of fire in Protected Areas in order to ensure that these ecosystems can contribute significantly to biodiversity conservation while taking into account the socioeconomic development of the country and its people.

This document sets out the overall framework for fire management in Namibia's Protected Areas for the future. Included in this Fire Management Strategy for Namibia's Protected Areas is a controlled fire management programme specific to each Protected Area that has been developed through a structured planning, implementation and monitoring process.

This Strategy has been developed through consultations with a broad range of stakeholders, including line ministries, freehold farmers, conservancies, non-governmental organisations, and researchers. The Strategy is based on the ideas of these stakeholders and the Ministry of Environment and Tourism, arising out of many years of collective experience.

The Ministry of Environment and Tourism would like to thank all its partners and stakeholders who participated in developing this Strategy.


Malan Lindeque
PERMANENT SECRETARY



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ACRONYMS

AFIS	Advanced Fire Information System
AMESD	African Monitoring of the Environment for Sustainable Development
CBFM	Community-based Fire Management
DoF	Directorate of Forestry
FMZ	Fire Management Zone
GIS	geographic information system
GP	Game Park
GWLCA	Greater Waterberg Landscape Conservation area
KNC	Khaudum North Complex
MET	Ministry of Environment and Tourism
MNC	Mudumu North Complex
MSC	Mudumu South Complex
NAM-PLACE	Namibia Protected Landscape Conservation Areas Initiative
NP	National Park
PAs	Protected Areas
PPE	Personal Protective Equipment
SADC	Southern African Development Community
SOPs	Standard Operating Procedures

1. INTRODUCTION

Namibia has an impressive network of Protected Areas (PAs) that forms the cornerstone of biodiversity conservation across the country and represents an important drawcard for the tourism industry nationally. The 22 PAs cover 17 per cent of Namibia's total area and provide protection to some of the country's most important habitats and species. It is the responsibility of the Ministry of Environment and Tourism (MET) to manage and develop PAs and their critical habitats and animal species in order to preserve biological diversity and ecosystems for use by present and future generations of Namibians, to generate global benefits, and to maintain water quality.

Management of fire is one of the challenges in Protected Areas and this Fire Management Strategy for Namibia's Protected Areas represents a framework to develop and implement programmes in the PA network to address this challenge.

Namibia's savanna ecosystems are characterised as fire-dependent, with vegetation structure and species composition having evolved with, and become inextricably linked to, fire activity. Fire maintains these ecosystems' characteristic form and functioning, and the species they contain are highly adapted to regular fire events. Fire has a variety of direct and indirect impacts upon these ecosystems that can be beneficial or negative depending on fire frequency, intensity, and timing. The human use of fire has been instrumental in developing these ecosystems and continues today in the form of traditional burning practices.

In savanna habitats, highly seasonal rainfall (represented by distinct wet and dry seasons) typically determines fire conditions, behaviour, and effects on an individual ecosystem. Fires occur almost exclusively in the dry season, with early dry season (May to July) fires characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Late dry season (August to November) fires are characterised by high intensity, a low level of patchiness, and a tendency to spread due to the hot, dry and windy conditions.

2. Background To The Development Of The Fire Management Strategy For Namibia's Protected Areas

This Fire Management Strategy for Namibia's Protected Areas is based on the use of controlled fire management practices to balance the beneficial and harmful effects of fire on the Protected Areas' management objectives.

Controlled fire management is the strategic implementation of controlled burning to manipulate ecosystems to enhance land use. The timing, extent, intensity, and frequency of fire is managed to enhance a PA's management objectives through:

- minimising uncontrolled wildfires;
- enhancing park use; and
- enhancing ecosystems and habitats.

The Strategy is founded on collaboration with PAs' residents and neighbours to improve livelihoods through the coordination of fire management objectives and the sharing of resources and workloads.

The Strategy provides guidelines and a development framework for each PA to plan, implement, and monitor a controlled fire management programme specific to the ecosystems and fire issues in the individual PA.

2.1 Management of Uncontrolled Wildfires

To minimise uncontrolled wildfires in the PAs, controlled burning is used to reduce – and importantly fragment – fuel loads to create a landscape of burnt and un-burnt patches. Interlinked burnt patches and corridors – representing 35 to 50 per cent coverage – create an early dry season firebreak network that minimises the occurrence and extent of late dry season fires. The result is a reduction in the total area burnt annually and a shift to a predominantly early dry season regime of low intensity fires.



Figure 1: Reducing the area burnt



Figure 2: Reducing the fire intensity

This approach effectively dispenses with the need for costly machinery and the hazardous conventional fire management practices of creating firebreak networks and engaging in wildfire suppression.

2.2 Controlled Burning in the Protected Areas Network

Controlled fire management can be used to protect life and property in the PAs by strategically reducing fuel loads around infrastructure – including management and tourism facilities – to minimise the fire hazard at specific locations. Controlling the timing, location, intensity, and extent of managed fires facilitates the planning and efficient use of PAs' management resources and capacity.

The economic benefits of PAs can be enhanced through establishing fire regimes that improve ecosystems' function and productivity, which provide the resource base for ecotourism, wildlife production, and professional hunting.

2.3 Habitats and Ecosystems

Fire regimes can be tailored to protect, maintain, and enhance high-value and vulnerable PA habitats and their flora and fauna species:

- The extent, intensity, and frequency of fires can be manipulated to increase or decrease woody vegetation cover by modifying seedling establishment, recruitment, and tree mortality.
- Fire regimes can increase ecosystem function, nutrient status, and productivity through manipulating plant biomass and soil surface cover, fertility, and moisture content.
- Variation in the timing, intensity, and frequency of fire increases the spatial and temporal heterogeneity of ecosystems to maximise overall habitat diversity and biological diversity.

3. Protected Areas' Fire Management in Namibia

Namibia's Protected Areas have a variety of different ecosystems, land uses, management capacities, and fire issues. Three broad categories of PAs are used in this Strategy, characterised by different fire management considerations:

1. Larger PAs in the savanna ecosystems of the north and north-east, where higher rainfall causes more frequent fires. Specific controlled fire management programmes have been developed for each of these PAs.
2. Smaller PAs in savanna ecosystems, where more intensive recreational use and management activities create less need for detailed fire management strategies.
3. PAs in very low rainfall desert biomes, where fires rarely occur and therefore do not represent a major issue or management priority.

3.1 Controlled Fire Management Programmes in the Larger Protected Areas

Controlled fire management programmes have been developed for each of the larger PAs in the savanna ecosystems of the north and north-east of Namibia, where higher rainfall cause more frequent fires.

The controlled fire management programmes are specific to each PA and have been developed through a structured annual planning, implementation, and monitoring process.

Controlled fire management is tailored to the specific requirements of fire management zones (FMZs) within each PA. This enables adaptable and robust implementation that addresses diversity of habitat and use, while achieving overall PA fire management objectives.

A dedicated fire management team in each PA carries out the planning, implementation, and monitoring process based on local landscape features, weather conditions, fire behaviour, and local knowledge in order to safely and efficiently implement the PA's individual controlled fire management programme. The training and resource needs of each fire management team are based on existing fire management capacity and the resources each PA requires to safely and efficiently implement controlled fire management activities.

(The fire management programmes for these PAs are detailed in the Appendices 1 to 7.)

3.2 Fire Management in Small Protected Areas

Some smaller PAs in savanna ecosystems have less need for detailed fire management strategies due to their more intensive recreational use and management activities.

Existing strategies to manage uncontrolled wildfires in smaller Protected Areas are strengthened with basic equipment, training, collaboration with neighbours, and road/firebreak maintenance.

(Details of small PAs' fire management are attached as Appendix 8.)

3.3 Fire Management in Desert Biome Protected Areas

In Protected Areas in desert biomes with very low rainfall, fires are not a major issue or management priority. In wet cycles, uncontrolled fires occasionally occur and existing strategies are to be strengthened when required.

(Details of desert biome PAs' fire management are attached as Appendix 9.)

4. Collaborative Fire Management

Fire management collaboration and coordination with Protected Areas' residents and neighbours sharing boundaries with PAs comprise an essential component of this Fire Management Strategy for Namibia's Protected Areas.

Alignment of fire management objectives and the sharing of resources and workloads with residents and neighbours facilitate effective fire management while reducing effort and the need for resources.

The Directorate of Forestry, through the Forest Act of 2001, regulates fire management in the areas neighbouring PAs, where authorisation is required to implement controlled burning. Development of structured controlled fire management in collaboration with neighbours – and in partnership with the DoF – is strongly recommended.

4.1 Communal Areas, Conservancies and Community Forests

The integrated community-based fire management (CBFiM) programmes established in conservancies and community forests neighbouring the north-east parks provide an operational framework for the approaches described in this Strategy.

Community fire management committees prepare a fire management plan and apply for a Permit to Burn from the DoF. The committee implements a controlled fire management programme to provide tangible livelihood benefits to the community through a reduction in uncontrolled fires and improved land use and environmental management. The traditional authority oversees the functions of the committee and arbitrates on fire-related disputes. The CBFiM programmes require technical and financial support from government and international and local development non-governmental organisations, e.g., the Namibia Protected Landscape Conservation Areas Initiative (NAM-PLACE) project, the Strengthening the Capacity of the Protected Area System to Address New Management Challenges (PASS) project, and the GIZ Community Forest Programme.

4.2 Private Farms

Controlled fire management collaboration and coordination with neighbouring privately owned farms should build upon existing relationships. Communication with neighbouring farms regarding planned burning and facilitating the sharing of resources, equipment and workloads – particularly when implementing protection burns along shared boundaries – is encouraged.

Development of private-farm-controlled fire management programmes is to be promoted and supported



by the Directorate of Wildlife and National Parks through joint training, sharing of technical skills and knowledge, and implementation activities.

4.3 Fire Management Awareness and Education

Development of fire management awareness and education programmes is recommended to foster the commitment and support of key role players for the Fire Management Strategy for Namibia's Protected Areas. These programmes are to be delivered to MET staff, residents, neighbours, and visitors to facilitate safe and efficient fire management activity implementation.

Pre-season multi-stakeholder meetings and/or agreements will establish the foundation for awareness. Public multi-media education using electronic and print media (newspaper, posters and brochures) will notify civil society, government and non-government institutions of the objectives and achievements of controlled fire management programmes.

4.4 International Collaboration

Cross-border collaboration with Botswana and Angola should build on existing regional initiatives, including the Southern African Development Committee (SADC) Regional Fire Management Programme and the SADC Protocol on Forestry (2002).

Integration with the Kavango Zambezi Trans-frontier Conservation Area (KAZA TFCA) should be facilitated through existing structures.

5. Fire Information Systems Use in Protected Areas

An integrated fire information system is utilised to collate, organise, and disseminate fire-related information to facilitate effective fire management implementation in Namibia's PAs network. The African Monitoring of the Environment for Sustainable Development (AMESD) Project's Advanced Fire Information System (AFIS) terminal provides the foundation of the system in use in Namibia. Information from the AFIS is integrated with local monitoring sources in a geographic information system (GIS) to develop PA-specific information and maps that assist the fire management teams to plan and implement their fire management programmes. The information and maps are disseminated as monthly fire reports to the PAs' wardens and fire management teams' bases during the fire season. Early warnings of active fires, or the high probability of fire presenting a danger, are communicated to the PAs' wardens and fire management teams when required.

5.1 The African Monitoring of the Environment for Sustainable Development's Advanced Fire Information System

The AMESD Project has installed AFIS terminals at the Etosha Ecological Institute of the MET and at the National Remote Sensing Centre of the DoF.

Information from Geonetcast (via satellite) provides information for the prediction of fires, detection and monitoring of fires, and the assessment of fires.

Prediction of Fires

The Fire Danger Index is updated daily and provides a 3-day forecast (Fire Weather Index and Lowveld Fire Danger Index).

Fuel Condition data are updated every 10 days and provide a Vegetation Condition Index and a Vegetation Productivity Index. Data also indicate the number of days since the last burn for each area.

Detection and Monitoring of Fires

MODIS Active Fire detection data are updated regularly on a daily basis (10am and 3pm).
MSG Active Fire detection data are updated every 15 minutes.

Assessment of Fires

MODIS Burned Area Product data are updated every month, mapping and documenting the extent of fires throughout the year. MODIS Imagery is updated every 6 hours and provides visual fire monitoring and assessment. The AFIS has limitations in terms of scale and much of the information is too coarse for use at the PAs' level. The data on prediction of fires and the monthly Burned Area Product are developed using automated processes at a scale that can only monitor broad parameters.

5.2 The Protected Areas' Integrated Fire Information System

To provide useful, PA-specific information and maps, the AFIS data are integrated with local-level information in a GIS. ArcGIS is a suitable software package to accomplish this and is the program currently in use. The detection and monitoring of fire through the MODIS and MSG Active Fire detection data and the assessment of fires through the visual interpretation of MODIS 250 m Imagery provide the most useful information. Maps specific to each PA – with information such as boundaries, roads and firebreaks, fire management zones, infrastructure, habitats, and local monitoring sources – are used in conjunction with this AFIS information to assist the fire management teams to plan and implement their fire management programmes.

A trained operator with appropriate GIS and remote sensing skills and knowledge operates the integrated fire information system from the Etosha Ecological Institute.

5.3 Communication and Early Warning

The information and maps developed from the integrated fire information system are disseminated as monthly fire reports to the PAs' wardens and fire management teams' bases during the fire season.

Early warnings of active fires, or the high probability of fire presenting a danger based on weather forecasts from the Bureau of Meteorology, are communicated to the PAs' wardens and fire management teams when necessary. This will require a staff member with GIS and remote sensing skills and knowledge to be trained and tasked with this responsibility at the various PAs.

Park officials can independently access much of the available AFIS data from the Internet through the:

- Advanced Fire Information System (AFIS) - <http://afis.meraka.org.za>
- Global Fire Information Management System (GFIMS) - <http://www.fao.org/nr/gfims/en>.

The GFIMS provides a fire email alert service whereby registered users of the MODIS-derived Active Fire detection product receive an email whenever a fire is detected in a specified area of interest.

5.4 Local Burned Area Information Processing

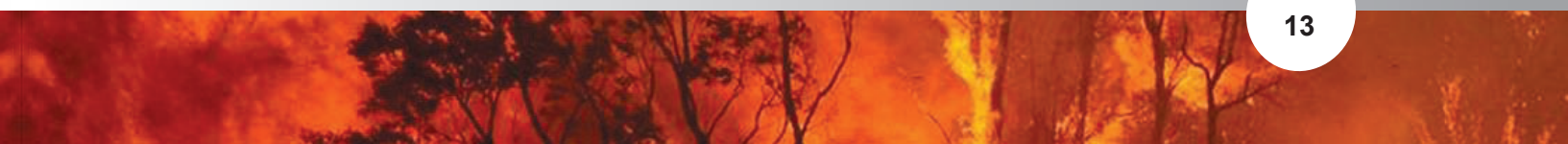
To implement a Controlled Fire Management Programme strategically and efficiently it is important to have updates on controlled burning progress. To overcome the limitations of scale of the AFIS MODIS Burned Area Product in the assessment of fires, there two recommended manual mapping options:



Option 1: processing of the MODIS 250 m resolution satellite imagery that is provided twice daily (from AFIS or GFIMS) is a cost-effective option, although limits of scale still exist. To map controlled burns effectively using this imagery, processing of images on a fortnightly or monthly basis is recommended.

Option 2: Processing of Landsat 5 30 m resolution satellite imagery is another option, although cost is a factor and availability/ordering of images over remote areas is unreliable. To map controlled burns effectively, a series of images (to cover each PA) is required at the end of July, and again at the end of the October/November to map the late dry season fires.

Both options will require training of an operator to achieve a high level of GIS and remote sensing skills and knowledge.



APPENDIX 1

CONTROLLED FIRE MANAGEMENT PROGRAMME,
BWABWATA NATIONAL PARK



1. Introduction

Bwabwata National Park covers approximately 6,274 km² of the Zambezi and Kavango-East regions in north-east Namibia. It completely covers the Caprivi Strip, which extends between the Kavango and Kwando rivers and is bordered by Angola to the north and Botswana to the south.

Kalahari woodlands of the remnant sand dunes, interspersed with fossil drainage lines called omurumba, dominate the savanna woodland ecosystem. The rivers support a diverse array of habitats, including riparian woodlands and extensive floodplains. The semi-arid tropical climate provides a variable and highly seasonal (October to April) average annual rainfall of 550 mm.

This large park includes three core areas designated for special protection and controlled tourism – Kwando Core Area (1,345 km²), Buffalo Core Area (629 km²), and Mahango Core Area (245 km²). The Multiple Use Area (4,055 km²) is zoned for community-based tourism, trophy hunting, and resident communities.

Collaborative wildlife and natural resource management involving the resident communities is facilitated through the Kyaramacan Association and the neighbouring communal area conservancies.

2. Fire History and Management

Fire is a natural and regular occurrence in savanna ecosystems and fire plays a fundamental role in the vegetation, wildlife, and land use management of Bwabwata National Park. Fires occur almost exclusively in the dry season, with early dry season (May to July) fires characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Late dry season (August to November) fires, on the other hand, are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions.

The management of fires in Bwabwata NP has been dominated by fire exclusion, protection, and suppression policies dating back several decades. These resulted in extensive uncontrolled wildfires occurring over most of the core areas' land every year in the late dry season (see Figure 1). In the Multiple Use Area, the resident communities have maintained an early dry season fire regime, particularly in the eastern half of the area.

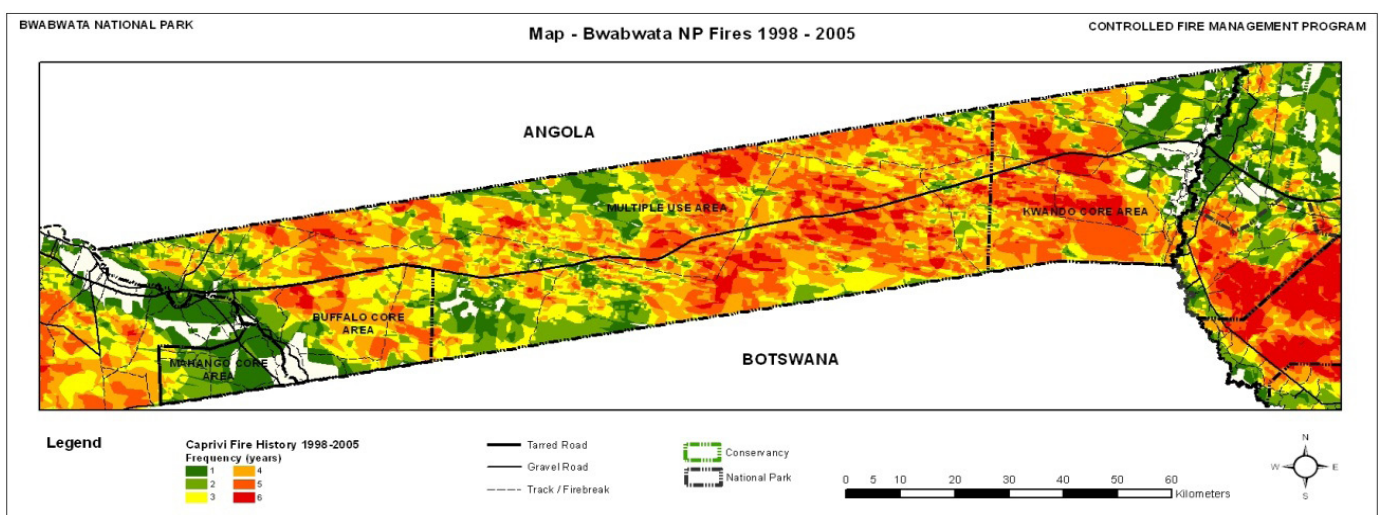


Figure 1: Bwabwata National Park fire history, 1998 to 2005

The vast majority of late dry season fires are caused by the uncoordinated use of fire as a land management tool, both in the park itself and in neighbouring Botswana and Angola. This includes its use in relation to slash-and-burn agriculture, natural product harvesting, livestock grazing, and hunting.

This dry season fire regime has contributed to the degradation of the park's ecosystem and management objectives by affecting vegetation structure and species composition; reducing ecosystem function and productivity; threatening high-value and vulnerable ecosystems; and reducing habitat and biological diversity.

In 2007, Bwabwata NP commenced an integrated controlled burning programme in the Caprivi Region (as it was then, now the Zambezi Region) and the Mudumu North Complex (MNC). One of the activities central to that fire management approach is early burning.

This approach was formalised in the management plan for the park, which states that: 'Fire should be used as a management tool to actively maintain and rehabilitate all habitats in the Parks.'

A controlled burning programme aimed to use controlled burning to reduce the area burnt annually and shift the seasonality of burning to the early dry season. Collaborative implementation with the Kyaramacan Association was structured to share resources and workloads in order to jointly manage uncontrolled wildfires, enhance land use, and improve the environment of the area.

Planning involves designating preparation, controlled burning, and assessment activities through the park's core areas' annual work planning process. Strategic planning currently relies on a very small number of key staff members, is not documented, and lacks a structured process – thus it falls short of achieving some of the fire management objectives.

Implementation of controlled burning is of an ad hoc nature at present and varies throughout the areas in question (see Table 1 and Figure 2).

- **Kwando Core Area** relies on the skills and knowledge of a few key staff members; implementation is, however, generally consistent and well organised. Controlled burning has been successfully established over approximately 30 to 40 per cent of the park in the early dry season of each year as a result. Uncontrolled fires have been significantly reduced.
- **Buffalo Core Area** has a distinct lack of skills and knowledge among the staff and implementation of controlled burning is inconsistent and disorganised as a result. Insufficient controlled burning allows extensive uncontrolled wildfires from Botswana to affect the majority of the core area annually.
- **Mahango Core Area** has a distinct lack of skills and knowledge among the staff too, and implementation is similarly inconsistent and disorganised. Insufficient controlled burning allows extensive uncontrolled wildfires from Botswana to affect the western half of this core area annually.
- **Multiple Use Area.** There is considerable knowledge of controlled burning among the resident communities and fire management implementation is consistent in the east, where the majority of the population reside. In the eastern half of the park, the integrated fire management approach contributes significantly to rehabilitating and maintaining the park's ecosystem and habitats, protecting infrastructure, and enhancing economic benefits from consumptive and non-consumptive tourism. The western half is still dominated by extensive late dry season fires, however. Uncontrolled wildfires continue to occur – arising from both external and internal sources – and a large proportion of the western half of the park continues to burn annually as a consequence.



Table 1: Bwabwata National Park fires, 2007 to 2011

	FIRE SEASON									
	2007*		2008*		2009^		2010^		2011^	
	Area (km ²)	% of park	Area (km ²)	% of park	Area (km ²)	% of park	Area (km ²)	% of park	Area (km ²)	% of park
Controlled burn	2,350	37	2,040	32	217	3	>795	>13	2,328	37
Uncontrolled fire	1,490	24	2,926	47	2,005	32	1,864	30	1,185	19
Total	3,840	61	4,966	79	2,222	35	2,659	43	3,513	56

* Fires mapped manually from monthly MODIS 250 m satellite imagery.

^ Automated fire mapping of MODIS Burned Area Product 500 m detects large intense fires only.

Monitoring is limited to ground observations of fire intensity, extent, and the effect on habitats; limited records are kept of controlled burning implementation or uncontrolled fire occurrences. Remotely sensed fire maps giving an overview of fire distribution are provided to park management only inconsistently.

3. Resources and Capacity

Collaborative fire management implementation with the Kyaramacan Association is structured to share resources and workloads in order to jointly manage uncontrolled wildfires, enhance land use, and improve the environment of the area.

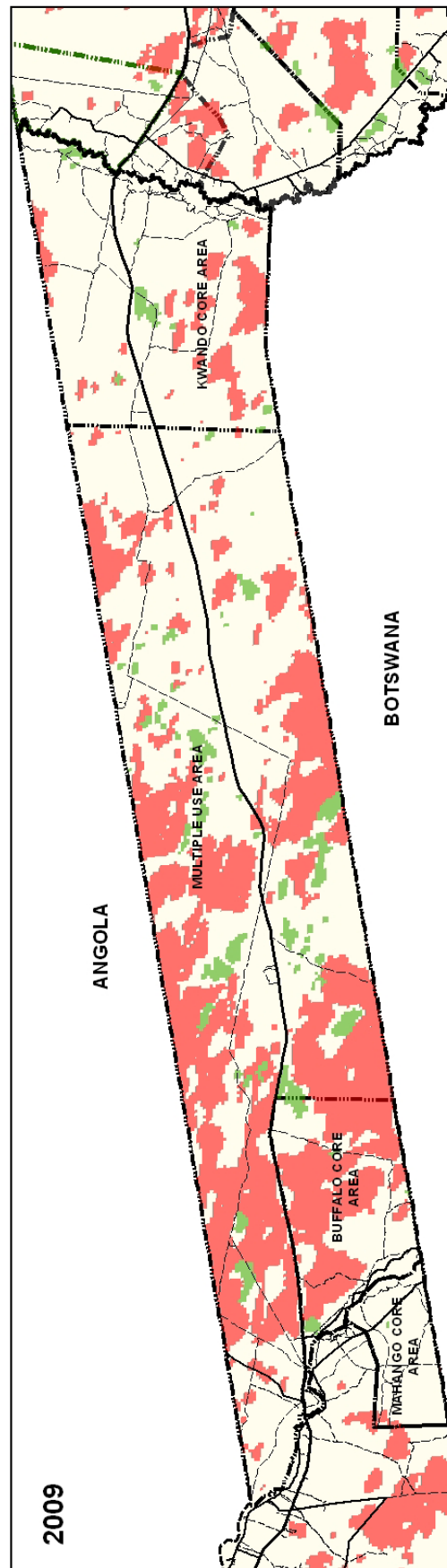
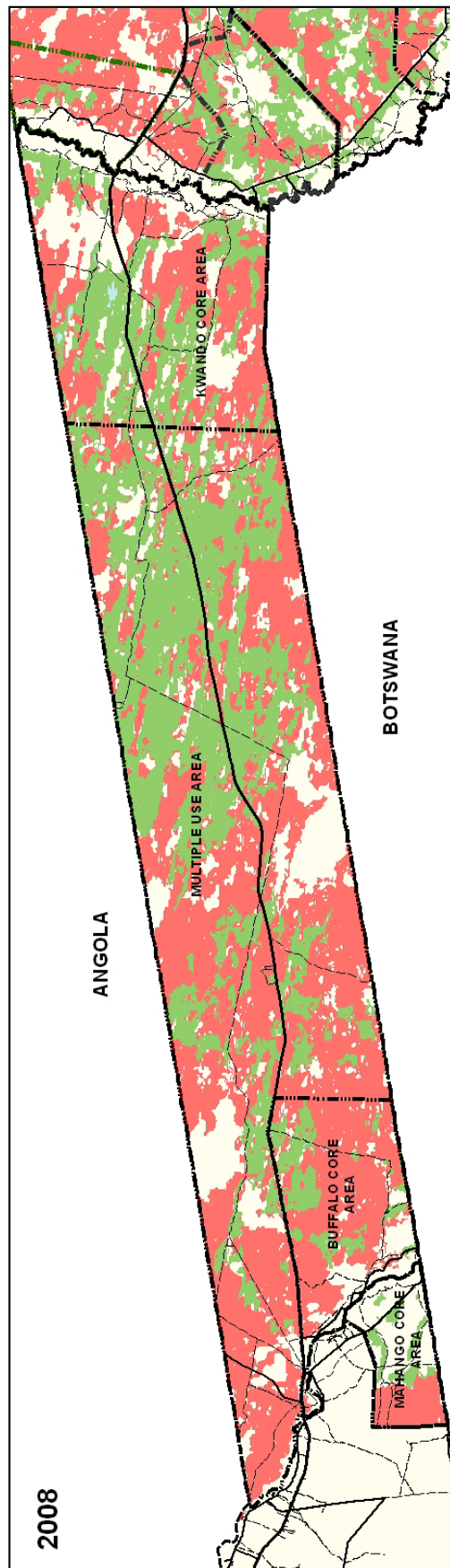
Skills and Knowledge: In respect of controlled fire management planning, implementation, and monitoring are limited to a few key staff members stationed in the Kwando Core Area. Other staff members have developed some fire management experience over the years but they lack the leadership skills required to coordinate a controlled fire programme.

Vehicles and Drivers: Availability limits implementation of the programme throughout all areas of the park. This is most acute in Buffalo Core Area.

Fire Management Equipment: Each core area has a drip torch, at least one water knapsack spray unit, and 5 to 10 fire beaters – representing the absolute minimum required for controlled fire management operations. Mahango Core Area has two 500 litre bakkie skid units.

Personal Protective Equipment: The existing staff uniforms are the minimum required for controlled fire management operations.

Communication: The existing park VHF radio communication network is adequate and must be maintained as such.



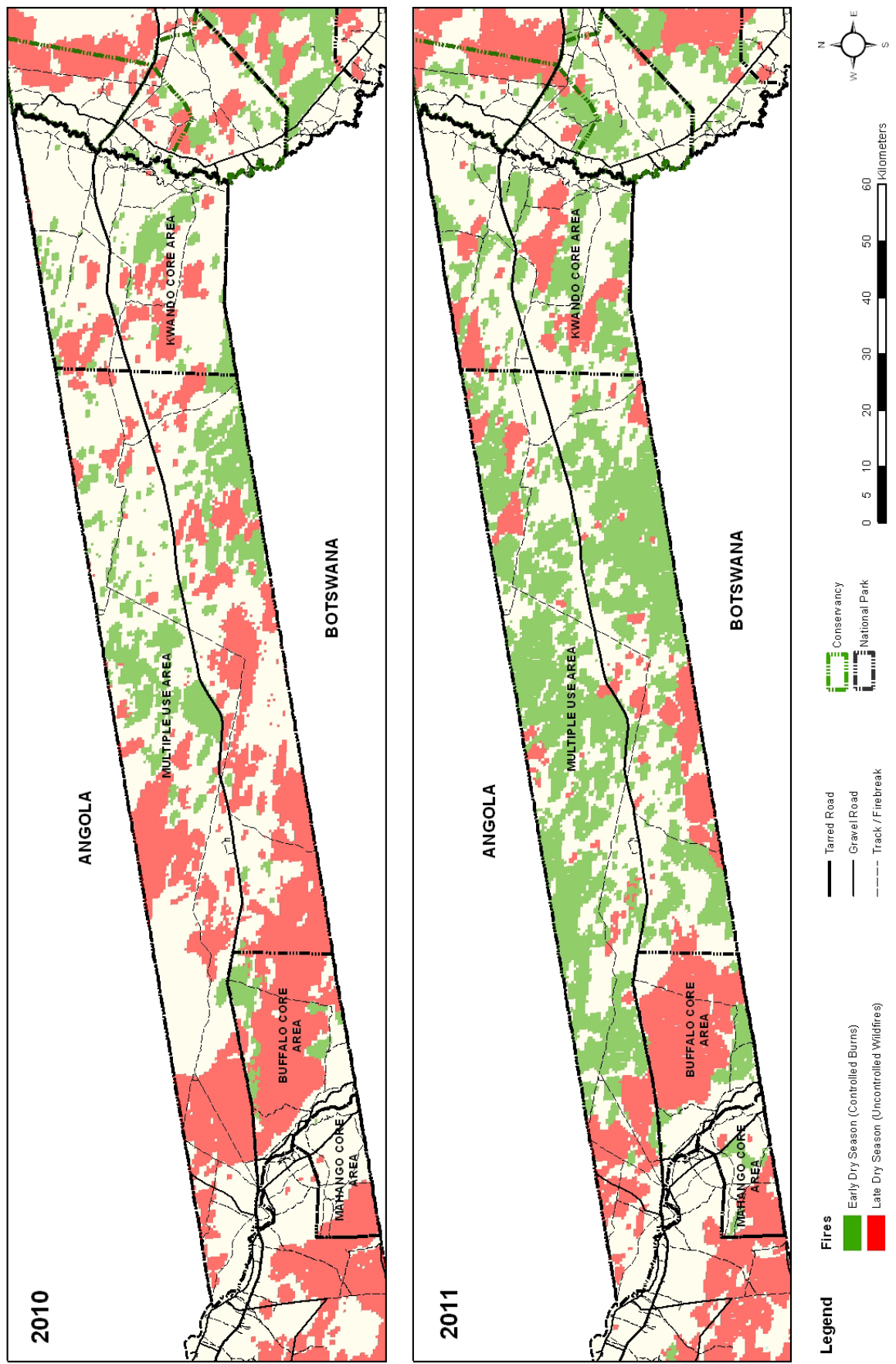


Figure 2: Bwabwata National Park fire occurrences, 2008 to 2011

Residents and Neighbours: in the Multiple Use Area, the resident communities have maintained an early dry season fire regime, particularly in the eastern half.

Collaborative fire management implementation with the Kyaramacan Association is structured around sharing resources and workloads to jointly manage uncontrolled wildfires, enhance land use, and improve the environment of the area. Minimal implementation occurs in the western half of the Multiple Use Area, however.

In the communal area surrounding Mahango Core Area, the uncoordinated use of fire as a land management tool is prevalent and communities rely on the DoF to provide fire management support in the form of firebreak maintenance and wildfire suppression in the event of uncontrolled fires.

Botswana has a prevention and suppression management policy and maintains extensive firebreaks along its borders. Nevertheless, these have no effect on impeding the spread of uncontrolled fires into Bwabwata NP.

Minimal to no fire management takes place in the remote areas of Angola to the north of the park.

SWOT Analysis: This analysis evaluates and summarises the implementation of the existing integrated fire management approach for Bwabwata National Park.

Table 2: SWOT analysis of existing fire management practices at Bwabwata NP

	Beneficial	Detrimental
Internal	STRENGTHS - Dedicated staff with 5 years’ controlled fire experience in Kwando Core Area - Established as a core park management activity in Kwando Core Area - Consistent planning and implementation with substantial controlled burning every year - Understanding of fire and habitat management	WEAKNESSES - Strategic planning lacks a structured process - Implementation ad hoc and repetitive, thus falling short of some fire management objectives - Minimal experience and implementation in Buffalo and Mahango core areas - No implementation in vast areas with limited access (particularly the Multiple Use Area) - Vehicle and driver availabilities limit implementation - Local and remote monitoring insufficient
External	OPPORTUNITIES - Considerable fire management skills, knowledge, and experience in resident population - Established resident collaboration, particularly in eastern half of Multiple Use Area - Multi-stakeholder collaborative management forums well established (MNC) - Strengthen partnership with DoF to facilitate controlled burning in neighbouring areas - High-profile area for development aid funding support	THREATS - Uncontrolled wildfires from neighbouring areas, particularly Angola and Botswana - Uncontrolled fires from residents, particularly in western half of park - No implementation in vast areas with limited access or population in the Multiple Use Area - Management fires impacting neighbours - Weak partnership with DoF hinders facilitation of controlled burning in neighbouring areas

4. The Controlled Fire Management Programme

The Controlled Fire Management Programme for Bwabwata National Park represented by this document is based on the strategic implementation of controlled burning to manipulate the park's ecosystem to enhance park management objectives.

The programme builds on the existing integrated fire management approach that has been utilised since 2007 to safely and efficiently implement controlled fire management activities that:

- minimise wildfire hazard;
- enhance park use; and
- improve ecosystems and habitats.

The programme implements an overarching strategic early dry season burning regime to establish a mosaic of interlinked burned patches and corridors, representing approximately 50 per cent of the park area. This creates an early dry season firebreak network that minimises the occurrence and extent of late dry season fires subsequently. The specific location, timing, and method of controlled burning are all guided by park land-use and habitat requirements.

A dedicated fire management team carries out a structured annual planning, implementation, and monitoring process based on fire management zones (FMZs). This enables adaptable and robust fire management activities that address the diversity of habitats and park use while achieving overall park fire management objectives.

This effectively dispenses with the need for the costly and hazardous conventional fire management practices of establishing firebreak networks and undertaking wildfire suppression.

4.1 Fire Management Teams

A fire management team is established in each core area (Kwando, Buffalo and Mahango) and in the Multiple Use Area.

Each fire management team comprises the functional group of five MET staff members based in a core area or the Multiple Use Area, whose role is to carry out the structured planning, implementation and monitoring of the programme within the FMZs under their jurisdiction every year. Resource monitors from the Kyaramacan Association should form part of the team in the Multiple Use Area.

A team leader within each fire management team works in close collaboration with the Warden to coordinate and supervise the activities of the Controlled Fire Management Programme.

It is important that the team members are maintained in their roles for extended periods, in order to build their fire management skills and knowledge and ensure development and improvement of the programme.

4.2 Fire Management Zones

Bwabwata NP has been divided into a number of FMZs, based on habitat, infrastructure, park use, and fire history/issues:

- **Kwando Core Area:** the Kwando River FMZ incorporates the floodplains and adjacent areas. The woodlands extending west are divided by the Trans-Caprivi highway into the Northern and Southern FMZs.
- **Buffalo Core Area:** the Kavango River FMZ incorporates the floodplains and adjacent areas. The Eastern FMZ consists of the woodlands extending east.

- **Mahango Core Area:** the Kavango River FMZ incorporates the floodplains and adjacent areas east of the gravel road. The Western FMZ consists of the woodlands extending west.
- **Multiple Use Area:** Omega I FMZ, Western FMZ, Eastern FMZ. Controlled fire management is tailored to the specific requirements of each FMZ through a structured planning, implementation and monitoring process.

4.3 Planning

Controlled fire management planning is carried out at the end of the summer/wet season (March/April) every year. A planning process involving information collection and analysis at the FMZ level is required to determine the key planning factors for developing the controlled fire management requirements for the following fire season.

Fire Management Zone Analysis

A combination of satellite data, field surveys, and liaison with park staff, concessionaires and neighbours is used to collect and analyse key information. A series of questions highlights the information required, as follows:

Fire History and Issues - When did the fires occur in the last three years? What was the source of fire (controlled or uncontrolled)? Where did the fires come from? How much rain fell last year? How much fuel load exists? What and where are the fire hazards?

Park Use - Where is the infrastructure (boreholes, lodges, fences, etc.) located? What are the locations and condition of roads, tracks and firebreaks? What areas do the park visitors frequent? What important events took place in the park and what are management's objectives?

Ecosystem and Habitat Maintenance - What and where are the different habitats? What are the wildlife habitat requirements? How have the habitats been affected by fire? Is there bush encroachment? What are the important habitat functions and services for the park?

Controlled Fire Management Requirements

Specific controlled fire management requirements for each FMZ are to be developed to minimise wildfire hazard, enhance park use, and improve habitats.

Wildfire hazard reduction - Strategic controlled burns are to be employed to reduce fuel loads along boundaries, roads and infrastructure. Patchwork mosaic burning is to be used to fragment fuel loads throughout the park area.

Enhance park use – Strategic controlled burns are to be employed to protect life and property; facilitate efficient management; enhance visitor aesthetics and experience; and enhance economic benefits.

Improving ecosystems and habitats - Strategic controlled burns are to be employed to maintain and enhance habitats; reduce fire intensity; improve ecosystem productivity and function; and establish a less repetitious annual burning regime to promote diversity of habitats and increase biodiversity.

The Controlled Fire Management Plan

The Controlled Fire Management Plan for Bwabwata National Park documents the analysis and the controlled fire management requirements for each FMZ for the fire season to come.



Development of the plan focuses on the operational aspects of implementation by employing a 'what, when, where and how' approach, as follows:

- What implementation activity is most suitable?
- When should the activity be employed?
- Where is the implementation activity needed?
- How should the activity be implemented?

Operating procedures for implementation and monitoring activities are detailed in the plan.

Standard Operating Procedures

Development of standard operating procedures (SOPs) for implementation and monitoring activities takes place during the planning process and the SOPs form an integral part of the Controlled Fire Management Plan for Bwabwata National Park.

SOPs will document and detail the work processes to be followed by the fire management teams and other park staff to facilitate consistent compliance with fire management implementation.

4.4 Implementation

Controlled fire management is implemented in the early dry season (May to July) when fires are characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Light winds, cool temperatures, and fuel (grass, litter etc.) that is only partly dried out all limit the intensity and extent of these fires.

Controlled Burning Coordination

The fire management team leader supervises and coordinates all controlled burning operations of the fire management team in each FMZ. The Controlled Fire Management Plan guides the specific location, timing, and method of controlled burning.

Many factors influence fire behaviour and controlled burning operations. Flexibility is required to adapt to prevailing conditions and preparation considers the following:

Coordination liaison and awareness: it is important to coordinate, collaborate, and increase awareness of controlled burning objectives and activities with park staff and residents, neighbours and visitors.

Access: knowledge of locations and condition of roads, tracks and firebreaks to provide access and a safe environment to implement controlled burning.

Start early in the season: early controlled burning is the key to an effective controlled fire management programme. Areas should be targeted as soon as they are dry enough to burn.

Start late in the day: in the afternoon, winds stabilise and the extent of the fire is under control before nightfall, thus reducing the possible uncontrolled spread of the burned area.

Conditions: weather conditions; fuel type and load; landscape features; and fire behaviour.

Control: active suppression of controlled burns is not desirable. Prior to ignition, the fire controller must be prepared for the control of burn in all possible eventualities.

Controlled Protection Burning

Strategic controlled burns within each FMZ are required to establish buffer zones of burnt areas around infrastructure, resources, and boundaries in order to protect life and property; prevent controlled fires escaping; and minimise the uncontrolled wildfire hazard.

Removing fire fuel load: resources and infrastructure are secured by removing fuel loads through controlled burning to establish buffer zones. Fires later in the year will not be able to reach resources or infrastructure with no fuel to burn around them.

Burn against the wind: protection burns, where possible, are implemented against the wind to create a fire of low intensity and low rate of spread that is easier to control.

Boundary protection: protection burns are established, where required, along park or internal boundaries to ensure controlled mosaic burns or uncontrolled fires do not traverse these boundaries.

Controlled Mosaic Burning

Strategic controlled burns are required to establish a mosaic of burnt and un-burnt patches throughout each FMZ. The protection burns enable controlled mosaic burning to be implemented safely and efficiently. The specific location, timing, and method of controlled mosaic burning are guided by the FMZ controlled fire management requirements developed in the ***Controlled Fire Management Plan***: controlled burns of varying extent, intensity, patch scale, and timing are implemented throughout the early dry season.

Combined with the protection burns, a mosaic of interlinked burnt patches and corridors representing approximately 50 per cent of the FMZs creates an effective early dry season firebreak network that minimises the occurrence and extent of late dry season fires.

Controlled burning is implemented by igniting fires from vehicles along roads and tracks or by walking across country.

Uncontrolled Wildfire Management

The management of uncontrolled wildfire relies on the patchwork of interlinked burnt patches and corridors established during the controlled protection and mosaic burning, which creates an early dry season firebreak network.

Uncontrolled wildfire management is important in the late dry season (August to November) when fires are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions and fully cured fuel.

Prevention: coordination, liaison and awareness are critical to inform park staff, residents, tour operators, visitors and neighbours about the hazards of wildfires and the use of prevention measures against wildfires.

Detection and communication: the detection of wildfires is the responsibility of all park staff members and they are required to communicate the location to the fire management team in the FMZ in question. Early detection and swift communication are essential.

Assessment: the fire management team needs to assess the uncontrolled wildfire and determine if it is a hazard to life and property, a threat to park use, or jeopardises ecosystems and habitats. The fire controller must have reliable knowledge and understanding of the area, particularly previously burnt areas, for safe and efficient management.

Response: minimising the threat or hazard of a wildfire involves focusing resources and tactics to protect life and property, contain the wildfire within a controlled area, and suppress and extinguish the wildfire (as a last resort).

Collaboration: if containing the wildfire is beyond the capabilities of the fire management team and park staff, they must notify neighbours and request assistance from them.

4.5 Monitoring

Monitoring of the Controlled Fire Management Programme activities is required throughout the year and involves a combination of park-level observations and remotely sensed information.

Park Level

At the park level, the fire management teams are responsible for monitoring within their own FMZs.

Controlled burning monitoring: follow-up surveillance and monitoring is conducted to ensure that controlled protection and mosaic burning achieve the intended objective(s).

Controlled fire management register: a register of location, timing, and nature of controlled burns and wildfires in each FMZ must be maintained.

Vegetation monitoring: monitoring sites (in a range of habitats) in each FMZ are to be marked and surveyed annually towards the end of the wet season (i.e., in April). Photographs, data on basic vegetation structure, and fire information are to be collected.

Remote Monitoring

The Bwabwata Ecological Institute is responsible for the remote monitoring of fires through the AMESD Advanced Fire Information System. Relevant information and maps are to be developed and communicated, with training in GIS and remote sensing skills and knowledge undertaken. Use of AFIS products will incorporate:

Prediction of fires is of limited use in the park due to scale and should be used in conjunction with local-level observations and weather forecasts. The Fire Danger Index is updated daily and provides a 3-day forecast of the Fire Weather Index and Lowveld Fire Danger Index. Fuel Condition data are updated every 10 days and provide a Vegetation Condition Index and a Vegetation Productivity Index, as well as indicating the number of days since the last burn for each area.

Detection and monitoring of fires: MODIS Active Fire detection data are updated regularly on a daily basis (10am and 3pm). MSG Active Fire detection data are updated every 15 minutes.

Assessment of fires: the MODIS Burned Area Product that maps and documents the extent of fires throughout the year is updated every month. MODIS Imagery is updated every 6 hours and provides visual fire monitoring and assessment.

The wardens and fire management teams should access the available information from the Internet at the: AFIS website - <http://afis.meraka.org.za> and the Global Fire Information Management System (GFIMS) website - <http://www.fao.org/nr/gfims/en>.

The GFIMS provides a fire email alert service whereby registered users of the MODIS-derived Active Fire detection product receive an email whenever a fire is detected in a specified area of interest.

The Controlled Fire Management Report

The fire management teams prepare a Controlled Fire Management Report based on the monitoring information; the report will address objectives and document the outcomes of implementation. Assessments of programme effectiveness and identification of inadequacies will provide fire management recommendations for the following year.

5. Capacity and Resource Requirements

To build the capacity of Bwabwata National Park management to safely and efficiently implement its Controlled Fire Management Programme requires a long-term process aimed at developing the skills and knowledge acquired only through firsthand experience. Most of this training takes place in the course of ongoing fire management implementation and requires strategic and regular trainer intervention.

5.1 Training and Skills Transfer

On-the-job training is required throughout the duration of the initial fire season to build the capacity of the fire management teams to implement the Controlled Fire Management Programme. Ideally, annual follow-up training should be provided in subsequent fire seasons.

A fire management specialist will be required to deliver three training sessions in Bwabwata National Park. Training of fire management teams, other park staff, neighbours, and other key stakeholders will coincide with the planning, implementing and monitoring of the programme in its first year.

Session 1: Controlled Fire Management Planning

A Controlled Fire Management Planning Session of two weeks’ duration is required in April/May and will cover all theoretical aspects of the programme; develop the Controlled Fire Management Plan; and commence implementation training.

Part 1, Controlled Fire Management Programme Theory: covering all aspects of the Controlled Fire Management Programme, including:

- Fire issues	- Planning
- Existing fire management	- Implementation
- The Controlled Fire Management Programme	- Monitoring
- Fire management teams - roles and responsibilities	- Reporting

Part 2, Fire Management Plan Development: a finalised draft of the Controlled Fire Management Plan is developed for Bwabwata NP through the training process whereby data collection, liaison activities, FMZ analyses, and controlled fire management requirement assessments will be carried out.

Part 3, Implementation Activity Introduction: practical introductory training of implementation techniques to achieve the FMZs’ controlled fire management requirements.

Session 2: Controlled Fire Management Implementation

A Controlled Fire Management Implementation Session of four weeks’ duration is required in July to train the fire management teams in operational/practical aspects of controlled burning coordination; controlled protection burning; controlled mosaic burning; and uncontrolled wildfire management. July is a critical time for the teams to acquire these skills, in order to ensure sufficient protection and controlled mosaic burning is implemented prior to the wildfire season.

Session 3: Controlled Fire Management Monitoring and Reporting

A Controlled Fire Management Monitoring and Reporting Session of one week’s duration is required in November to train the fire management teams in collating park-level and remote monitoring information to assess programme effectiveness to date and develop recommendations.

5.2 Equipment

Implementing an effective controlled fire management programme dispenses with the need for costly fire suppression equipment. The following equipment is recommended for the satisfactory implementation of the Controlled Fire Management Programme for Bwabwata National Park (Table 3 to Table 5, below).

Vehicles and Drivers

Vehicle and driver availabilities underpin the effectiveness of successful implementation of the Controlled Fire Management Programme throughout this large and remote park. Suitable 4x4 bakkie vehicles (with competent drivers) are to be prioritised for use in controlled burning operations during the months of May, June and July. A 4x4 bakkie fitted with a twin impellor skid unit constitutes the most suitable mobile fire management equipment for the rough terrain of this remote park.

Personal Protective Equipment (PPE)

Personal equipment for fighting fires should be provided as follows:

Table 3: Personal protective equipment (PPE) required at Bwabwata NP

Description	Fire retardant Conti suit	
Specifications	100% cotton	
No. of units, Bwabwata National Park: Kwando Core Area		X 75
No. of units, Bwabwata National Park: Buffalo Core Area		X 75
No. of units, Bwabwata National Park: Mahango Core Area		X 75
No. of units, Bwabwata National Park: Multiple Use Area		X 75
Total = 300		
Description	Safety boots (pair)	
Specifications	Leather, non-steel-capped	
No. of units, Bwabwata National Park: Kwando Core Area		X 75
No. of units, Bwabwata National Park: Buffalo Core Area		X 75
No. of units, Bwabwata National Park: Mahango Core Area		X 75
No. of units, Bwabwata National Park: Multiple Use Area		X 75
Total = 300		
Description	Broad-brimmed hat	
Specifications	100% cotton	
No. of units, Bwabwata National Park: Kwando Core Area		X 75
No. of units, Bwabwata National Park: Buffalo Core Area		X 75
No. of units, Bwabwata National Park: Mahango Core Area		X 75
No. of units, Bwabwata National Park: Multiple Use Area		X 75
Total = 300		

Description	Gloves (pair)	
Specifications	Heavy duty leather	
No. of units, Bwabwata National Park: Kwando Core Area		X 75
No. of units, Bwabwata National Park: Buffalo Core Area		X 75
No. of units, Bwabwata National Park: Mahango Core Area		X 75
No. of units, Bwabwata National Park: Multiple Use Area		X 75
Total = 300		
Description	UV protection safety glasses	
No. of units, Bwabwata National Park: Kwando Core Area		X 75
No. of units, Bwabwata National Park: Buffalo Core Area		X 75
No. of units, Bwabwata National Park: Mahango Core Area		X 75
No. of units, Bwabwata National Park: Multiple Use Area		X 75
Total = 300		
Description	First aid burn kit	
No. of units, Bwabwata National Park: Kwando Core Area		X 12
No. of units, Bwabwata National Park: Buffalo Core Area		X 12
No. of units, Bwabwata National Park: Mahango Core Area		X 12
No. of units, Bwabwata National Park: Multiple Use Area		X 12
Total = 48		

PPE can be sourced from safety equipment distributors in Namibia.

Fire Management Equipment

The following fire management equipment is required for controlled burning implementation:

Table 4: Mobile vehicle fire management units required at Bwabwata NP

Description	Twin impeller skid unit	
Model	Honda GX160/Davey 2-stage pump	
Specifications	600 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length - 60 m; weight +/- 125 kg	
No. of units, Bwabwata National Park: Kwando Core Area		X 3
No. of units, Bwabwata National Park: Buffalo Core Area		X 3
No. of units, Bwabwata National Park: Mahango Core Area		X 3
No. of units, Bwabwata National Park: Multiple Use Area		X 3
Total = 12		
Description	Dry chemical powder fire extinguishers	
Specifications	9 kg size conforming to minimum fire ratings specified in SANS 1910:2009 for A, B or C class fires or a combination of ABC class fires	
No. of units, Bwabwata National Park: Kwando Core Area		X 3
No. of units, Bwabwata National Park: Buffalo Core Area		X 3
No. of units, Bwabwata National Park: Mahango Core Area		X 3
No. of units, Bwabwata National Park: Multiple Use Area		X 3
Total = 12		
Description	Off-road trailer unit	
Model	Honda GX160/Davey 2-stage pump	
Specifications	5.5 HP engine capacity, 40 mm suction diameter, pump flow rate 400 LMP/10 bar, hose length – 60 m	
No. of units, Bwabwata National Park: Kwando Core Area		X 2
No. of units, Bwabwata National Park: Buffalo Core Area		X 2
No. of units, Bwabwata National Park: Mahango Core Area		X 2
No. of units, Bwabwata National Park: Multiple Use Area		X 2
Total = 8		

Description	Support 2500 L truck skid unit	
Model	Honda GX 160/Davey 2-stage pump	
Specifications	2500 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length – 60 m	
No. of units, Bwabwata National Park: Kwando Core Area		X 2
No. of units, Bwabwata National Park: Buffalo Core Area		X 2
No. of units, Bwabwata National Park: Mahango Core Area		X 2
No. of units, Bwabwata National Park: Multiple Use Area		X 2
Total = 8		
Description	Off-road 4x4 2-tonne fire truck	
Model	Samil 20 or Unimong truck 2 x Honda GX160/Davey 2-stage pump	
Specifications	1500 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length – 60 m	
No. of units, Bwabwata National Park: Kwando Core Area		X 1
No. of units, Bwabwata National Park: Buffalo Core Area		X 1
No. of units, Bwabwata National Park: Mahango Core Area		X 1
No. of units, Bwabwata National Park: Multiple Use Area		X 1
Total = 4		

Table 5: Personnel fire management equipment required at Bwabwata NP

Description	Aluminium drip torch	
Specifications	5 L lightweight high-strength aluminium alloy tank with full-length handle, smooth domed-top to prevent fuel accumulation, and check valve in cover to protect against flashbacks	
No. of units, Bwabwata National Park: Kwando Core Area		X 8
No. of units, Bwabwata National Park: Buffalo Core Area		X 8
No. of units, Bwabwata National Park: Mahango Core Area		X 6
No. of units, Bwabwata National Park: Multiple Use Area		X 10
Total = 32		
Description	Knapsack spray units 20 L high-density polythene tank with double action plunger valve pump	
Specifications	20 L high-density polythene tank with double action plunger valve pump	
No. of units, Bwabwata National Park: Kwando Core Area		X 6
No. of units, Bwabwata National Park: Buffalo Core Area		X 6
No. of units, Bwabwata National Park: Mahango Core Area		X 10
No. of units, Bwabwata National Park: Multiple Use Area		X 10
Total = 32		
Description	Fire beaters	
Specifications	Steel or aluminium handled with rubber head	
No. of units, Bwabwata National Park: Kwando Core Area		X 35
No. of units, Bwabwata National Park: Buffalo Core Area		X 35
No. of units, Bwabwata National Park: Mahango Core Area		X 35
No. of units, Bwabwata National Park: Multiple Use Area		X 40
Total = 145		
Description	Jerry cans	
Specifications	25 L steel	
No. of units, Bwabwata National Park: Kwando Core Area		X 8
No. of units, Bwabwata National Park: Buffalo Core Area		X 8
No. of units, Bwabwata National Park: Mahango Core Area		X 6
No. of units, Bwabwata National Park: Multiple Use Area		X 10
Total = 32		

Specialised fire management equipment can be sourced from distributors in Namibia and/or South Africa.

Communications Equipment

The existing park VHF radio communication network should be strengthened with five handheld VHF radios for each fire management team.

6. Collaboration With Residents and Neighbours of Bwabwata National Park

Coordination and collaboration with park residents and neighbours sharing a boundary with the park constitute an important component of the Controlled Fire Management Programme for Bwabwata National Park.

6.1 Park Residents

Collaborative fire management implementation with Bwabwata NP residents should be strengthened through the Kyaramacan Association in order to share resources and workloads to jointly manage uncontrolled wildfires, enhance land use, and improve the environment of the area.

Residents in the Eastern FMZ of the Multiple Use Area have well-established early dry season controlled fire management. Strategic planning and implementation could be strengthened through collaborative efforts with the Kwando Core Area Fire Management Team in the remote areas along the borders of Angola and Botswana.

Considerable strengthening of controlled fire management implementation in the Omega I and Western FMZs of the Multiple Use Area is required. Different land use in Omega I FMZ (predominantly cattle grazing) and sparse populations in the Western FMZ are hindering implementation. Collaborative efforts with the Buffalo Core Area Fire Management Team are required throughout this part of the park.

6.2 Communal Areas and Conservancies

The DoF, through the Forest Act of 2001, regulates fire management in the communal areas neighbouring the park and authorisation is required to implement controlled burning.

Structured, controlled fire management collaboration and coordination between the park's fire management teams and neighbouring conservancies and communal areas will enable alignment of fire management objectives and the sharing of resources and workloads. The MNC is an ideal collaborative forum for management of fire-related activities in the east of the park, for example.

Integrated community-based fire management (CBFiM) programmes established in conservancies neighbouring the north-east parks between 2007 and 2009 provide an operational framework for coordinated activities: conservancies register a fire management committee to prepare a fire management plan and apply for a Permit to Burn from the DoF. The fire management committee implements a controlled fire management programme to provide tangible livelihood benefits to the community through reducing uncontrolled fires, improving land use, and enhancing environmental management. The traditional authority oversees the functions of the committee and arbitrates on fire-related disputes within the community.

The CBFiM programmes will require technical and financial support from government and international and local development non-governmental organisations such as the Namibia Protected Landscape Conservation Areas Initiative (NAM-PLACE) project, the Strengthening the Capacity of the Protected Area System to Address New Management Challenges (PASS) Project, and the GIZ Community Forest Programme.



6.3 International Collaboration

International collaboration with Botswana and Angola should build on existing regional initiatives, including the Southern African Development Committee (SADC) Regional Fire Management Programme and the SADC Protocol on Forestry (2002).

Integration with the Kavango Zambezi Trans-frontier Conservation Area (KAZA TFCA) should be facilitated through the Namibian Integrated Development Plan and the Master Integrated Development Plan, as well as other existing structures.

6.4 Awareness and Education

Increasing the awareness and understanding of the Controlled Fire Management Programme for Bwabwata National Park in park staff, residents, neighbours, and visitors is required to facilitate safe and efficient implementation. Pre-season multi-stakeholder meetings and/or agreements establish the foundation for this awareness.

Awareness and education programmes are recommended to foster the commitment and support of key role players and clearly describe their roles and responsibilities within the programme. A multi-media public awareness campaign (electronic and print media such as newspapers, posters and brochures) is required to inform park visitors, civil society, government, and non-government institutions of the programme's objectives and achievements.

APPENDIX 2

CONTROLLED FIRE MANAGEMENT PROGRAMME,
ETOSHA NATIONAL PARK



1. Introduction

Etosha National Park is centred on the Etosha Pan, an extensive salt pan remnant of a fossil lake surrounded by an open tree and shrub savanna ecosystem. The semi-arid to arid climate provides a highly variable and seasonal (October to April) average annual rainfall ranging from 300 mm in the south-west to 500 mm in the north-east.

The park contains a diversity of large game and other wildlife species, including rare and endangered species. It is the flagship of Namibia's Protected Areas network and an important tourist destination.

Communal lands with rural populations neighbour the fenced park to the north and west, while privately owned farms border the southern and eastern boundaries. Collaborative efforts to coordinate wildlife and natural resource management with neighbouring communities and private farms is well established in the form of formal arrangements with communal area conservancies and neighbouring farms.

2. Fire History, Management and Issues

Fire is a natural and regular occurrence in savanna ecosystems and fire plays a fundamental role in the vegetation, wildlife, and land use management of Etosha National Park. The management of fires in Etosha NP has been dominated by fire exclusion, protection, and suppression policies dating back to the 1970s. Over the decades, the intentional use of fire developed to simulate lightning-ignited fire regimes and to modify grazing pressure. Annual burning targets of 12 per cent were set and implemented through block-burning techniques. All other anthropogenic fires were suppressed.

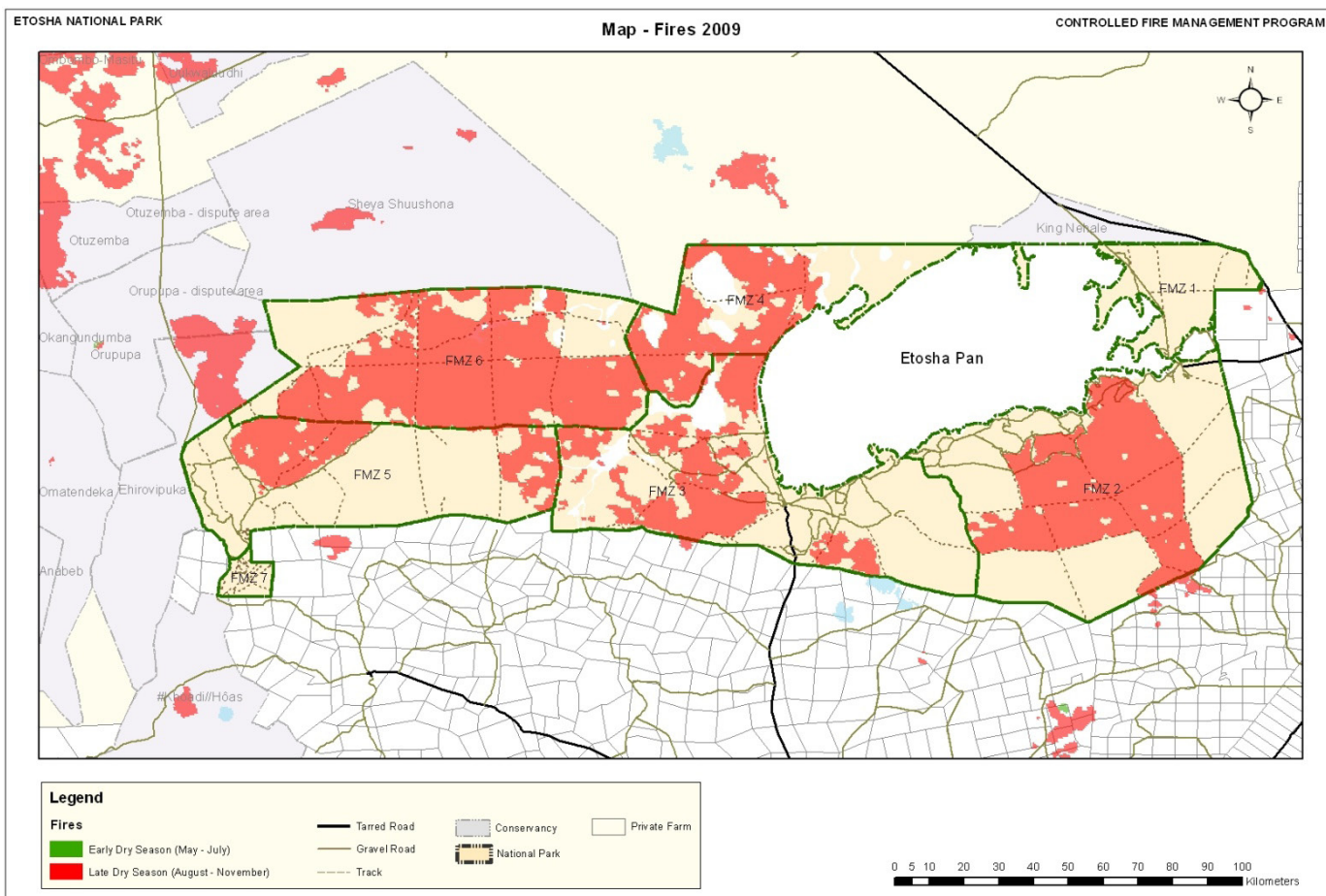
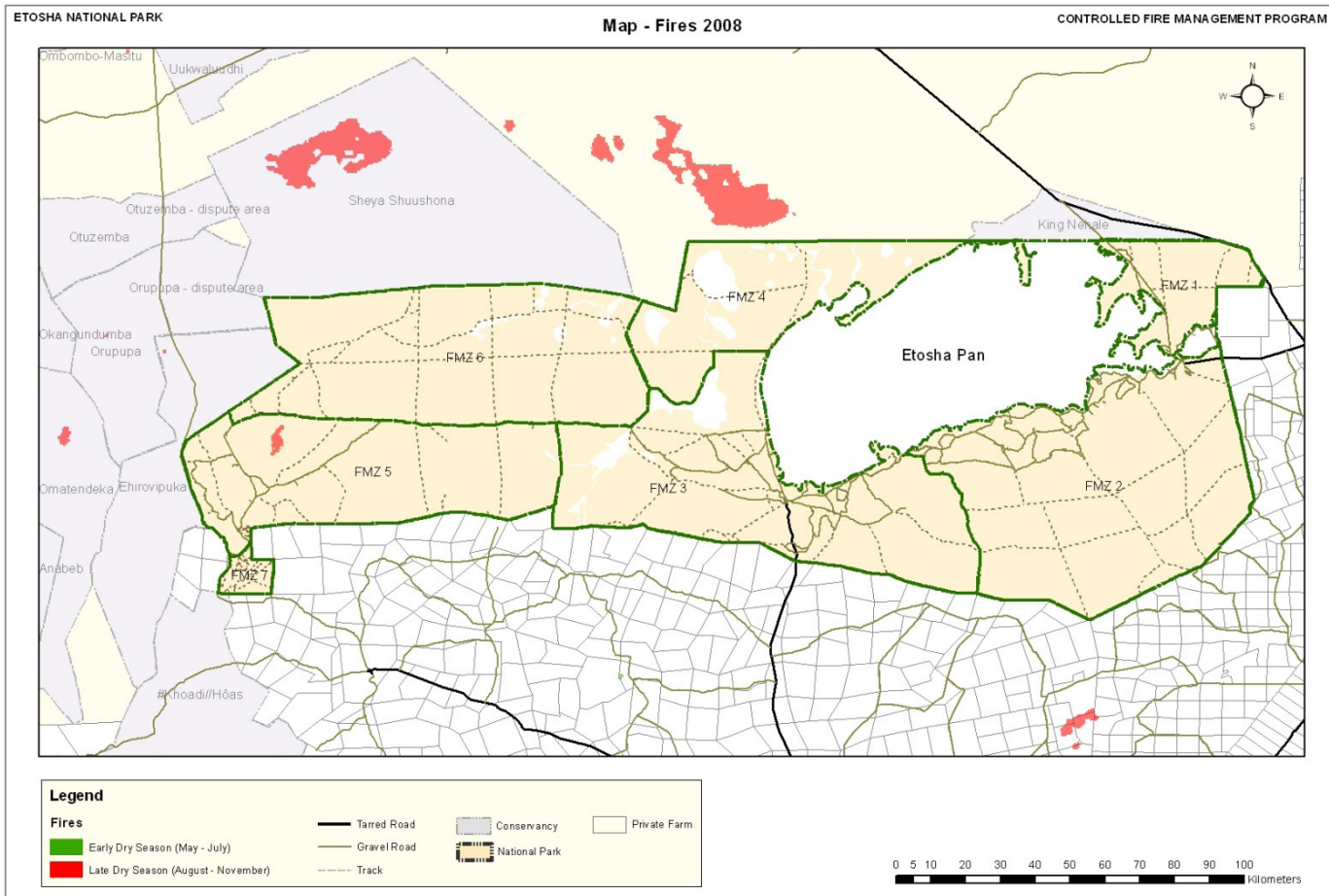
Fire occurrence in Etosha NP has been dominated, almost exclusively, by a late dry season (August to November) regime of high-intensity fires started by neighbours and park visitors; intentional management fires; and lightning. The annual occurrence and distribution of these fires is highly variable: over the past few years, the proportion of the park burnt has varied from 10 to 50% per cent annually.

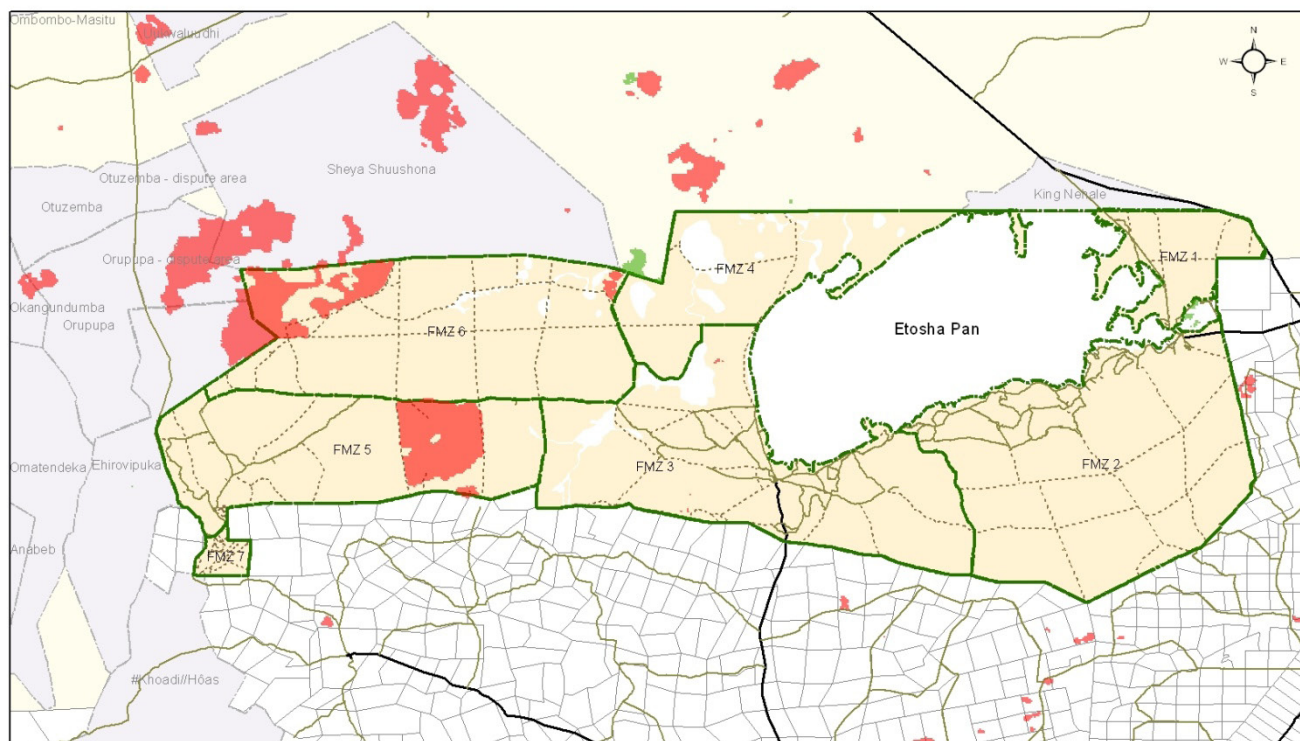
Table 1: Etosha National Park fires, 2008 to 2011

	FIRE SEASON							
	2008		2009		2010		2010	
	Area (km ²)	% of park	Area (km ²)	% of park	Area (km ²)	% of park	Area (km ²)	% of park
Early dry season	-	-	-	-	15	<1	106	<1
Late dry season	18	<1	7,161	38	897	5	6,727	36
Total	18	<1	7,161	38	912	5	6,833	36

% of park area is calculated excluding Etosha Pan.

The relative success of fire exclusion in the park, however, enabled the accumulation of fuel loads that have resulted in extensive, high-intensity uncontrolled fires over recent years (see Figure 1 maps and Table 1, above). Ineffective suppression activities enabled uncontrolled fires in 2008 and 2011 to significantly impact management objectives through the degradation of the park's ecosystems, with fires affecting vegetation structure and species composition; reducing ecosystem function and productivity; threatening high-value and vulnerable ecosystems; and reducing habitat and biological diversity.





Legend

Fires

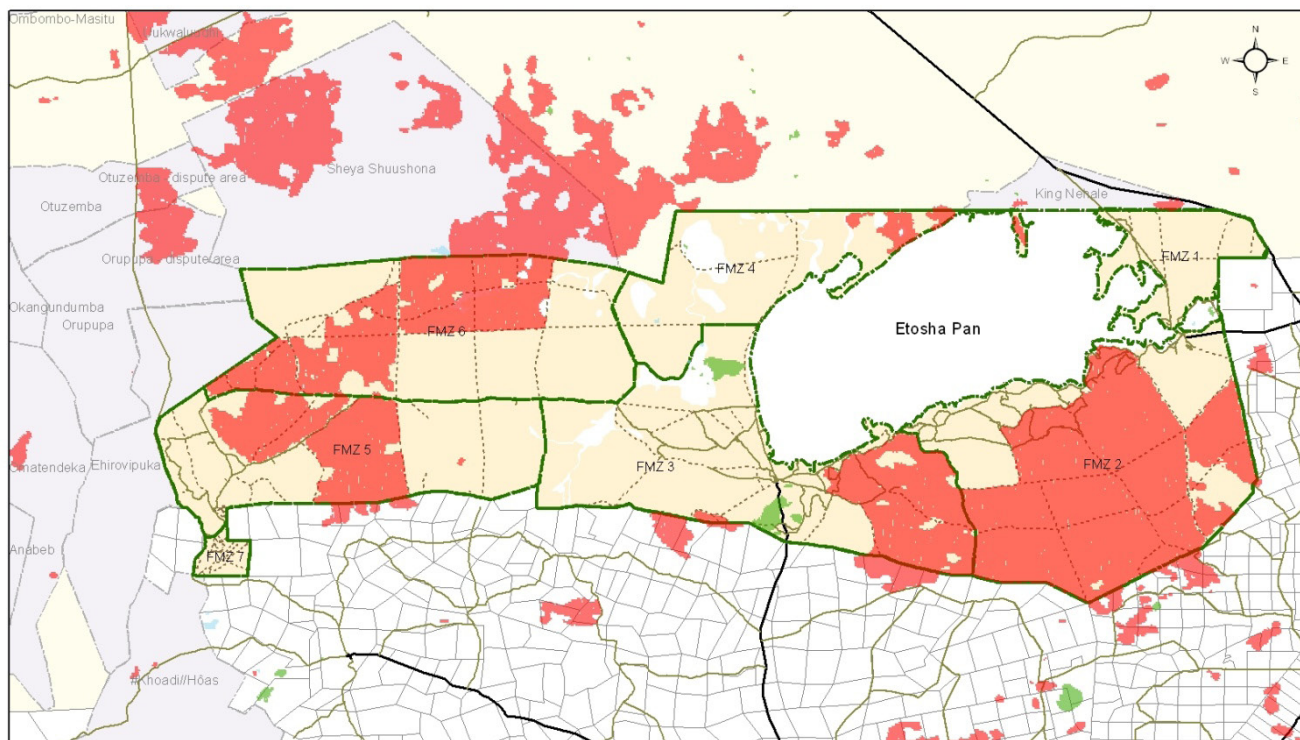
- Early Dry Season (May - July)
- Late Dry Season (August - November)

- Tarred Road
- Gravel Road
- Track

- Conservancy
- National Park

- Private Farm

0 5 10 20 30 40 50 60 70 80 90 100
Kilometers



Legend

Fires

- Early Dry Season (May - July)
- Late Dry Season (August - November)

- Tarred Road
- Gravel Road
- Track

- Conservancy
- National Park

- Private Farm

0 5 10 20 30 40 50 60 70 80 90 100
Kilometers

Figure 1: Etosha National Park fire occurrences, 2008 to 2011

An abrupt change in Etosha NP's fire management policy and strategy was deemed necessary in 2012, in response to the uncontrolled fires of 2009 and 2011.

In line with the 2007 Etosha Management Plan, the revised policy recognises: 'Fire as an integral part of the environment in Etosha and will be applied primarily to reduce fuel loads to limit the size and scale of fires, especially wildfires as and when they occur.'

The Patchwork Mosaic Burn Programme is at a very early stage of development, designed to establish a diversity of fires throughout the dry season (autumn and winter) as part of the implementation of the revised policy. Fire management zones have been demarcated based on vegetation, rainfall, and fire history.

Planning involves setting burning quotas based on biomass and rainfall for each zone. Fires are ignited randomly and allowed to burn freely until they reach the planned quota or pose a threat to property.

Implementation has commenced, but has not yet been institutionalised throughout park management. Considerable gaps exist among park staff in the controlled fire management experience, skills and knowledge required to safely and efficiently implement the programme. Controlled burning attempts in 2012 required suppression intervention and suppression of uncontrolled fires continues as a knee-jerk reaction to some extent.

Monitoring is limited to ground observations of fire intensity and extent, but no records are kept. Remotely sensed maps provide an overview of the distribution of fires.

3. Resources and Capacity

Skills and knowledge of fire management planning, implementation and monitoring are limited among staff.

Vehicles and drivers: availabilities limit implementation of the programme.

Fire management equipment: the existing 15 gas burners, 10 bakkie skid units, and 200 fire beaters are inadequate to implement safe and efficient controlled fire management operations.

Personal protective equipment: the existing staff uniforms are the minimum required for controlled fire management operations.

Communication: the existing park VHF radio communication network is adequate.

Neighbours: the occurrence and management of fire in the areas neighbouring Etosha NP, particularly those sharing a boundary with the park, comprise an integral aspect of park fire management.

There are four communal area conservancies (Ehrovipuka, King Nehale, Sheya Shuushona, and #Khoodi-/Hôas) that share boundaries with Etosha NP in the west and north. Communal lands not registered as conservancies are situated along the northern boundary of the park.

Use of fire as a land management tool is not widespread in the communal lands although uncontrolled wildfires are increasing in the conservancies in the north-west, with up to 20 per cent of areas affected.



Uncontrolled wildfires regularly cross boundaries, with fires entering and exiting the park adjacent to these areas.

Fire management as it pertains to communal areas consists mainly of prevention and suppression approaches. Collaborative management exists, with park management providing human resources and equipment to assist in wildfire suppression in bordering areas. The DoF provides limited support in the form of firebreak maintenance and wildfire suppression.

There are numerous (approximately 54) privately owned farms that border Etosha NP along the southern and eastern boundaries. The majority of farms (80 to 90 per cent) are involved in livestock production, with the remainder operating ecotourism ventures associated with the park.

Management of fire on these farms consists of exclusion and suppression approaches. Land tenure of relatively small blocks and intensive management capacity (firebreaks, fencing, and intensive stocking) have limited the extent and occurrence of fires in the past.

In recent years, however, uncontrolled fires have been increasing in these areas and entering the park. Collaborative management exists with park management engaging in wildfire suppression in areas bordering the farms. Structured collaboration forums are limited to meetings with the Namibia Agricultural Union (NAU).

SWOT analysis: this analysis evaluates and summarises the implementation of the existing Patchwork Mosaic Burn Programme at Etosha National Park.

Table 2: SWOT analysis of the existing Patchwork Mosaic Burn Programme at Etosha NP

	Beneficial	Detrimental
Internal	STRENGTHS - Recent policy change from entrenched fire exclusion and suppression approach - Well-established and dedicated management structure - Strong Scientific Services support - Good access roads throughout - Other departmental support, i.e., maintenance - Consistent remotely sensed monitoring	WEAKNESSES - Insufficient controlled fire management experience, skills and knowledge in staff - Random ignition and ‘open’ burning systems becoming uncontrolled - Vehicle availability limits implementation - Insufficient equipment to implement controlled fires safely - Local habitat monitoring insufficient
External	OPPORTUNITIES - Established collaboration with neighbouring community and conservancy structures - Existing fire management collaboration with neighbours to build upon - Strengthen partnership with DoF to facilitate controlled burning in neighbouring areas - High-profile area for development aid funding support	THREATS - Uncontrolled wildfires from neighbouring areas, i.e., from charcoal production - Park management fires impacting neighbouring properties, particularly private farms - Deeply entrenched exclusion and suppression management on private farms - Numerous neighbouring landholders and no established collaborative management forums - Weak partnership with DoF hinders facilitation of controlled burning in neighbouring areas - Management influenced by public perception and political motivations

4. The Controlled Fire Management Programme

The Controlled Fire Management Programme for Etosha National Park represented by this document is based on the strategic implementation of controlled burning to manipulate the park's ecosystems to enhance park management objectives.

The programme builds on the existing patchwork mosaic burning fire management approach developed in 2012 to safely and efficiently implement controlled fire management activities that:

- minimise wildfire hazard;
- enhance park use; and
- improve ecosystems and habitats.

The programme implements an overarching strategic early dry season burning regime to establish a mosaic of interlinked burnt patches and corridors representing approximately 35 per cent of the park (excluding the Etosha Pan). This creates an early dry season firebreak network that minimises the occurrence and extent of late dry season fires.

The specific location, timing, and method of controlled burning is guided by park use and habitat requirements. An important consideration will be to ensure controlled management fires do not cross into private farms on the southern and eastern boundaries.

A dedicated fire management team carries out a structured annual planning, implementation and monitoring process based on fire management zones (FMZs). This enables adaptable and robust fire management that addresses the diversity of habitats and park use while achieving overall park fire management objectives.

This effectively dispenses with the need for the costly and hazardous conventional fire management practices of establishing firebreak networks and undertaking wildfire suppression.

4.1 Fire Management Teams

A fire management team is established for each park management area (Halali, Namutoni, Okaukuejo, and Otjovazandu).

Each fire management team comprises a functional group of five MET staff members based at a different site within Etosha NP, whose role is to carry out the structured planning, implementation and monitoring of the programme within the FMZs under their jurisdiction every year.

The team leader (the park area warden) within each team works in close collaboration with the Etosha Ecological Institute to coordinate and supervise the Controlled Fire Management Programme.

It is important that the team members are maintained in their roles for at least three years, in order to build fire management skills and knowledge and ensure development and improvement of the programme.

4.2 Fire Management Zones

Etosha is currently divided into seven FMZs, based on habitat, infrastructure, park use, and fire history/ issues (see Figure 1). It is recommended these FMZs are adjusted to align with the existing park management areas of Halali, Namutoni, Okaukuejo and Otjovazandu.



Controlled fire management is tailored to the specific requirements of each FMZ through a structured planning, implementation and monitoring process.

4.3 Planning

Controlled fire management planning is carried out at the end of the summer/wet season (March/April) every year. A planning process of information collection and analysis at the FMZ level is required to determine the key planning factors to develop the controlled fire management requirements for the following fire season.

Fire Management Zone Analysis

A combination of satellite data, field surveys, and liaison with park staff, concessionaires and neighbours is used to collect and analyse key information. A series of questions highlights the information required, as follows:

Fire history and issues - When did the fires occur in the last three years? What was the source of fire (controlled or uncontrolled)? Where did the fires come from? How much rain fell last year? How much fuel load exists? What and where are the fire hazards?

Park use - Where is the infrastructure (boreholes, lodges, fences, etc.) located? What are the locations and condition of roads, tracks and firebreaks? What areas do the park visitors frequent? What important events took place in the park and what are management's objectives?

Ecosystem and habitat maintenance - What and where are the different habitats? What are the wildlife habitat requirements? How have the habitats been affected by fire? Is there bush encroachment? What are the important habitat functions and services for the park?

Controlled fire management requirements

Specific controlled fire management requirements for each FMZ are to be developed to minimise wildfire hazard, enhance park use, and improve habitats.

Wildfire hazard reduction - Strategic controlled burns are to be employed to reduce fuel loads along boundaries, roads and infrastructure. Patchwork mosaic burning is to be used to fragment fuel loads throughout the park area.

Enhance park use - Strategic controlled burns are to be employed to protect life and property; facilitate efficient management; enhance visitor aesthetics and experience; and enhance economic benefits.

Improving ecosystems and habitats - Strategic controlled burns are to be employed to maintain and enhance habitats; reduce fire intensity; improve ecosystem productivity and function; and establish a less repetitious annual burning regime to promote diversity of habitats and increase biodiversity.

The Controlled Fire Management Plan

The *Controlled Fire Management Plan for Etosha National Park* documents the analysis and the controlled fire management requirements for each FMZ for the fire season to come. Development of the plan focuses on the operational aspects of implementation by employing a 'what, when, where and how' approach, as follows:

- What implementation activity is most suitable?
- When should the activity be employed?
- Where is the implementation activity needed?
- How should the activity be implemented?

Operating procedures for implementation and monitoring activities are detailed in the plan.

Standard Operating Procedures

Development of standard operating procedures (SOPs) for implementation and monitoring activities takes place during the planning process and the SOPs form an integral part of the Controlled Fire Management Plan for Etosha National Park.

SOPs will document and detail the work processes to be followed by the fire management teams and other park staff to facilitate consistent compliance with fire management implementation.

4.4 Implementation

Controlled fire management is implemented in the early dry season (May to July) when fires are characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Light winds, cool temperatures, and fuel (grass, litter etc.) that is only partly dried all out limit the intensity and extent of these fires.

Controlled Burning Coordination

The fire management team leader supervises and coordinates all controlled burning operations of the fire management team in its FMZs. The Controlled Fire Management Plan guides the specific location, timing, and method of controlled burning.

Many factors influence fire behaviour and controlled burning operations. Flexibility is required to adapt to prevailing conditions and preparation considers the following:

Coordination liaison and awareness: it is important to coordinate, collaborate, and increase awareness of controlled burning objectives and activities with park staff and residents, neighbours and visitors.

Access: knowledge of locations and condition of roads, tracks and firebreaks to provide access and a safe environment to implement controlled burning.

Start early in the season: early controlled burning is the key to an effective controlled fire management programme. Areas should be targeted as soon as they are dry enough to burn.

Start late in the day: in the afternoon, winds stabilise and the extent of the fire is under control before nightfall, thus reducing the possible uncontrolled spread of the burned area.

Conditions: weather conditions; fuel type and load; landscape features; and fire behaviour.



Control: active suppression of controlled burns is not desirable. Prior to ignition, the fire controller must be prepared for the control of burn in all possible eventualities.

Controlled Protection Burning

Strategic controlled burns within each FMZ are required to establish buffer zones of burnt areas around infrastructure, resources, and boundaries in order to protect life and property; prevent controlled fires escaping; and minimise the uncontrolled wildfire hazard.

Removing fire fuel load: resources and infrastructure are secured by removing fuel loads through controlled burning to establish buffer zones. Fires later in the year will not be able to reach resources or infrastructure with no fuel to burn around them.

Burn against the wind: protection burns, where possible, are implemented against the wind to create a fire of low intensity and low rate of spread that is easier to control.

Boundary protection: protection burns are established, where required, along park or internal boundaries to ensure controlled mosaic burns or uncontrolled fires do not traverse these boundaries.

Controlled Mosaic Burning

Strategic controlled burns are required to establish a mosaic of burnt and un-burnt patches throughout each FMZ. The protection burns enable controlled mosaic burning to be implemented safely and efficiently. The specific location, timing, and method of controlled mosaic burning are guided by the FMZ controlled fire management requirements developed in the Controlled Fire Management Plan: controlled burns of varying extent, intensity, patch scale, and timing are implemented throughout the early dry season.

Combined with the protection burns, a mosaic of interlinked burnt patches and corridors representing approximately 35 per cent of the FMZs creates an effective early dry season firebreak network that minimises the occurrence and extent of late dry season fires.

Controlled burning is implemented by igniting fires from vehicles along roads and tracks or by walking across country.

Uncontrolled Wildfire Management

The management of uncontrolled wildfire relies on the patchwork of interlinked burnt patches and corridors established during the controlled protection and mosaic burning, which creates an early dry season firebreak network.

Uncontrolled wildfire management is important in the late dry season (August to November) when fires are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions and fully cured fuel.

Prevention: coordination, liaison and awareness are critical to inform park staff, residents, tour operators, visitors and neighbours about the hazards of wildfires and the use of prevention measures against wildfires.

Detection and communication: the detection of wildfires is the responsibility of all park staff members and they are required to communicate the location to the fire management team in the FMZ in question. Early detection and swift communication are essential.

Assessment: the fire management team needs to assess the uncontrolled wildfire and determine if it is a hazard to life and property, a threat to park use, or jeopardises ecosystems and habitats. The fire controller must have reliable knowledge and understanding of the area, particularly previously burnt areas, for safe and efficient management.

Response: minimising the threat or hazard of a wildfire involves focusing resources and tactics to protect life and property, contain the wildfire within a controlled area, and suppress and extinguish the wildfire (as a last resort).

Collaboration: if containing the wildfire is beyond the capabilities of the fire management team and park staff, they must notify neighbours and request assistance from them.

4.5 Monitoring

Monitoring of the Controlled Fire Management Programme activities is required throughout the year and involves a combination of park-level observations and remotely sensed information.

Park Level

At the park level, the fire management teams are responsible for monitoring within their own FMZs.

Controlled burning monitoring: follow-up surveillance and monitoring is conducted to ensure that controlled protection and mosaic burning achieve the intended objective(s).

Controlled fire management register: a register of location, timing, and nature of controlled burns and wildfires in each FMZ must be maintained.

Vegetation monitoring: monitoring sites (in a range of habitats) in each FMZ are to be marked and surveyed annually towards the end of the wet season (i.e., in April). Photographs, data on basic vegetation structure, and fire information are to be collected.

Remote Monitoring

The Etosha Ecological Institute is responsible for the remote monitoring of fires through the AMESD Advanced Fire Information System. Relevant information and maps are to be developed and communicated, with training in GIS and remote sensing skills and knowledge undertaken. Use of AFIS products will incorporate:

Prediction of fires is of limited use in the park due to scale and should be used in conjunction with local-level observations and weather forecasts. The Fire Danger Index is updated daily and provides a 3-day forecast of the Fire Weather Index and Lowveld Fire Danger Index. Fuel Condition data are updated every 10 days and provide a Vegetation Condition Index and a Vegetation Productivity Index, as well as indicating the number of days since the last burn for each area.

Detection and monitoring of fires: MODIS Active Fire detection data are updated regularly on a daily basis (10am and 3pm). MSG Active Fire detection data are updated every 15 minutes.



Assessment of fires: the MODIS Burned Area Product that maps and documents the extent of fires throughout the year is updated every month. MODIS Imagery is updated every 6 hours and provides visual fire monitoring and assessment.

The wardens and fire management teams should access the available information from the Internet at the: AFIS website - <http://afis.meraka.org.za> and the Global Fire Information Management System (GFIMS) website - <http://www.fao.org/nr/gfims/en>.

The GFIMS provides a fire email alert service whereby registered users of the MODIS-derived Active Fire detection product receive an email whenever a fire is detected in a specified area of interest.

The Controlled Fire Management Report

The fire management teams prepare a *Controlled Fire Management Report* based on the monitoring information; the report will address objectives and document the outcomes of implementation. Assessments of programme effectiveness and identification of inadequacies will provide fire management recommendations for the following year.

5. Capacity and Resource Requirements

To build the capacity of Etosha National Park management to safely and efficiently implement its Controlled Fire Management Programme requires a long-term process aimed at developing the skills and knowledge acquired only through firsthand experience. Most of this training takes place in the course of ongoing fire management implementation and requires strategic and regular trainer intervention.

5.1 Training and Skills Transfer

On-the-job training is required throughout the duration of the initial fire season to build the capacity of the fire management teams to implement the Controlled Fire Management Programme. Ideally, annual follow-up training should be provided in subsequent fire seasons.

A fire management specialist will be required to deliver three training sessions in Etosha National Park. Training of fire management teams, other park staff, neighbours, and other key stakeholders will coincide with the planning, implementing and monitoring of the programme in its first year.

Session 1: Controlled Fire Management Planning

A Controlled Fire Management Planning Session of two weeks’ duration is required in April/May and will cover all theoretical aspects of the programme; develop the Controlled Fire Management Plan; and commence implementation training.

Part 1, Controlled Fire Management Programme Theory: covering all aspects of the Controlled Fire Management Programme, including:

- Fire issues	- Planning
- Existing fire management	- Implementation
- The Controlled Fire Management Programme	- Monitoring
- Fire management teams - roles and responsibilities	- Reporting

Part 2, Fire Management Plan Development: a finalised draft of the Controlled Fire Management Plan is developed for Etosha NP through the training process whereby data collection, liaison activities, FMZ analyses, and controlled fire management requirement assessments will be carried out.

Part 3, Implementation Activity Introduction: practical introductory training of implementation techniques to achieve the FMZs' controlled fire management requirements.

Session 2: Controlled Fire Management Implementation

A Controlled Fire Management Implementation Session of four weeks' duration is required in July to train the fire management teams in operational/practical aspects of controlled burning coordination; controlled protection burning; controlled mosaic burning; and uncontrolled wildfire management.

July is a critical time for the teams to acquire these skills, in order to ensure sufficient protection and controlled mosaic burning is implemented prior to the wildfire season.

Session 3: Controlled Fire Management Monitoring and Reporting

A Controlled Fire Management Monitoring and Reporting Session of one week's duration is required in November to train the fire management teams in collating park-level and remote monitoring information to assess programme effectiveness to date and develop recommendations.

5.2 Equipment

Implementing an effective controlled fire management programme dispenses with the need for costly fire suppression equipment. The following equipment is recommended for the satisfactory implementation of the Controlled Fire Management Programme for Etosha National Park (Table 3 to Table 5, below).

Vehicles and Drivers

Vehicle and driver availabilities underpin the effectiveness of successful implementation of the Controlled Fire Management Programme throughout this large and remote park.

Suitable 4x4 bakkie vehicles (with competent drivers) are to be prioritised for use in controlled burning operations during the months of May, June and July. A 4x4 bakkie fitted with a twin impellor skid unit constitutes the most suitable mobile fire management equipment for the rough terrain of this remote park.



Personal Protective Equipment (PPE)

Personal equipment for fighting fires should be provided as follows:

Table 3: Personal protective equipment (PPE) required at Etosha NP

Description	Fire retardant Conti suit	
Specifications	100% cotton	
No. of units, Etosha National Park: Halali park management area		X 200
No. of units, Etosha National Park: Okaukuejo park management area		X 200
No. of units, Etosha National Park: Namutoni park management area		X 200
No. of units, Etosha National Park: Otjovazandu park management area		X 200
Total = 800		
Description	Safety boots (pair)	
Specifications	Leather, non-steel-capped	
No. of units, Etosha National Park: Halali park management area		X 200
No. of units, Etosha National Park: Okaukuejo park management area		X 200
No. of units, Etosha National Park: Namutoni park management area		X 200
No. of units, Etosha National Park: Otjovazandu park management area		X 200
Total = 800		
Description	Broad-brimmed hat	
Specifications	100% cotton	
No. of units, Etosha National Park: Halali park management area		X 200
No. of units, Etosha National Park: Okaukuejo park management area		X 200
No. of units, Etosha National Park: Namutoni park management area		X 200
No. of units, Etosha National Park: Otjovazandu park management area		X 200
Total = 800		
Description	Gloves (pair)	
Specifications	Heavy duty leather	
No. of units, Etosha National Park: Halali park management area		X 200
No. of units, Etosha National Park: Okaukuejo park management area		X 200
No. of units, Etosha National Park: Namutoni park management area		X 200
No. of units, Etosha National Park: Otjovazandu park management area		X 200
Total = 800		
Description	UV protection safety glasses	
No. of units, Etosha National Park: Halali park management area		X 200
No. of units, Etosha National Park: Okaukuejo park management area		X 200
No. of units, Etosha National Park: Namutoni park management area		X 200
No. of units, Etosha National Park: Otjovazandu park management area		X 200
Total = 800		
Description	First aid burn kit	
No. of units, Etosha National Park: Halali park management area		X 20
No. of units, Etosha National Park: Okaukuejo park management area		X 20
No. of units, Etosha National Park: Namutoni park management area		X 20
No. of units, Etosha National Park: Otjovazandu park management area		X 20
Total = 80		

PPE can be sourced from safety equipment distributors in Namibia.

Fire Management Equipment

The following fire management equipment is required for controlled burning implementation:

Table 4: Mobile vehicle fire management units required at Etosha NP

Description	Twin impeller skid unit	
Model	Honda GX160/Davey 2-stage pump	
Specifications	600 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length - 60 m; weight +/- 125 kg.	
No. of units, Etosha National Park: Halali park management area		X 3
No. of units, Etosha National Park: Okaukuejo park management area		X 3
No. of units, Etosha National Park: Namutoni park management area		X 3
No. of units, Etosha National Park: Otjovazandu park management area		X 3
Total = 12		
Description	Dry chemical powder fire extinguishers	
Specifications	9 kg size conforming to minimum fire ratings specified in SANS 1910:2009 for A, B or C class fires or a combination of ABC class fires	
No. of units, Etosha National Park: Halali park management area		X 3
No. of units, Etosha National Park: Okaukuejo park management area		X 3
No. of units, Etosha National Park: Namutoni park management area		X 3
No. of units, Etosha National Park: Otjovazandu park management area		X 3
Total = 12		
Description	Off-road trailer unit	
Model	Honda GX160/Davey 2-stage pump	
Specifications	5.5 HP engine capacity, 40 mm suction diameter, pump flow rate 400 LMP/10 bar, hose length – 60 m	
No. of units, Etosha National Park: Halali park management area		
No. of units, Etosha National Park: Okaukuejo park management area		
No. of units, Etosha National Park: Namutoni park management area		
No. of units, Etosha National Park: Otjovazandu park management area		
Total = 10		
Description	Support 2500 L truck skid unit	
Model	Honda GX 160/Davey 2-stage pump	
Specifications	2500 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length – 60 m	
No. of units, Etosha National Park: Halali park management area		
No. of units, Etosha National Park: Okaukuejo park management area		
No. of units, Etosha National Park: Namutoni park management area		
No. of units, Etosha National Park: Otjovazandu park management area		
Total = 10		
Description	Off-road 4x4 2-tonne fire truck	
Model	Samil 20 or Unimong truck 2 x Honda GX160/Davey 2-stage pump	
Specifications	1500 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length – 60 m	
No. of units, Etosha National Park: Halali park management area		X 2
No. of units, Etosha National Park: Okaukuejo park management area		X 2
No. of units, Etosha National Park: Namutoni park management area		X 2
No. of units, Etosha National Park: Otjovazandu park management area		X 2
Total = 8		
Description	Off-road 4x4 5-tonne fire truck	
Model	Samil 50 2 x Honda GX160/Davey 2-stage pump	
Specifications	4000 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10bar; hose length – 60 m	
No. of units, Etosha National Park: Halali park management area		X 2
No. of units, Etosha National Park: Okaukuejo park management area		X 2
No. of units, Etosha National Park: Namutoni park management area		X 2
No. of units, Etosha National Park: Otjovazandu park management area		X 2
Total = 4		

Table 5: Personnel fire management equipment required at Etosha NP

Description	Aluminium drip torch	
Specifications	5 L lightweight high-strength aluminium alloy tank with full-length handle, smooth domed-top to prevent fuel accumulation, and check valve in cover to protect against flashbacks	
No. of units, Etosha National Park: Halali park management area		X 10
No. of units, Etosha National Park: Okaukuejo park management area		X 10
No. of units, Etosha National Park: Namutoni park management area		X 10
No. of units, Etosha National Park: Otjovazandu park management area		X 10
Total = 40		
Description	Knapsack spray units 20 L high-density polythene tank with double action plunger valve pump	
Specifications	20 L high-density polythene tank with double action plunger valve pump	
No. of units, Etosha National Park: Halali park management area		X 10
No. of units, Etosha National Park: Okaukuejo park management area		X 10
No. of units, Etosha National Park: Namutoni park management area		X 10
No. of units, Etosha National Park: Otjovazandu park management area		X 10
Total = 40		
Description	Fire beaters	
Specifications	Steel or aluminium handled with rubber head	
No. of units, Etosha National Park: Halali park management area		X 100
No. of units, Etosha National Park: Okaukuejo park management area		X 100
No. of units, Etosha National Park: Namutoni park management area		X 100
No. of units, Etosha National Park: Otjovazandu park management area		X 100
Total = 400		
Description	Jerry cans	
Specifications	25 L steel	
No. of units, Etosha National Park: Halali park management area		X 10
No. of units, Etosha National Park: Okaukuejo park management area		X 10
No. of units, Etosha National Park: Namutoni park management area		X 10
No. of units, Etosha National Park: Otjovazandu park management area		X 10
Total = 40		

Specialised fire management equipment can be sourced from distributors in Namibia and/or South Africa.

Communication

The existing Park VHF radio communication network should be strengthened with five handheld VHF radios for each fire management team.

6. Collaboration with Residents and Neighbours of Etosha National Park

Coordination and collaboration with park residents and neighbours sharing a boundary with the park constitute an important component of the Controlled Fire Management Programme for Etosha National Park.

The DoF, through the Forest Act of 2001, regulates fire management in the neighbouring private farms, communal area conservancies and communal lands. To implement controlled burning these landholders/land managers require authorisation from the DoF.

Structured, controlled fire management collaboration and coordination between the park's fire management teams and neighbouring conservancies and communal areas will enable alignment of fire management objectives and the sharing of resources and workloads.

The park fire management teams implement direct liaison coordination, joint training, and implementation with the neighbouring landholders/land managers.

6.1 Communal Areas and Conservancies

Structured community-based fire management (CBFiM) programmes should be established in the four communal area conservancies (Ehrovipuka, King Nehale, #Khoadi-/Hôas and Sheya Shuushona) that share a boundary with Etosha National Park.

Integrated CBFiM programmes established in conservancies neighbouring the north-east parks between 2007 and 2009 provide an operational framework for coordinated activities: conservancies register a fire management committee to prepare a fire management plan and apply for a Permit to Burn from the DoF. The fire management committee implements a controlled fire management programme to provide tangible livelihood benefits to the community through reducing uncontrolled fires, improving land use, and enhancing environmental management. The traditional authority oversees the functions of the committee and arbitrates on fire-related disputes within the community.

The CBFiM programmes will require technical and financial support from government and international and local development non-governmental organisations such as the Namibia Protected Landscape Conservation Areas Initiative (NAM-PLACE) project, the Strengthening the Capacity of the Protected Area System to Address New Management Challenges (PASS) Project, and the GIZ Community Forest Programme.

6.2 Private Farms

Controlled fire management collaboration and coordination with the privately owned farms bordering Etosha NP along the south and east boundaries should build upon existing relationships. Progressive efforts will be required to change the deeply entrenched fire exclusion approaches to those based on controlled fire management practices.

The park fire management teams need to notify neighbours and facilitate the sharing of resources, equipment, and workloads, particularly when implementing protection burns along shared boundaries. In the short term it is particularly important to ensure the park controlled fire management activities do not extend into farms that maintain fire exclusion management.



Development of private-farm-controlled fire management programmes is to be encouraged and supported by park management through joint training, sharing of technical skills and knowledge, and implementation activities. Controlled fire management programmes and permits are administered through the DoF.

6.3 Awareness and Education

Increasing the awareness and understanding of the Controlled Fire Management Programme for Etosha National Park in park staff, residents, neighbours, and visitors is required to facilitate safe and efficient implementation. Pre-season multi-stakeholder meetings and/or agreements establish the foundation for this awareness.

Awareness and education programmes are recommended to foster the commitment and support of key role players and clearly describe their roles and responsibilities within the programme. A multi-media public awareness campaign (electronic and print media such as newspapers, posters and brochures) is required to inform park visitors, civil society, government, and non-government institutions of the programme's objectives and achievements.

Appendix 3

CONTROLLED FIRE MANAGEMENT PROGRAMME,
KHAUDUM NATIONAL PARK



1. Introduction

Khaudum National Park covers approximately 3,845 km² of Kavango-East Region in north-east Namibia. Kalahari woodlands of the remnant sand dunes, interspersed by fossil drainage lines called omurumba, dominate the savanna woodland ecosystem. The semi-arid climate provides a variable and highly seasonal (October to April) average annual rainfall of approximately 450 mm.

This large, remote and essentially unfenced park contains a diversity of large game and other wildlife species, and includes rare species such as the African wild dog and roan antelope.

Botswana borders the park to the east and communal lands with rural populations surround the park to the north, west and south. Collaborative efforts to coordinate wildlife and natural resource management with neighbouring communities are established through communal area conservancies and the Khaudum North Complex (KNC).

2. Fire History, Management and Issues

Fire is a natural and regular occurrence in savanna ecosystems and fire plays a fundamental role in the vegetation, wildlife, and land use management of Khaudum National Park.

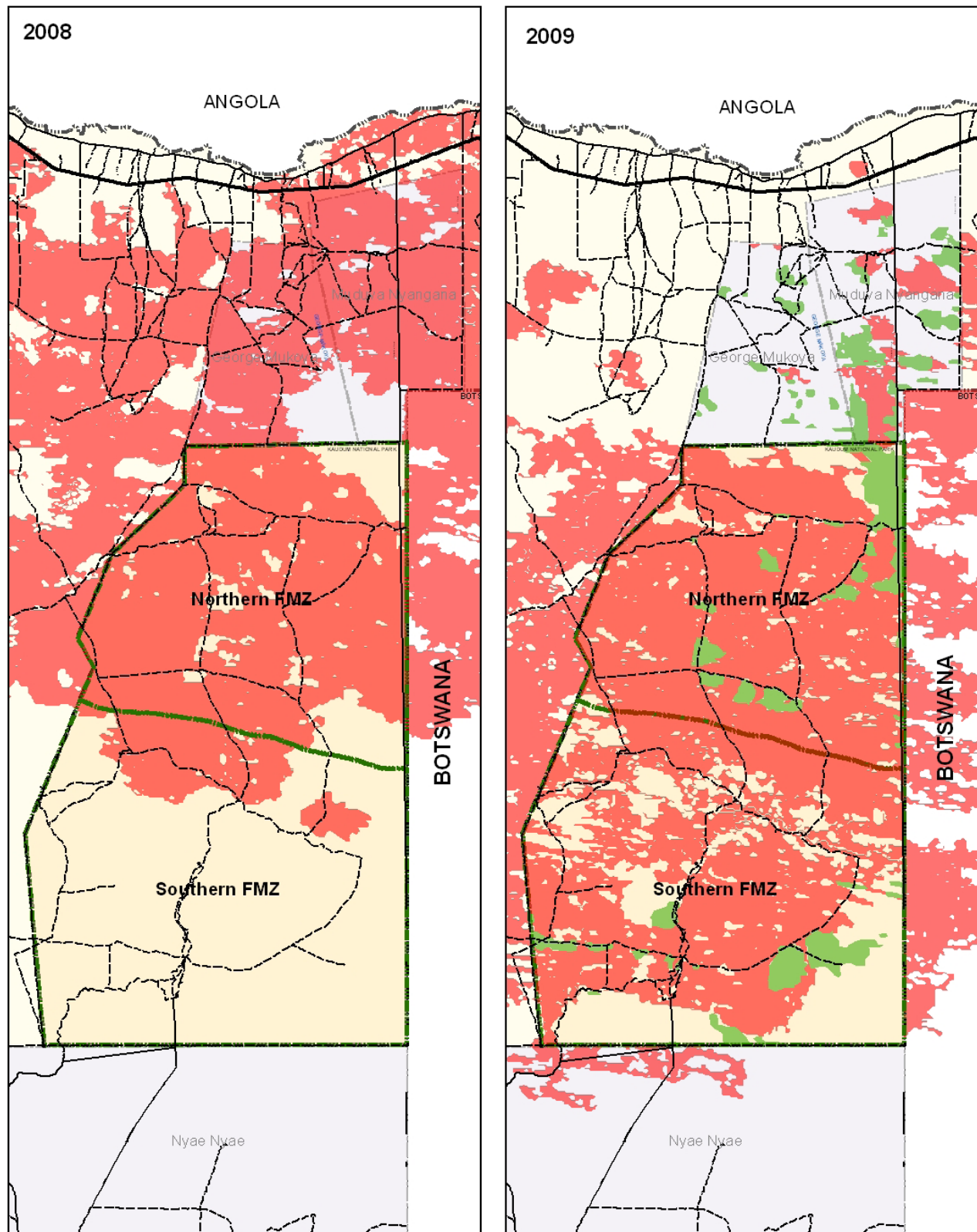
Fires occur almost exclusively in the dry season, with early dry season (May to July) fires characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Late dry season (August to November) fires, on the other hand, are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions.

Management of fires in Khaudum NP has been dominated by fire exclusion, protection, and suppression policies dating back several decades. This has resulted in extensive uncontrolled wildfires in the late dry season occurring over most (50 to 85 per cent) of the park area every year. These high-intensity fires enter from neighbouring areas, particularly Botswana – where border firebreaks 50 m wide fail to arrest the spread of fires.

Over the past two decades, the annual fire regime (outbreaks originating in Botswana primarily) has contributed to the degradation of the park's ecosystem and failure of management objectives by affecting vegetation structure and species composition; reducing ecosystem function and productivity; threatening high value and vulnerable ecosystems; and reducing habitat and biological diversity.

Table 1: Khaudum National Park fires, 2008 to 2011

	FIRE SEASON							
	2008		2009		2010		2011	
	Area (km ²)	% of park	Area (km ²)	% of park	Area (km ²)	% of park	Area (km ²)	% of park
Early dry season	-	-	273	7	395	10	10	<1
Late dry season	1,633	42	2,636	69	887	23	3,283	85
Total	1,633	42	2,909	76	1,282	33	3,293	85



Legend

Fires

- Early Dry Season (Controlled Burns)
- Late Dry Season (Uncontrolled Wildfires)

— Tarred Road

— Gravel Road

— Track

Conservancy

National Park

0 5 10 20 30 Kilometers



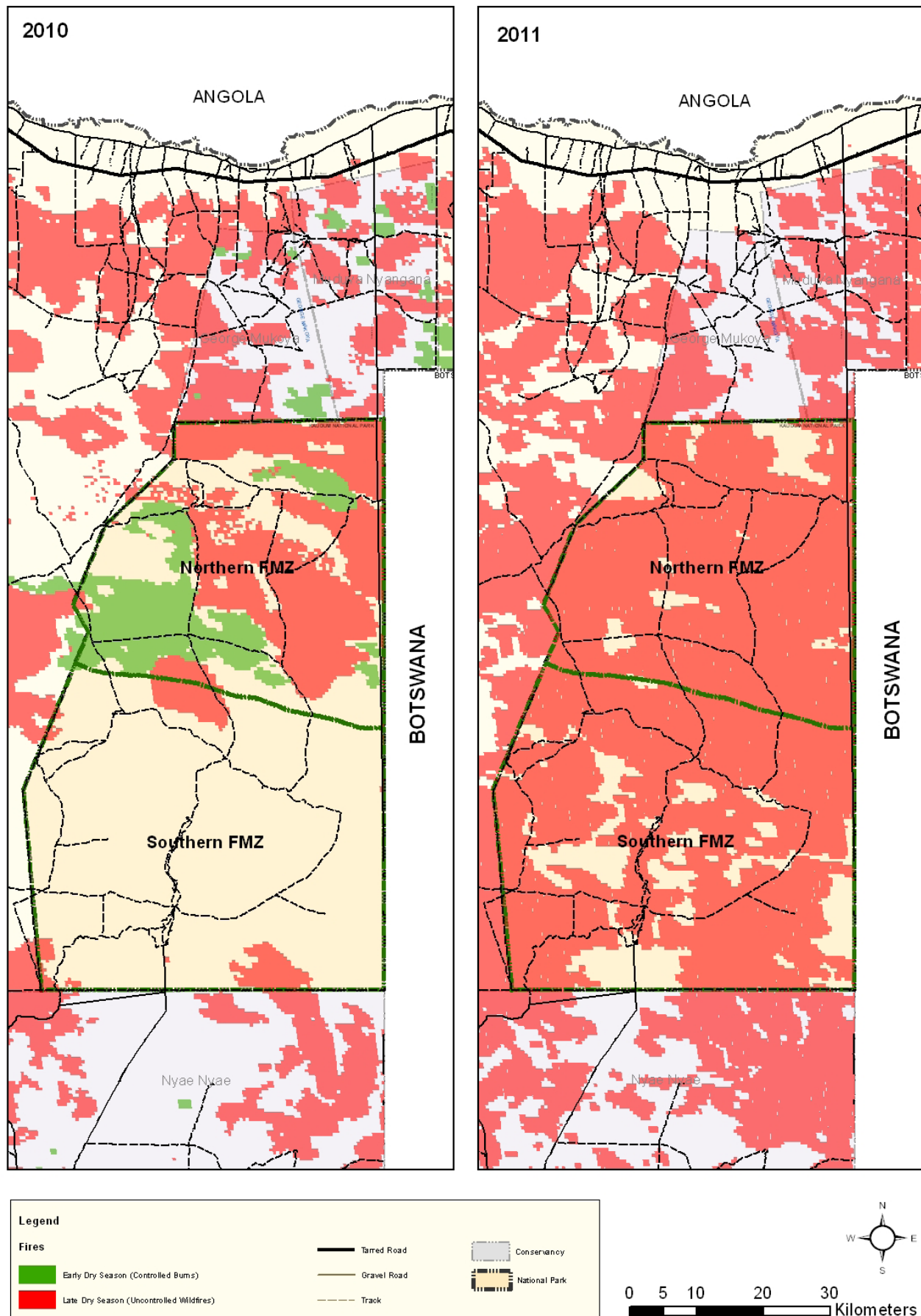


Figure 1: Khudum National Park fire occurrences, 2008 to 2011

In 2009, Khaudum NP embarked on a controlled burning programme through the Khaudum North Complex Integrated Fire Management Programme.

This programme aims to use controlled burning to reduce the area burnt annually and shift the seasonality of burning to the early dry season. Collaboration with the Maduva Nyangana and George Makoya conservancies is structured in order to share resources and workloads to manage uncontrolled wildfires, enhance land use, and improve the environment of the KNC.

Planning involves designating preparation, controlled burning, and assessment activities through the park's annual work planning process. Strategic planning is minimal, undocumented, and lacks a structured process and thus generally fails to meet the objective of guiding fire management implementation.

Implementation is of an ad hoc nature, being inconsistent and disorganised. Insufficient controlled burning is implemented to achieve the fire management objectives effectively. Consequently, uncontrolled wildfires continue to affect most of the park's area every year.

Monitoring is limited to ground observations of fire intensity and extent, but no records are kept. Remotely sensed maps provide an overview of the distribution of fires.

3. Resources and Capacity

Skills and knowledge of fire management planning, implementation and monitoring are limited.

Vehicles and drivers: availabilities limit implementation of the programme.

Fire management equipment: is limited to 2 drip torches, one 500 litre bakkie skid unit, and a handful of fire beaters.

Personal protective equipment: the existing staff uniforms are the minimum required for controlled fire management operations.

Communication: the existing park VHF radio communication network is inadequate.

Neighbours: in 2009, the George Mukoya and Maduva Nyangana conservancies/community forests embarked on a controlled burning programme through the KNC Integrated Fire Management Programme. A fire management committee was established to prepare a fire management plan and apply for a Permit to Burn from the DoF. The fire management committee implemented the controlled fire management programme to provide tangible livelihood benefits to the communities through a reduction in uncontrolled fires and an improvement in land use and environmental management. The traditional authority oversees the functions of the committee and arbitrates on fire-related disputes.

In Nyae Nyae Conservancy and the communal areas in the west, the uncoordinated use of fire as a land management tool is prevalent in a range of activities that include slash-and-burn agriculture, natural product harvesting, livestock grazing, protection measures against wildlife, hunting, and honey gathering. Communities rely on the DoF to provide fire management support in the form of firebreak maintenance and wildfire suppression in the event that these fires become uncontrolled.



Botswana has a fire prevention and suppression management policy and maintains extensive firebreaks along its borders. These have no effect on impeding the spread of uncontrolled fires into Khaudum NP, however.

SWOT analysis: this analysis evaluates and summarises the implementation of the existing Khaudum North Complex Integrated Fire Management Programme.

Table 2: SWOT analysis of the implementation of the existing Khaudum North Complex Integrated Fire Management Programme

	Beneficial	Detrimental
Internal	STRENGTHS - Developing as a core park management activity - Dedicated and motivated management and staff - Previous CBFiM programme experience with neighbouring conservancies/community forests	WEAKNESSES - Strategic planning lacks a structured process - Implementation minimal and not achieving fire management objectives - Minimal fire management experience among staff - Remote and large park with difficult access and demanding logistics - Vehicle and driver availabilities limit implementation - Local and remote monitoring insufficient
External	OPPORTUNITIES - Established collaboration with neighbouring community and conservancy structures - Multi-stakeholder collaborative management forums well established (e.g., KNC) - Previous CBFiM programmes in neighbouring conservancies/community forests can be revitalized - Strengthen partnership with DoF to facilitate controlled burning in neighbouring areas - High-profile area for development aid funding support	THREATS - Uncontrolled wildfires from neighbouring areas, particularly Botswana - Poachers igniting fires inside park - Management fires impacting neighbouring areas - Weak partnership with DoF hinders facilitation of controlled burning in neighbouring areas

4. The Controlled Fire Management Programme

The Controlled Fire Management Programme for Khaudum National Park represented by this document is based on the strategic implementation of controlled burning to manipulate the park ecosystem to enhance park management objectives.

The programme builds on the existing Khaudum North Complex Integrated Fire Management Programme established in 2009 to safely and efficiently implement controlled fire management activities that:

- minimise wildfire hazard;
- enhance park use; and
- improve ecosystems and habitats.

The programme implements an overarching strategic early dry season burning regime to establish a mosaic of interlinked burned patches and corridors, representing approximately 50 per cent of the park area. This creates an early dry season firebreak network that minimises the occurrence and extent of late dry season fires subsequently. The specific location, timing, and method of controlled burning is guided by park land-use and habitat requirements.

A dedicated fire management team carries out a structured annual planning, implementation, and monitoring process based on the two fire management zones (FMZs). This enables adaptable and robust fire management activities that address the diversity of habitats and park use while achieving overall park fire management objectives.

This effectively dispenses with the need for the costly and hazardous conventional fire management practices of establishing firebreak networks and undertaking wildfire suppression.

4.1 Fire Management Teams

The fire management teams comprise the two functional groups of five MET staff members based at Khaudum and Sikerretti respectively, whose role is to carry out the structured planning, implementation, and monitoring of the programme within their respective FMZs every year.

Each fire management team leader works in close collaboration with a park warden to coordinate and supervise the Controlled Fire Management Programme.

It is important that the team members are maintained in their roles for at least three years in order to build fire management skills and knowledge and ensure development and improvement of the programme.

4.2 Fire Management Zones

Khaudum NP has been divided into two fire management zones based on habitat, infrastructure, park use, and fire history/issues (see Figure 1):

- The Northern Fire Management Zone covers 1,623km² (42 per cent) of Khaudum NP and is bounded by the George Makoya and Maduva Nyangana conservancies/community forests to the north, Botswana to the east, and the Southern FMZ to the south.
- The Southern Fire Management Zone covers 2,215km² (58 per cent) of Khaudum NP and is bounded to the north by the Northern FMZ, Botswana to the east, and Nyae Nyae Conservancy to the south.



Controlled fire management is tailored to the specific requirements of each FMZ through a structured planning, implementation, and monitoring process.

4.3 Planning

Controlled fire management planning is carried out at the end of the summer/wet season (March/April) every year. A planning process involving information collection and analysis at the FMZ level is required to determine the key planning factors for developing the controlled fire management requirements for the following fire season.

Fire Management Zone Analysis

A combination of satellite data, field surveys, and liaison with park staff, concessionaires and neighbours is used to collect and analyse key information. A series of questions highlights the information required, as follows:

Fire history and issues - When did the fires occur in the last three years? What was the source of fire (controlled or uncontrolled)? Where did the fires come from? How much rain fell last year? How much fuel load exists? What and where are the fire hazards?

Park use - Where is the infrastructure (boreholes, lodges, fences, etc.) located? What are the locations and condition of roads, tracks and firebreaks? What areas do the park visitors frequent? What important events took place in the park and what are management's objectives?

Ecosystem and habitat maintenance - What and where are the different habitats? What are the wildlife habitat requirements? How have the habitats been affected by fire? Is there bush encroachment? What are the important habitat functions and services for the park?

Controlled Fire Management Requirements

Specific controlled fire management requirements for each FMZ are to be developed to minimise wildfire hazard, enhance park use, and improve habitats.

Wildfire hazard reduction - Strategic controlled burns are to be employed to reduce fuel loads along boundaries, roads and infrastructure. Patchwork mosaic burning is to be used to fragment fuel loads throughout the park area.

Enhance park use – Strategic controlled burns are to be employed to protect life and property; facilitate efficient management; enhance visitor aesthetics and experience; and enhance economic benefits.

Improving ecosystems and habitats - Strategic controlled burns are to be employed to maintain and enhance habitats; reduce fire intensity; improve ecosystem productivity and function; and establish a less repetitious annual burning regime to promote diversity of habitats and increase biodiversity.

The Controlled Fire Management Plan

The *Controlled Fire Management Plan for Khaudum National Park* documents the analysis and the controlled fire management requirements for each FMZ for the fire season to come. Development of the plan focuses on the operational aspects of implementation by employing a 'what, when, where and how' approach, as follows:

- What implementation activity is most suitable?
- When should the activity be employed?
- Where is the implementation activity needed?
- How should the activity be implemented?

Operating procedures for implementation and monitoring activities are detailed in the plan.

Standard Operating Procedures

Development of standard operating procedures (SOPs) for implementation and monitoring activities takes place during the planning process and the SOPs form an integral part of the Controlled Fire Management

Plan for Khaudum National Park.

SOPs will document and detail the work processes to be followed by the fire management teams and other park staff to facilitate consistent compliance with fire management implementation.

4.4 Implementation

Controlled fire management is implemented in the early dry season (May to July) when fires are characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Light winds, cool temperatures, and fuel (grass, litter etc.) that is only partly dried all out limit the intensity and extent of these fires.

Controlled Burning Coordination

A fire management team leader supervises and coordinates all controlled burning operations of the fire management team in each of the two FMZs. The *Controlled Fire Management Plan* guides the specific location, timing, and method of controlled burning.

Many factors influence fire behaviour and controlled burning operations. Flexibility is required to adapt to prevailing conditions and preparation considers the following:

Coordination liaison and awareness: it is important to coordinate, collaborate, and increase awareness of controlled burning objectives and activities with park staff and residents, neighbours and visitors.

Access: knowledge of locations and condition of roads, tracks and firebreaks to provide access and a safe environment to implement controlled burning.

Start early in the season: early controlled burning is the key to an effective controlled fire management programme. Areas should be targeted as soon as they are dry enough to burn.

Start late in the day: in the afternoon, winds stabilise and the extent of the fire is under control before nightfall, thus reducing the possible uncontrolled spread of the burned area.



Conditions: weather conditions; fuel type and load; landscape features; and fire behaviour.

Control: active suppression of controlled burns is not desirable. Prior to ignition, the fire controller must be prepared for the control of burn in all possible eventualities.

Controlled Protection Burning

Strategic controlled burns within each of the two FMZs are required to establish buffer zones of burnt areas around infrastructure, resources, and boundaries in order to protect life and property; prevent controlled fires escaping; and minimise the uncontrolled wildfire hazard.

Removing fire fuel load: resources and infrastructure are secured by removing fuel loads through controlled burning to establish buffer zones. Fires later in the year will not be able to reach resources or infrastructure with no fuel to burn around them.

Burn against the wind: protection burns, where possible, are implemented against the wind to create a fire of low intensity and low rate of spread that is easier to control.

Boundary protection: protection burns are established, where required, along park or internal boundaries to ensure controlled mosaic burns or uncontrolled fires do not traverse these boundaries.

Controlled Mosaic Burning

Strategic controlled burns are required to establish a mosaic of burnt and un-burnt patches throughout each FMZ. The protection burns enable controlled mosaic burning to be implemented safely and efficiently.

The specific location, timing, and method of controlled mosaic burning are guided by the FMZ controlled fire management requirements developed in the *Controlled Fire Management Plan*: controlled burns of varying extent, intensity, patch scale, and timing are implemented throughout the early dry season.

Combined with the protection burns, a mosaic of interlinked burnt patches and corridors representing approximately 50 per cent of the FMZs creates an effective early dry season firebreak network that minimises the occurrence and extent of late dry season fires.

Controlled burning is implemented by igniting fires from vehicles along roads and tracks or by walking across country.

Uncontrolled Wildfire Management

The management of uncontrolled wildfire relies on the patchwork of interlinked burnt patches and corridors established during the controlled protection and mosaic burning, which creates an early dry season firebreak network.

Uncontrolled wildfire management is important in the late dry season (August to November) when fires are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions and fully cured fuel.

Prevention: coordination, liaison and awareness are critical to inform park staff, residents, tour operators, visitors and neighbours about the hazards of wildfires and the use of prevention measures against wildfires.

Detection and communication: the detection of wildfires is the responsibility of all park staff members and they are required to communicate the location to the fire management team in the FMZ in question. Early detection and swift communication are essential.

Assessment: the fire management team needs to assess the uncontrolled wildfire and determine if it is a hazard to life and property, a threat to park use, or jeopardises ecosystems and habitats. The fire controller must have reliable knowledge and understanding of the area, particularly previously burnt areas, for safe and efficient management.

Response: minimising the threat or hazard of a wildfire involves focusing resources and tactics to protect life and property, contain the wildfire within a controlled area, and suppress and extinguish the wildfire (as a last resort).

Collaboration: if containing the wildfire is beyond the capabilities of the fire management team and park staff, they must notify neighbours and request assistance from them.

4.5 Monitoring

Monitoring of the Controlled Fire Management Programme activities is required throughout the year and involves a combination of park-level observations and remotely sensed information.

Park Level

At the park level, the fire management teams are responsible for monitoring within their own FMZ.

Controlled burning monitoring: follow-up surveillance and monitoring is conducted to ensure that controlled protection and mosaic burning achieve the intended objective(s).

Controlled fire management register: a register of location, timing, and nature of controlled burns and wildfires in each FMZ must be maintained.

Vegetation monitoring: monitoring sites (in a range of habitats) in each FMZ are to be marked and surveyed annually towards the end of the wet season (i.e., in April). Photographs, data on basic vegetation structure, and fire information are to be collected.

Remote Monitoring

The Bwabwata Ecological Institute is responsible for the remote monitoring of fires through the AMESD Advanced Fire Information System. Relevant information and maps are to be developed and communicated, with training in GIS and remote sensing skills and knowledge undertaken. Use of AFIS products will incorporate:

Prediction of fires is of limited use in the park due to scale and should be used in conjunction with local-level observations and weather forecasts. The Fire Danger Index is updated daily and provides a 3-day forecast of the Fire Weather Index and Lowveld Fire Danger Index. Fuel Condition data are updated every 10 days and provide a Vegetation Condition Index and a Vegetation Productivity Index, as well as indicating the number of days since the last burn for each area.



Detection and monitoring of fires: MODIS Active Fire detection data are updated regularly on a daily basis (10am and 3pm). MSG Active Fire detection data are updated every 15 minutes.

Assessment of fires: the MODIS Burned Area Product that maps and documents the extent of fires throughout the year is updated every month. MODIS Imagery is updated every 6 hours and provides visual fire monitoring and assessment.

The wardens and fire management teams should access the available information from the Internet at the: AFIS website - <http://afis.meraka.org.za> and the Global Fire Information Management System (GFIMS) website - <http://www.fao.org/nr/gfims/en>.

The GFIMS provides a fire email alert service whereby registered users of the MODIS-derived Active Fire detection product receive an email whenever a fire is detected in a specified area of interest.

The Controlled Fire Management Report

The fire management teams prepare a Controlled Fire Management Report based on the monitoring information; the report will address objectives and document the outcomes of implementation. Assessments of programme effectiveness and identification of inadequacies will provide fire management recommendations for the following year.

5. Capacity and Resource Requirements

To build the capacity of Khaudum National Park management to safely and efficiently implement its Controlled Fire Management Programme requires a long-term process aimed at developing the skills and knowledge acquired only through firsthand experience. Most of this training takes place in the course of ongoing fire management implementation and requires strategic and regular trainer intervention.

5.1 Training and Skills Transfer

On-the-job training is required throughout the duration of the initial fire season to build the capacity of the fire management teams to implement the Controlled Fire Management Programme. Ideally, annual follow-up training should be provided in subsequent fire seasons.

A fire management specialist will be required to deliver three training sessions in Khaudum National Park. Training of fire management teams, other park staff, neighbours, and other key stakeholders will coincide with the planning, implementing and monitoring of the programme in its first year.

Session 1: Controlled Fire Management Planning

A Controlled Fire Management Planning Session of two weeks' duration is required in April/May and will cover all theoretical aspects of the programme; develop the Controlled Fire Management Plan; and commence implementation training.

Part 1, Controlled Fire Management Programme Theory: covering all aspects of the Controlled Fire Management Programme, including:

- Fire issues	- Planning
- Existing fire management	- Implementation
- The Controlled Fire Management Programme	- Monitoring
- Fire management teams - roles and responsibilities	- Reporting

Part 2, Fire Management Plan Development: a finalised draft of the Controlled Fire Management Plan is developed for Khaudum NP through the training process whereby data collection, liaison activities, FMZ analyses, and controlled fire management requirement assessments will be carried out.

Part 3, Implementation Activity Introduction: practical introductory training of implementation techniques to achieve the FMZs’ controlled fire management requirements.

Session 2: Controlled Fire Management Implementation

A Controlled Fire Management Implementation Session of four weeks’ duration is required in July to train the fire management teams in operational/practical aspects of controlled burning coordination; controlled protection burning; controlled mosaic burning; and uncontrolled wildfire management. July is a critical time for the teams to acquire these skills, in order to ensure sufficient protection and controlled mosaic burning is implemented prior to the wildfire season.

Session 3: Controlled Fire Management Monitoring and Reporting

A Controlled Fire Management Monitoring and Reporting Session of one week’s duration is required in November to train the fire management teams in collating park-level and remote monitoring information to assess programme effectiveness to date and develop recommendations.

5.2 Equipment

Implementing an effective controlled fire management programme dispenses with the need for costly fire suppression equipment. The following equipment is recommended for the satisfactory implementation of the Controlled Fire Management Programme for Khaudum NP (Table 3 to Table 5, below).

Vehicles and Drivers

Vehicle and driver availabilities underpin the effectiveness of successful implementation of the Controlled Fire Management Programme throughout this large and remote park.

Suitable 4x4 bakkie vehicles (with competent drivers) are to be prioritised for use in controlled burning operations during the months of May, June and July. A 4x4 bakkie fitted with a twin impellor skid unit constitutes the most suitable mobile fire management equipment for the rough terrain of this remote park.



Personal Protective Equipment (PPE)

Personal equipment for fighting fires should be provided as follows:

Table 3: Personal protective equipment (PPE) required at Khaudum NP

Description	Fire retardant Conti suit	
Specifications	100% cotton	
No. of units, Khaudum National Park: Northern FMZ		X 40
No. of units, Khaudum National Park: Southern FMZ		X 40
Total = 80		
Description	Safety boots (pair)	
Specifications	Leather, non-steel-capped	
No. of units, Khaudum National Park: Northern FMZ		X 40
No. of units, Khaudum National Park: Southern FMZ		X 40
Total = 80		
Description	Broad-brimmed hat	
Specifications	100% cotton	
No. of units, Khaudum National Park: Northern FMZ		X 40
No. of units, Khaudum National Park: Southern FMZ		X 40
Total = 80		
Description	Gloves (pair)	
Specifications	Heavy duty leather	
No. of units, Khaudum National Park: Northern FMZ		X 40
No. of units, Khaudum National Park: Southern FMZ		X 40
Total = 80		
Description	UV protection safety glasses	
No. of units, Khaudum National Park: Northern FMZ		X 40
No. of units, Khaudum National Park: Southern FMZ		X 40
Total = 80		
Description	First aid burn kit	
No. of units, Khaudum National Park: Northern FMZ		X 4
No. of units, Khaudum National Park: Southern FMZ		X 4
Total = 8		

PPE can be sourced from safety equipment distributors in Namibia.

Fire Management Equipment

The following fire management equipment is required for controlled burning implementation:

Table 4: Mobile vehicle fire management units required at Khaudum NP

Description	Twin impeller skid unit	
Model	Honda GX160/Davey 2-stage pump	
Specifications	600 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length - 60 m; weight +/- 125 kg.	
No. of units, Khaudum National Park: Northern FMZ		X 5
No. of units, Khaudum National Park: Southern FMZ		X 5
Total = 10		
Description	Dry chemical powder fire extinguishers	
Specifications	9 kg size conforming to minimum fire ratings specified in SANS 1910:2009 for A, B or C class fires or a combination of ABC class fires	
No. of units, Khaudum National Park: Northern FMZ		X 5
No. of units, Khaudum National Park: Southern FMZ		X 5
Total = 10		

Table 5: Personnel fire management equipment required at Khaudum NP

Description	Aluminium drip torch	
Specifications	5 L lightweight high-strength aluminium alloy tank with full-length handle, smooth domed-top to prevent fuel accumulation, and check valve in cover to protect against flashbacks	
No. of units, Khaudum National Park: Northern FMZ		X 5
No. of units, Khaudum National Park: Southern FMZ		X 5
Total = 10		
Description	Knapsack spray units 20 L high-density polythene tank with double action plunger valve pump	
Specifications	20 L high-density polythene tank with double action plunger valve pump	
No. of units, Khaudum National Park: Northern FMZ		X 5
No. of units, Khaudum National Park: Southern FMZ		X 5
Total = 10		
Description	Fire beaters	
Specifications	Steel or aluminium handled with rubber head	
No. of units, Khaudum National Park: Northern FMZ		X 60
No. of units, Khaudum National Park: Southern FMZ		X 60
Total = 120		
Description	Jerry cans	
Specifications	25 L steel	
No. of units, Khaudum National Park: Northern FMZ		X 5
No. of units, Khaudum National Park: Southern FMZ		X 5
Total = 10		

Specialised fire management equipment can be sourced from distributors in Namibia and/or South Africa.

Communication

The existing park VHF radio communication network should be strengthened with five handheld VHF radios.

6. Collaboration with Residents and Neighbours of Khaudum National Park

Coordination and collaboration with park residents and neighbours sharing a boundary with the park constitute an important component of the Controlled Fire Management Programme for Khaudum National Park.

The DoF, through the Forest Act of 2001, regulates fire management in the neighbouring private farms, communal area conservancies and communal lands. To implement controlled burning these landholders/land managers require authorisation from the DoF.

6.1 Communal Areas and Conservancies

The integrated community-based fire management (CBFiM) programmes established in Maduva Nyangana and George Makoya conservancies/community forests in 2009 through the KNC Integrated Fire Management Programme provide a suitable operational framework and should be revitalized and extended to Nyae Nyae Conservancy (this should be carried out in partnership with the DoF). The KNC is an ideal forum for coordination to the north of the park.

The CBFiM programmes will require technical and financial support from government and international and local development non-governmental organisations such as the Namibia Protected Landscape Conservation Areas Initiative (NAM-PLACE) Project, the Strengthening the Capacity of the Protected Area System to Address New Management Challenges (PASS) Project, and the GIZ Community Forest Programme.

6.2 International Collaboration

International collaboration with Botswana and Angola should build on existing regional initiatives, including the Southern African Development Committee (SADC) Regional Fire Management Programme and the SADC Protocol on Forestry (2002).

Integration with the Kavango Zambezi Trans-frontier Conservation Area (KAZA TFCA) should be facilitated through the Namibian Integrated Development Plan as well as other existing structures.

6.3 Awareness and Education

Increasing the awareness and understanding of the Controlled Fire Management Programme for Khaudum National Park in park staff, residents, neighbours, and visitors is required to facilitate safe and efficient implementation. Pre-season multi-stakeholder meetings and/or agreements establish the foundation for this awareness.

Awareness and education programmes are recommended to foster the commitment and support of key role players and clearly describe their roles and responsibilities within the programme. A multi-media public awareness campaign (electronic and print media such as newspapers, posters and brochures) is required to inform park visitors, civil society, government, and non-government institutions of the programme's objectives and achievements.

Appendix 4

CONTROLLED FIRE MANAGEMENT PROGRAMME,
NKASA RUPARA NATIONAL PARK



1. Introduction

Nkasa Rupara National Park covers approximately 320 km² of the Zambezi Region at the confluence/delta of the Kwando and Linyanti rivers. It consists of extensive floodplains, lagoons and termitaria islands. The semi-arid tropical climate provides a variable and highly seasonal (October to April) average annual rainfall of approximately 600 mm.

The unfenced park is the largest protected wetland in Namibia, and an important buffalo habitat and wildlife corridor – particularly for elephant – between Botswana, Angola and Zambia.

Communal lands with rural populations neighbour the park to the north and Botswana is found to the south. Wildlife and natural resource management in collaboration with neighbouring communities is well established in the form of communal area conservancies and forests and the Mudumu South Complex (MSC).

2. Fire History, Management and Issues

Fire is a natural and regular occurrence in savanna ecosystems and fire plays a fundamental role in the vegetation, wildlife, and land use management of Nkasa Rupara NP.

Fires occur almost exclusively in the dry season, with early dry season (May to July) fires characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Late dry season (August to November) fires, on the other hand, are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions.

Management of fires in Nkasa Rupara NP has been dominated by fire exclusion, protection, and suppression policies dating back several decades (to the time when the area was known as Malimi National Park). Unseasonal flooding is characteristic of the Kwando/Linyanti river system, which strongly determines the timing and extent of annual fires. Typically, uncontrolled wildfires in the late dry season occur throughout the dry areas of the park every year (see Figure 1). The majority of these fires are caused by uncoordinated use of fire as a land management tool in neighbouring communal areas and through the activities of poachers within the park.

Over the past two decades, the annual fire regime has contributed to the degradation of the park's ecosystem and limited management objectives by affecting vegetation structure and species composition; reducing ecosystem function and productivity; threatening high value and vulnerable ecosystems; and reducing habitat and biological diversity.

In 2007 Nkasa Rupara National Park embarked on the controlled burning programme that is central to what is now known as the Zambezi Regional Integrated Fire Management Programme.

This integrated fire management programme aims to use controlled burning to reduce the area burnt annually and shift the seasonality of burning to the early dry season. Collaborative implementation with neighbouring conservancies is structured to share resources and workloads to jointly manage uncontrolled wildfires, enhance land use, and improve the environment of the area.

Since 2008, high water levels have inundated most of Nkasa Rupara NP, preventing access to the few islands still above water and bringing the existing fire management programme to a standstill in recent years (see Table 1 and Figure 2).

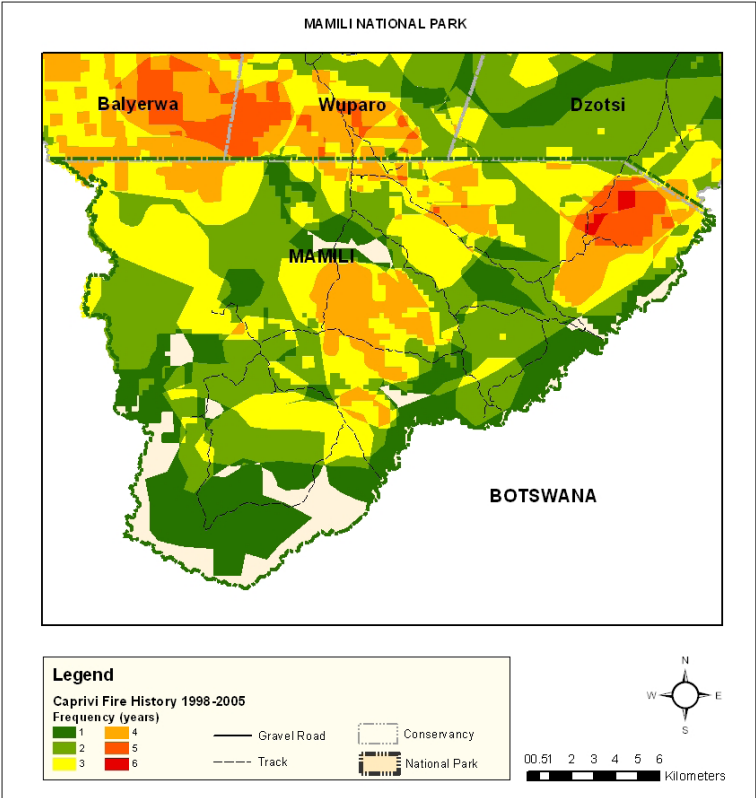


Figure 1: Nkasa Rupara (previously Mamili) National Park fire history, 1998 to 2005

Planning involves designating preparation, controlled burning, and assessment activities through the park’s annual work planning process.

Implementation and **monitoring** have ceased since 2008 due to inundation.

Table 1: Nkasa Rupara (previously Mamili) NP fires, 2007 and 2008

	FIRE SEASON			
	2007*		2008*	
	Area (km²)	% of park	Area (km²)	% of park
Controlled burn	84	25	11	3
Uncontrolled fire	34	10	1	<1
Total	118	35	2	4

* Fires mapped manually from monthly MODIS 250 m satellite imagery.

3. Resources and Capacity

Skills and knowledge of fire management planning, implementation and monitoring are limited.

Vehicles and drivers: availabilities limit implementation of the programme.

Fire management equipment: the existing single drip torch and fewer than 10 fire beaters represent the absolute minimum required for controlled fire management operations.

Personal protective equipment: the existing staff uniforms are the minimum required for controlled fire management operations.

Communication: the existing park VHF radio communication network is adequate.

Neighbours: Nkasa Rupara National Park's neighbouring conservancies/community forests are an integral component of what is now known as the Zambezi Regional Integrated Fire Management Programme, initiated in 2007. Community-based fire management (CBFiM) programmes in each conservancy/community forest were established with fire management committees to prepare a fire management plan and apply for a Permit to Burn from the DoF. The fire management committees implemented controlled fire management programmes that provided tangible livelihood benefits to communities through reducing uncontrolled fires and improving land use and environmental management. The traditional authority oversees the functions of each committee and arbitrates on fire-related disputes within the community.

Since withdrawal of support funding in 2010, the fire management initiatives have ceased to be implemented with any meaningful effect. Communities have reverted back to the uncoordinated use of fire and uncontrolled wildfires continue to impact negatively on livelihoods and natural resource management.

The DoF provides fire management support by means of discouraging burning through education and awareness campaigns, creating firebreak networks, and through community wildfire suppression training.

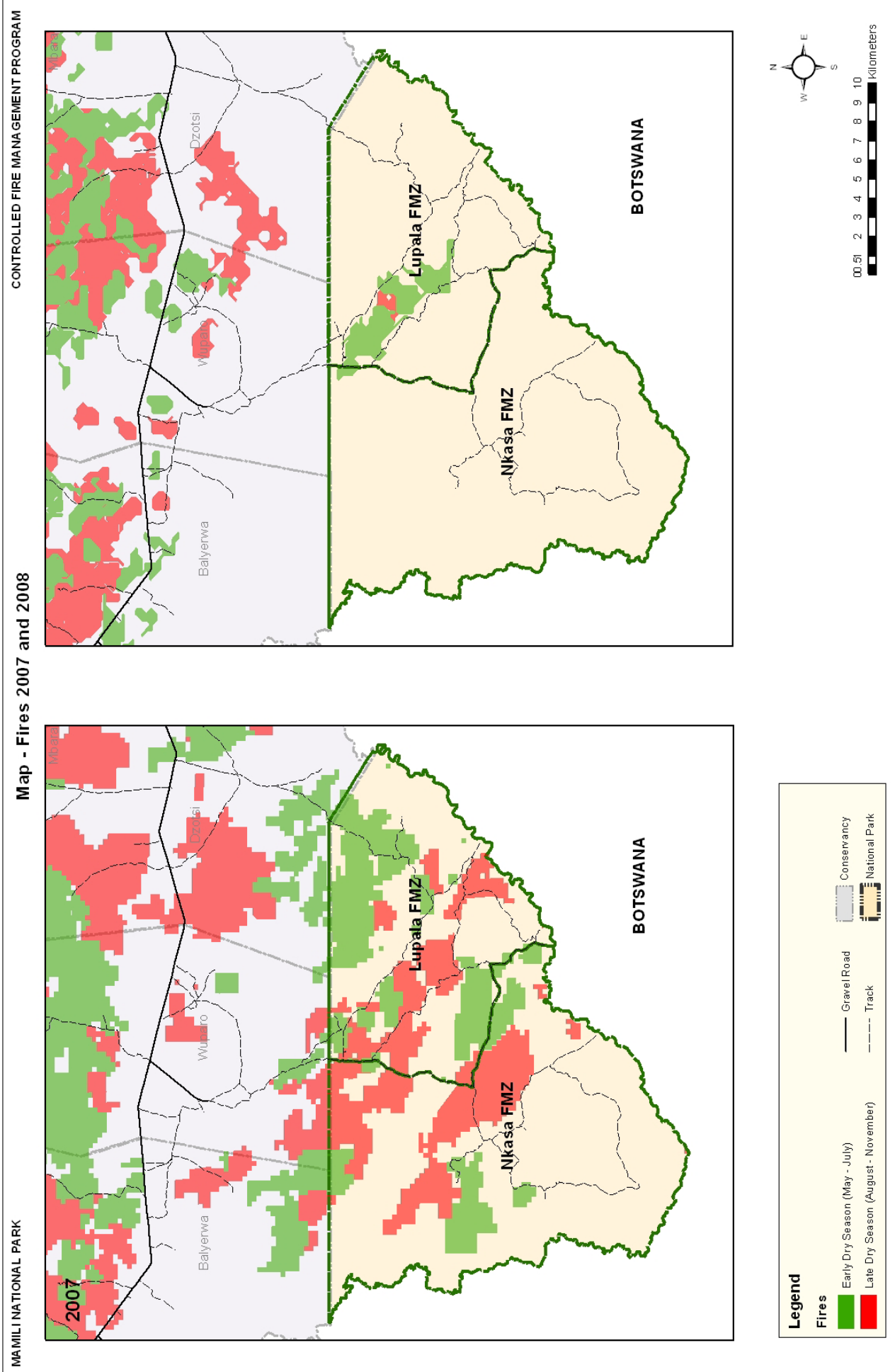


Figure 2: Nkasa Rupara National Park fire occurrences, 2007 (left) and 2008 (right)

SWOT analysis: this analysis evaluates and summarises the existing implementation of the Integrated Fire Management Programme at Nkasa Rupara NP.

Table 2: SWOT analysis for the existing implementation of the Integrated Fire Management Programme at Nkasa Rupara National Park

	Beneficial	Detrimental
Internal	STRENGTHS - Two years’ controlled burning programme experience - Dedicated staff members with some fire management skills and knowledge - Established as a core park management activity	WEAKNESSES - Strategic planning lacks a structured process - Implementation ad hoc and falls short of some fire management objectives - Vehicle and driver availabilities limit implementation - Access and implementation restricted due to flooding - Local and remote monitoring insufficient
External	OPPORTUNITIES - Well-established collaboration with neighbouring community and conservancy structures - Multi-stakeholder collaborative management forums well established (e.g., MSC) - Previous CBFiM programmes in neighbouring conservancies/community forests can easily be revitalized - Strengthen partnership with DoF to facilitate controlled burning in neighbouring areas - High-profile area for development aid funding support	THREATS - Uncontrolled wildfires from neighbouring areas - Poachers igniting fires inside park - Management fires impacting neighbours - No implementation due to flooding and lack of access - Weak partnership with DoF hinders facilitation of controlled burning in neighbouring areas

4. The Controlled Fire Management Programme

The *Controlled Fire Management Programme for Nkasa Rupara National Park* represented by this document is based on the strategic implementation of controlled burning to manipulate the park ecosystem to enhance park management objectives.

The programme builds on the existing integrated fire management approach that has been utilised since 2007 to safely and efficiently implement controlled fire management activities that:

- minimise wildfire hazard;
- enhance park use; and
- improve ecosystems and habitats.

Development of a controlled fire management programme will, however, be dependent on the subsiding of the high water levels that have inundated most of Nkasa Rupara NP since 2008.

It is intended that once the programme is underway it will implement an overarching strategic early dry season burning regime to establish a mosaic of interlinked burned patches and corridors, representing approximately 35 per cent of the park area. This creates an early dry season firebreak network that minimises the occurrence and extent of late dry season fires subsequently. The specific location, timing, and method of controlled burning are all guided by park land-use and habitat requirements.

The Fire Management Team will carry out a structured annual planning, implementation, and monitoring process based on fire management zones (FMZs). This enables adaptable and robust fire management activities that address the diversity of habitats and park use while achieving overall park fire management objectives.

This effectively dispenses with the need for the costly and hazardous conventional fire management practices of establishing firebreak networks and undertaking wildfire suppression.

4.1 The Fire Management Team

The Fire Management Team is the functional group of five key MET staff members based at Nkasa Rupara NP, whose role is to carry out the structured planning, implementation, and monitoring of the programme within the two FMZs every year once it is up and running.

The Fire Management Team Leader works in close collaboration with the Park Warden to coordinate and supervise the Controlled Fire Management Programme.

It is important that the team members are maintained in their roles for at least three years to build fire management skills and knowledge and ensure development and improvement of the programme.

4.2 Fire Management Zones

Nkasa Rupara NP has been divided into two fire management zones (FMZs) based on habitat, infrastructure, park use, and fire history/issues (see Figure 2):

- **Rupara FMZ:** the eastern half of the park, encompassing Rupara Island and the surrounding channels and floodplains.
- **Nkasa FMZ:** the western half of the park, encompassing Nkasa Island and the surrounding channels and floodplains.

Controlled fire management is tailored to the specific requirements of each of the two FMZs through a structured planning, implementation and monitoring process.

4.3 Planning

Controlled fire management planning is to be carried out at the end of the summer/wet season (March/April) every year. A planning process involving information collection and analysis at the FMZ level is required to determine the key planning factors for developing the controlled fire management requirements for the following fire season.



Fire management zone analysis

A combination of satellite data, field surveys, and liaison with park staff, concessionaires and neighbours is to be used to collect and analyse key information. A series of questions highlights the information required, as follows:

Fire history and issues - When did the fires occur in the last three years? What was the source of fire (controlled or uncontrolled)? Where did the fires come from? How much rain fell last year? How much fuel load exists? What and where are the fire hazards?

Park use - Where is the infrastructure (boreholes, lodges, fences, etc.) located? What are the locations and condition of roads, tracks and firebreaks? What areas do the park visitors frequent? What important events took place in the park and what are management's objectives?

Ecosystem and habitat maintenance - What and where are the different habitats? What are the wildlife habitat requirements? How have the habitats been affected by fire? Is there bush encroachment? What are the important habitat functions and services for the park?

Controlled fire management requirements

Specific controlled fire management requirements for both FMZs are to be developed to minimise wildfire hazard, enhance park use, and improve habitats.

Wildfire hazard reduction - Strategic controlled burns are to be employed to reduce fuel loads along boundaries, roads and infrastructure. Patchwork mosaic burning is to be used to fragment fuel loads throughout the park area.

Enhance park use – Strategic controlled burns are to be employed to protect life and property; facilitate efficient management; enhance visitor aesthetics and experience; and enhance economic benefits.

Improving ecosystems and habitats - Strategic controlled burns are to be employed to maintain and enhance habitats; reduce fire intensity; improve ecosystem productivity and function; and establish a less repetitious annual burning regime to promote diversity of habitats and increase biodiversity.

The Controlled Fire Management Plan

The *Controlled Fire Management Plan for Nkasa Rupara National Park* documents the analysis and the controlled fire management requirements for each of the two FMZs for the fire season to come. Development of the plan focuses on the operational aspects of implementation by employing a 'what, when, where and how' approach, as follows:

- What implementation activity is most suitable?
- When should the activity be employed?
- Where is the implementation activity needed?
- How should the activity be implemented?

Operating procedures for implementation and monitoring activities are detailed in the plan.

Standard Operating Procedures

Development of standard operating procedures (SOPs) for implementation and monitoring activities takes place during the planning process and the SOPs form an integral part of the Controlled Fire Management Plan for Nkasa Rupara National Park.

SOPs will document and detail the work processes to be followed by the Fire Management Team and other park staff to facilitate consistent compliance with fire management implementation.

4.4 Implementation

Controlled fire management is to be implemented in the early dry season (May to July) when fires are characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Light winds, cool temperatures, and fuel (grass, litter etc.) that is only partly dried out all limit the intensity and extent of these fires.

Controlled Burning Coordination

The Fire Management Team Leader supervises and coordinates all controlled burning operations of the Fire Management Team in the two FMZs. The Controlled Fire Management Plan guides the specific location, timing, and method of controlled burning.

Many factors influence fire behaviour and controlled burning operations. Flexibility is required to adapt to prevailing conditions and preparation considers the following:

Coordination liaison and awareness: it is important to coordinate, collaborate, and increase awareness of controlled burning objectives and activities with park staff and residents, neighbours and visitors.

Access: knowledge of locations and condition of roads, tracks and firebreaks to provide access and a safe environment to implement controlled burning.

Start early in the season: early controlled burning is the key to an effective controlled fire management programme. Areas should be targeted as soon as they are dry enough to burn.

Start late in the day: in the afternoon, winds stabilise and the extent of the fire is under control before nightfall, thus reducing the possible uncontrolled spread of the burned area.

Conditions: weather conditions; fuel type and load; landscape features; and fire behaviour.

Control: active suppression of controlled burns is not desirable. Prior to ignition, the fire controller must be prepared for the control of burn in all possible eventualities.

Controlled Protection Burning

Strategic controlled burns within each of the two FMZs are required to establish buffer zones of burnt areas around infrastructure, resources, and boundaries in order to protect life and property; prevent controlled fires escaping; and minimise the uncontrolled wildfire hazard.

Removing fire fuel load: resources and infrastructure are secured by removing fuel loads through controlled burning to establish buffer zones. Fires later in the year will not be able to reach resources or infrastructure with no fuel to burn around them.



Burn against the wind: protection burns, where possible, are implemented against the wind to create a fire of low intensity and low rate of spread that is easier to control.

Boundary protection: protection burns are established, where required, along park or internal boundaries to ensure controlled mosaic burns or uncontrolled fires do not traverse these boundaries.

Controlled Mosaic Burning

Strategic controlled burns are required to establish a mosaic of burnt and un-burnt patches throughout each FMZ. The protection burns enable controlled mosaic burning to be implemented safely and efficiently.

The specific location, timing, and method of controlled mosaic burning are guided by the FMZ controlled fire management requirements developed in the Controlled Fire Management Plan: controlled burns of varying extent, intensity, patch scale, and timing are implemented throughout the early dry season.

Combined with the protection burns, a mosaic of interlinked burnt patches and corridors representing approximately 35 per cent of the two FMZs creates an effective early dry season firebreak network that minimises the occurrence and extent of late dry season fires.

Controlled burning is implemented by igniting fires from vehicles along roads and tracks, by accessing certain areas by boat, or by walking across country.

Uncontrolled Wildfire Management

The management of uncontrolled wildfire relies on the patchwork of interlinked burnt patches and corridors established during the controlled protection and mosaic burning, which creates an early dry season firebreak network.

Uncontrolled wildfire management is important in the late dry season (August to November) when fires are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions and fully cured fuel.

Prevention: coordination, liaison and awareness are critical to inform park staff, residents, tour operators, visitors and neighbours about the hazards of wildfires and the use of prevention measures against wildfires.

Detection and communication: the detection of wildfires is the responsibility of all park staff members and they are required to communicate the location of the fire within the FMZ in question to the Fire Management Team. Early detection and swift communication are essential.

Assessment: the Fire Management Team needs to assess the uncontrolled wildfire and determine if it is a hazard to life and property, a threat to park use, or jeopardises ecosystems and habitats. The fire controller must have reliable knowledge and understanding of the area, particularly previously burnt areas, for safe and efficient management.

Response: minimising the threat or hazard of a wildfire involves focusing resources and tactics to protect life and property, contain the wildfire within a controlled area, and suppress and extinguish the wildfire (as a last resort).

Collaboration: if containing the wildfire is beyond the capabilities of the Fire Management Team and park staff, they must notify neighbours and request assistance from them.

4.5 Monitoring

Monitoring of the Controlled Fire Management Programme activities is required throughout the year and involves a combination of park-level observations and remotely sensed information.

Park Level

At the park level, the Fire Management Team is responsible for monitoring both of the FMZs.

Controlled burning monitoring: follow-up surveillance and monitoring is conducted to ensure that controlled protection and mosaic burning achieve the intended objective(s).

Controlled fire management register: a register of location, timing, and nature of controlled burns and wildfires in each FMZ must be maintained.

Vegetation monitoring: monitoring sites (in a range of habitats) in each FMZ are to be marked and surveyed annually towards the end of the wet season (i.e., in April). Photographs, data on basic vegetation structure, and fire information are to be collected.

Remote Monitoring

The Bwabwata Ecological Institute is responsible for the remote monitoring of fires through the AMESD Advanced Fire Information System. Relevant information and maps are to be developed and communicated, with training in GIS and remote sensing skills and knowledge undertaken. Use of AFIS products will incorporate:

Prediction of fires is of limited use in the park due to scale and should be used in conjunction with local-level observations and weather forecasts. The Fire Danger Index is updated daily and provides a 3-day forecast of the Fire Weather Index and Lowveld Fire Danger Index. Fuel Condition data are updated every 10 days and provide a Vegetation Condition Index and a Vegetation Productivity Index, as well as indicating the number of days since the last burn for each area.

Detection and monitoring of fires: MODIS Active Fire detection data are updated regularly on the daily basis (10am and 3pm). MSG Active Fire detection data are updated every 15 minutes.

Assessment of fires: the MODIS Burned Area Product that maps and documents the extent of fires throughout the year is updated every month. MODIS Imagery is updated every 6 hours and provides visual fire monitoring and assessment.

The Park Warden and Fire Management Team should access the available information from the Internet at the: AFIS website - <http://afis.meraka.org.za> and the Global Fire Information Management System (GFIMS) website - <http://www.fao.org/nr/gfims/en>.

The GFIMS provides a fire email alert service whereby registered users of the MODIS-derived Active Fire detection product receive an email whenever a fire is detected in a specified area of interest.



The Controlled Fire Management Report

The Fire Management Team prepares a Controlled Fire Management Report based on the monitoring information; the report will address objectives and document the outcomes of implementation. Assessments of programme effectiveness and identification of inadequacies will provide fire management recommendations for the following year.

5. Capacity and Resource Requirements

To build the capacity of Nkasa Rupara National Park management to safely and efficiently implement its Controlled Fire Management Programme requires a long-term process aimed at developing the skills and knowledge acquired only through firsthand experience. Most of this training takes place in the course of ongoing fire management implementation and requires strategic and regular trainer intervention.

5.1 Training and skills transfer

On-the-job training is required throughout the duration of the initial fire season to build the capacity of the Fire Management Team to implement the Controlled Fire Management Programme. Ideally, annual follow-up training should be provided in subsequent fire seasons.

A fire management specialist will be required to deliver three training sessions in Nkasa Rupara National Park. Training of the Fire Management Team, other park staff, neighbours, and other key stakeholders will coincide with the planning, implementing and monitoring of the programme in its first year.

Session 1: Controlled Fire Management Planning

A Controlled Fire Management Planning Session of two weeks' duration is required in April/May and will cover all theoretical aspects of the programme; develop the Controlled Fire Management Plan; and commence implementation training.

Part 1, Controlled Fire Management Programme Theory: covering all aspects of the Controlled Fire Management Programme, including:

- Fire issues	- Planning
- Existing fire management	- Implementation
- The Controlled Fire Management Programme	- Monitoring
- The Fire management Team - roles and responsibilities	- Reporting

Part 2, Fire Management Plan Development: a finalised draft of the Controlled Fire Management Plan is developed for Nkasa Rupara NP through the training process whereby data collection, liaison activities, FMZ analyses, and controlled fire management requirement assessments will be carried out.

Part 3, Implementation Activity Introduction: practical introductory training of implementation techniques to achieve the FMZs' controlled fire management requirements.

Session 2: Controlled Fire Management Implementation

A Controlled Fire Management Implementation Session of four weeks' duration is required in July to train

the Fire Management Team in operational/practical aspects of controlled burning coordination; controlled protection burning; controlled mosaic burning; and uncontrolled wildfire management.

July is a critical time for the Fire Management Team to acquire these skills, in order to ensure sufficient protection and controlled mosaic burning is implemented prior to the wildfire season.

Session 3: Controlled Fire Management Monitoring and Reporting

A Controlled Fire Management Monitoring and Reporting Session of one week’s duration is required in November to train the Fire Management Team in collating park-level and remote monitoring information to assess programme effectiveness to date and develop recommendations.

5.2 Equipment

Implementing an effective controlled fire management programme dispenses with the need for costly fire suppression equipment. The following equipment is recommended for the satisfactory implementation of the Controlled Fire Management Programme for Nkasa Rupara National Park (Table 3 to Table 5, below).

Vehicles and Drivers

Vehicle/boat and driver availabilities underpin the effectiveness of successful implementation of the Controlled Fire Management Programme throughout this inundated and remote park.

Suitable 4x4 bakkie vehicles and boats with competent drivers are to be prioritised for use in controlled burning operations during the months of May, June and July, prior to the river system flooding.

Personal Protective Equipment (PPE)

The Fire Management Team will require:

Table 3: Personal protective equipment (PPE) required at Nkasa Rupara NP

Description	Fire retardant Conti suit
Specifications	100% cotton
Total = 30	
Description	Safety boots (pair)
Specifications	Leather, non-steel-capped
Total = 30	
Description	Broad-brimmed hat
Specifications	100% cotton
Total = 30	
Description	Gloves (pair)
Specifications	Heavy duty leather
Total = 30	
Description	UV protection safety glasses
Total = 30	
Description	First aid burn kit
Total = 4	

PPE can be sourced from safety equipment distributors in Namibia.

Fire Management Equipment



The following fire management equipment is required for controlled burning implementation:

Table 4: Mobile vehicle fire management units required at Nkasa Rupara NP

Description	Twin impeller skid unit	
Model	Honda GX160/Davey 2-stage pump	
Specifications	600 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length - 60 m; weight +/- 125 kg.	
No. of units, Nkasa Rupara National Park: Rupara FMZ		X 2
No. of units, Nkasa Rupara National Park: Nkasa FMZ		X 2
Total = 4		
Description	Dry chemical powder fire extinguishers	
Specifications	9 kg size conforming to minimum fire ratings specified in SANS 1910:2009 for A, B or C class fires or a combination of ABC class fires	
No. of units, Nkasa Rupara National Park: Rupara FMZ		X 5
No. of units, Nkasa Rupara National Park: Nkasa FMZ		X 5
Total = 10		

Table 5: Personnel fire management equipment required at Nkasa Rupara NP

Description	Aluminium drip torch	
Specifications	5 L lightweight high-strength aluminium alloy tank with full-length handle, smooth domed-top to prevent fuel accumulation, and check valve in cover to protect against flashbacks	
No. of units, Nkasa Rupara National Park: Rupara FMZ		X 10
No. of units, Nkasa Rupara National Park: Nkasa FMZ		X 10
Total = 20		
Description	Knapsack spray units 20 L high-density polythene tank with double action plunger valve pump	
Specifications	20 L high-density polythene tank with double action plunger valve pump	
No. of units, Nkasa Rupara National Park: Rupara FMZ		X 5
No. of units, Nkasa Rupara National Park: Nkasa FMZ		X 5
Total = 10		

Description	Fire beaters	
Specifications	Steel or aluminium handled with rubber head	
No. of units, Nkasa Rupara National Park: Rupara FMZ		X 20
No. of units, Nkasa Rupara National Park: Nkasa FMZ		X 20
Total = 40		
Description	Jerry cans	
Specifications	25 L steel	
No. of units, Nkasa Rupara National Park: Rupara FMZ		X 5
No. of units, Nkasa Rupara National Park: Nkasa FMZ		X 5
Total = 10		

Specialised fire management equipment can be sourced from distributors in Namibia and/or South Africa.

Communication

The existing park VHF radio communication network should be strengthened with five handheld VHF radios.

6. Collaboration with Residents and Neighbours of Nkasa Rupara National Park

Coordination and collaboration with park residents and neighbours sharing a boundary with the park constitute an important component of the Controlled Fire Management Programme for Nkasa Rupara National Park.

6.1 Communal area Conservancies

Structured, controlled fire management collaboration and coordination between the park's Fire Management Team and neighbouring conservancies/community forests and communal areas enables alignment of fire management objectives and the sharing of resources and workloads

The DoF, through the Forest Act of 2001, regulates fire management in the neighbouring communal area conservancies/community forests and communal lands. To implement controlled burning, these landholders/land managers require authorisation from the DoF.

The community-based fire management (CBFiM) programmes established in the neighbouring conservancies/community forests can provide the operational framework for future collaborations and should be revitalized. The park Fire Management Team implements direct liaison coordination, joint training and implementation through the CBFiM institutional structures.

The CBFiM programmes will require technical and financial support from government and international and local development non-governmental organisations.

6.2 International Collaboration

International collaboration with Botswana and Angola should build on existing regional initiatives, including the Southern African Development Committee (SADC) Regional Fire Management Programme and the SADC Protocol on Forestry (2002).

Integration with the Kavango Zambezi Trans-frontier Conservation Area (KAZA TFCA) should be facilitated through the Namibian Integrated Development Plan as well as other existing structures.

6.3 Awareness and Education

Increasing the awareness and understanding of the Controlled Fire Management Programme for Nkasa Rupara National Park in park staff, residents, neighbours, and visitors is required to facilitate safe and efficient implementation. Pre-season multi-stakeholder meetings and/or agreements establish the foundation for this awareness.

Awareness and education programmes are recommended to foster the commitment and support of key role players and clearly describe their roles and responsibilities within the programme. A multi-media public awareness campaign (electronic and print media such as newspapers, posters and brochures) is required to inform park visitors, civil society, government, and non-government institutions of the programme's objectives and achievements.



Appendix 5

CONTROLLED FIRE MANAGEMENT PROGRAMME,
MANGETTI NATIONAL PARK

1. Introduction

Mangetti National Park covers approximately 420 km² of the Kavango-West Region in the north-east of Namibia. Kalahari woodlands of the remnant sand dunes, interspersed by fossil drainage lines called omurumba, dominate the savanna woodland ecosystem. The semi-arid climate provides a consistent and highly seasonal (October to April) average annual rainfall of approximately 500 – 550 mm.

The fenced park is under development as a tourist destination with the aim of reducing rural poverty through tourism development, joint management, and benefit sharing with local communities.

Communal areas surround most of the park's boundary, with private farms situated along the southern border.

2. Fire history, Management and Issues

Fire is a natural and regular occurrence in savanna ecosystems and fire plays a fundamental role in the vegetation, wildlife, and land use management of Mangetti National Park.

Fires occur almost exclusively in the dry season, with early dry season (May to July) fires characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Late dry season (August to November) fires, on the other hand, are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions.

Management of fires in Mangetti NP has been dominated by exclusion, protection and suppression policies, dating back several decades. This has resulted in uncontrolled wildfires in the late dry season occurring throughout extensive areas of the park each year. The majority of these fires are caused by the uncoordinated use of fire as a land management tool in neighbouring communal areas, including for the purposes of slash-and-burn agriculture, natural product harvesting, and livestock grazing.

Table 1: Mangetti National Park fires, 2009 to 2011

	FIRE SEASON					
	2009		2010		2011	
	Area (km ²)	% of park	Area (km ²)	% of park	Area (km ²)	% of park
Early dry season	-	-	-	-	-	-
Late dry season	114	27	135	32	295	70
Total	114	27	135	32	295	70

Over the past two decades, the annual fire regime has contributed to the degradation of the park's ecosystem and has limited management objectives by affecting vegetation structure and species composition; reducing ecosystem function and productivity; threatening high value and vulnerable ecosystems; and reducing habitat and biological diversity.

The Mangetti National Park Management Plan details the implementation of a controlled burning programme based on rotational block burning techniques in the late dry season to control bush encroachment in the park.

Planning involves a decision-making process based on assessments of rainfall and fuel loads, with one of three blocks burnt per year.

Implementation has not commenced and suppression of uncontrolled fires has continued as the mainstay of fire management in the park over the past several years. Gaps in controlled fire management experience, skills and knowledge exist among park staff members, which constrains their ability to safely and efficiently implement the programme as it stands.

Monitoring is limited to ground observations of fire intensity and extent, with basic records also kept on uncontrolled fires.

3. Resources and Capacity

Skills and knowledge of fire management planning, implementation and monitoring are limited.

Vehicles and drivers: availabilities limit implementation of the programme.

Fire management equipment: no fire management equipment exists in the park.

Personal protective equipment: the existing staff uniforms are not appropriate for controlled fire management operations.

Communication: the existing park VHF radio communication network is adequate.

Neighbours: in the communal areas to the west of the park, the uncoordinated use of fire as a land management tool is prevalent, including its role in slash-and-burn agriculture, natural product harvesting, livestock grazing, measures for protection from wildlife, hunting, and honey gathering. Communities rely on the DoF to provide fire management support in the form of firebreak maintenance and wildfire suppression in the event these fires become uncontrolled.

The majority of the private farms are involved in livestock production and management of fire consists of exclusion and suppression approaches. Uncontrolled fire occurrence affects these areas in a similar manner to the way it negatively impacts the park.

Map: Fire History 2008 - 2011

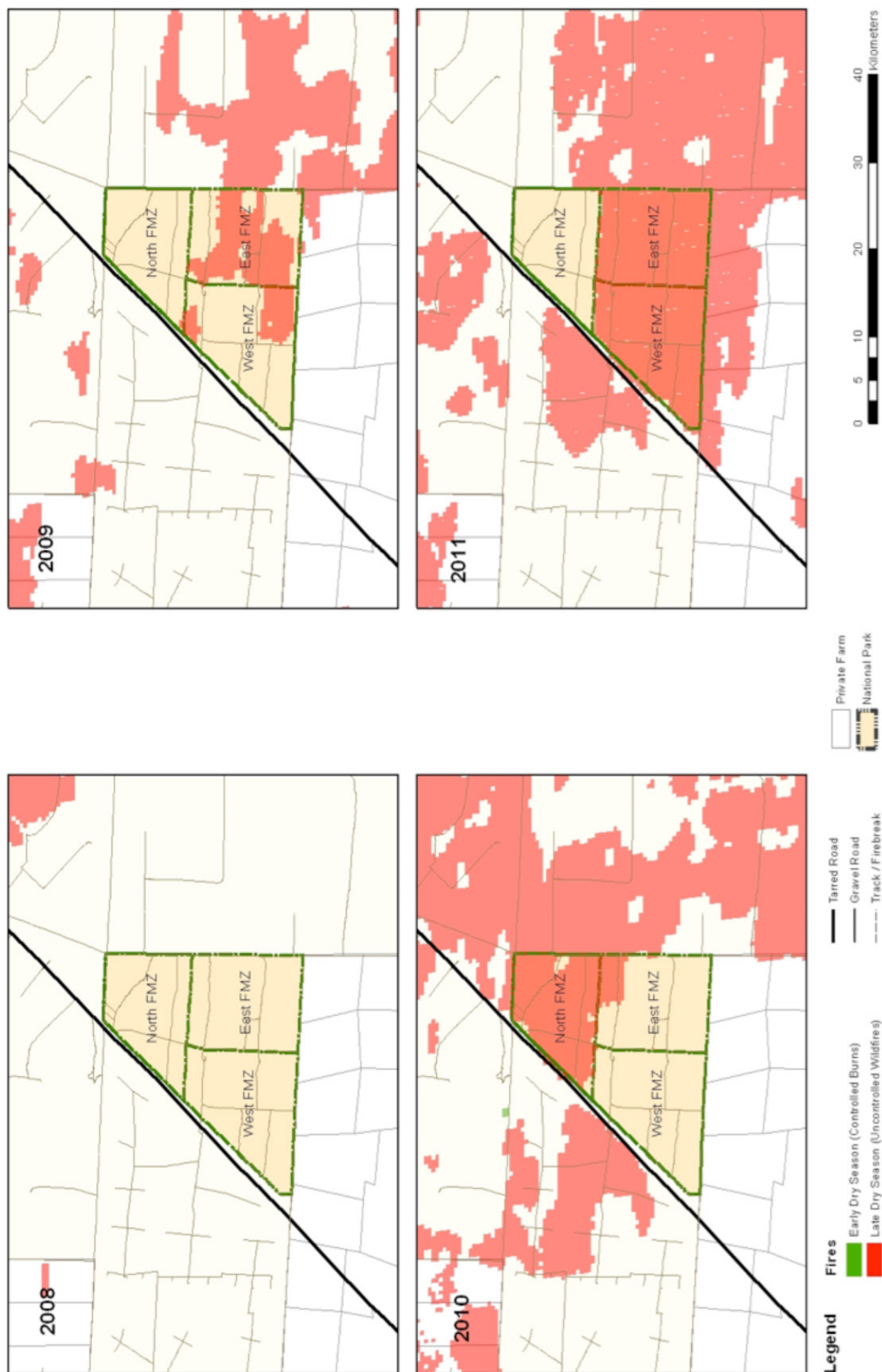


Figure 1: Mangetti National Park fire occurrences, 2008 to 2011

SWOT analysis: this analysis evaluates and summarises the implementation of the existing fire management programme at Mangetti NP.

Table 2: SWOT analysis of existing fire management practices at Mangetti NP

	Beneficial	Detrimental
Internal	STRENGTHS • Warden with controlled fire management experience • Dedicated and motivated management and staff • Good access throughout the park	WEAKNESSES • Minimal fire management experience among staff • Minimal strategic planning • Implementation currently involves suppression only • Vehicle and driver availabilities limit implementation • Local and remote monitoring insufficient
	OPPORTUNITIES • Established collaboration with neighbouring community and conservancy structures • Multi-stakeholder collaborative management forums established, e.g., Mudumu Mamili Complex (MMC) • Strengthen partnership with DoF to facilitate controlled burning in neighbouring areas • High-profile area for development funding	THREATS • Uncontrolled wildfires from neighbouring areas • Management fires impacting neighbours • Entrenched fire exclusion and suppression management on private farms • Weak partnership with DoF hinders facilitation of controlled burning in neighbouring areas

4. The Controlled Fire Management Programme

The Controlled Fire Management Programme for Mangetti National Park represented by this document is based on the strategic implementation of controlled burning to manipulate the park ecosystem to enhance park management objectives.

The programme builds on the existing rotational block burning fire management approach to safely and efficiently implement controlled fire management activities that:

- minimise wildfire hazard;
- enhance park use; and
- improve ecosystems and habitats.

The programme implements an overarching strategic early dry season burning regime to establish a mosaic of interlinked burned patches and corridors, representing approximately 40 per cent of the park area. This creates an early dry season firebreak network that minimises the occurrence and extent of late dry season fires subsequently. The specific location, timing, and method of controlled burning are all guided by park land-use and habitat requirements.

The Fire Management Team carries out a structured annual planning, implementation, and monitoring process based on fire management zones (FMZs). This enables adaptable and robust fire management activities that address the diversity of habitats and park use while achieving overall park fire management objectives.

This effectively dispenses with the need for the costly and hazardous conventional fire management practices of establishing firebreak networks and undertaking wildfire suppression.

4.1 The Fire Management Team

The Fire Management Team is the functional group of five key MET staff members based at Mangetti NP, whose role is to carry out the structured planning, implementation, and monitoring of the programme within the three FMZs every year.

The Fire Management Team Leader works in close collaboration with the Park Warden to coordinate and supervise the Controlled Fire Management Programme.

It is important that the team members are maintained in their roles for at least three years to build fire management skills and knowledge and ensure development and improvement of the programme.

4.2 Fire Management Zones

Mangetti NP has been divided into three fire management zones, based on habitat, infrastructure, park use, and fire history/issues and these will be maintained in the programme. The three zones are as follows:

- The North Fire Management Zone (NFMZ)

The North FMZ borders the area administered by the Uukwangali Traditional Authority in the northern part of the park. The zone extends from the northern boundary of the park to the centre track/firebreak that runs to the tourist entrance gate and extends to the north-eastern corner of the park.

- East Fire Management Zone (EFMZ)

This zone extends from the centre track/firebreak where the NFMZ ends, then continues to the south-eastern corner, to the point where the road from middle dam joins the veterinary fence.

- West Fire Management Zone (WFMZ)

This zone borders the veterinary fence along the commercial farmlands to Mururani Gate and stretches to the tourist entrance gate.

Controlled fire management is tailored to the specific requirements of each of the three FMZs through a structured planning, implementation and monitoring process.

4.3 Planning

Controlled fire management planning is carried out at the end of the summer/wet season (March/April) every year. A planning process involving information collection and analysis at the FMZ level is required to determine the key planning factors for developing the controlled fire management requirements for the following fire season.



Fire Management Zone Analysis

A combination of satellite data, field surveys, and liaison with park staff, concessionaires and neighbours is used to collect and analyse key information. A series of questions highlights the information required, as follows:

Fire history and issues - When did the fires occur in the last three years? What was the source of fire (controlled or uncontrolled)? Where did the fires come from? How much rain fell last year? How much fuel load exists? What and where are the fire hazards?

Park use - Where is the infrastructure (boreholes, lodges, fences, etc.) located? What are the locations and condition of roads, tracks and firebreaks? What areas do the park visitors frequent? What important events took place in the park and what are management's objectives?

Ecosystem and habitat maintenance - What and where are the different habitats? What are the wildlife habitat requirements? How have the habitats been affected by fire? Is there bush encroachment? What are the important habitat functions and services for the park?

Controlled Fire Management Requirements

Specific controlled fire management requirements for each FMZ are to be developed to minimise wildfire hazard, enhance park use, and improve habitats.

Wildfire hazard reduction - Strategic controlled burns are to be employed to reduce fuel loads along boundaries, roads and infrastructure. Patchwork mosaic burning is to be used to fragment fuel loads throughout the park area.

Enhance park use – Strategic controlled burns are to be employed to protect life and property; facilitate efficient management; enhance visitor aesthetics and experience; and enhance economic benefits.

Improving ecosystems and habitats - Strategic controlled burns are to be employed to maintain and enhance habitats; reduce fire intensity; improve ecosystem productivity and function; and establish a less repetitious annual burning regime to promote diversity of habitats and increase biodiversity.

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SOPs will document and detail the work processes to be followed by the Fire Management Team and other park staff to facilitate consistent compliance with fire management implementation.

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Controlled Burning Coordination

The Fire Management Team Leader supervises and coordinates all controlled burning operations of the Fire Management Team. The Controlled Fire Management Plan guides the specific location, timing, and method of controlled burning.

Many factors influence fire behaviour and controlled burning operations. Flexibility is required to adapt to prevailing conditions and preparation considers the following:

Coordination liaison and awareness: it is important to coordinate, collaborate, and increase awareness of controlled burning objectives and activities with park staff and residents, neighbours and visitors.

Access: knowledge of locations and condition of roads, tracks and firebreaks to provide access and a safe environment to implement controlled burning.

Start early in the season: early controlled burning is the key to an effective controlled fire management programme. Areas should be targeted as soon as they are dry enough to burn.

Start late in the day: in the afternoon, winds stabilise and the extent of the fire is under control before nightfall, thus reducing the possible uncontrolled spread of the burned area.

Conditions: weather conditions; fuel type and load; landscape features; and fire behaviour.

Control: active suppression of controlled burns is not desirable. Prior to ignition, the fire controller must be prepared for the control of burn in all possible eventualities.

Controlled Protection Burning

Strategic controlled burns within each FMZ are required to establish buffer zones of burnt areas around infrastructure, resources, and boundaries in order to protect life and property; prevent controlled fires escaping; and minimise the uncontrolled wildfire hazard.

Removing fire fuel load: resources and infrastructure are secured by removing fuel loads through controlled burning to establish buffer zones. Fires later in the year will not be able to reach resources or infrastructure with no fuel to burn around them.



Burn against the wind: protection burns, where possible, are implemented against the wind to create a fire of low intensity and low rate of spread that is easier to control.

Boundary protection: protection burns are established, where required, along park or internal boundaries to ensure controlled mosaic burns or uncontrolled fires do not traverse these boundaries.

Controlled Mosaic Burning

Strategic controlled burns are required to establish a mosaic of burnt and un-burnt patches throughout each FMZ. The protection burns enable controlled mosaic burning to be implemented safely and efficiently.

The specific location, timing, and method of controlled mosaic burning are guided by the FMZ controlled fire management requirements developed in the Controlled Fire Management Plan: controlled burns of varying extent, intensity, patch scale, and timing are implemented throughout the early dry season.

Combined with the protection burns, a mosaic of interlinked burnt patches and corridors representing approximately 40 per cent of the FMZs creates an effective early dry season firebreak network that minimises the occurrence and extent of late dry season fires.

Controlled burning is implemented by igniting fires from vehicles along roads and tracks or by walking across country.

Uncontrolled Wildfire Management

The management of uncontrolled wildfire relies on the patchwork of interlinked burnt patches and corridors established during the controlled protection and mosaic burning, which creates an early dry season firebreak network.

Uncontrolled wildfire management is important in the late dry season (August to November) when fires are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions and fully cured fuel.

Prevention: coordination, liaison and awareness are critical to inform park staff, residents, tour operators, visitors and neighbours about the hazards of wildfires and the use of prevention measures against wildfires.

Detection and communication: the detection of wildfires is the responsibility of all park staff members and they are required to communicate the location of the fire within the FMZ in question to the Fire Management Team. Early detection and swift communication are essential.

Assessment: the fire management team needs to assess the uncontrolled wildfire and determine if it is a hazard to life and property, a threat to park use, or jeopardises ecosystems and habitats. The fire controller must have reliable knowledge and understanding of the area, particularly previously burnt areas, for safe and efficient management.

Response: minimising the threat or hazard of a wildfire involves focusing resources and tactics to protect life and property, contain the wildfire within a controlled area, and suppress and extinguish the wildfire (as a last resort).

Collaboration: if containing the wildfire is beyond the capabilities of the Fire Management Team and park staff, they must notify neighbours and request assistance from them.

4.5 Monitoring

Monitoring of the Controlled Fire Management Programme activities is required throughout the year and involves a combination of park-level observations and remotely sensed information.

Park Level

At the park level, the Fire Management Team is responsible for monitoring within the three FMZs.

Controlled burning monitoring: follow-up surveillance and monitoring is conducted to ensure that controlled protection and mosaic burning achieve the intended objective(s).

Controlled fire management register: a register of location, timing, and nature of controlled burns and wildfires in each FMZ must be maintained.

Vegetation monitoring: monitoring sites (in a range of habitats) in each FMZ are to be marked and surveyed annually towards the end of the wet season (i.e., in April). Photographs, data on basic vegetation structure, and fire information are to be collected.

Remote Monitoring

The Bwabwata Ecological Institute is responsible for the remote monitoring of fires through the AMESD Advanced Fire Information System. Relevant information and maps are to be developed and communicated, with training in GIS and remote sensing skills and knowledge undertaken. Use of AFIS products will incorporate:

Prediction of fires is of limited use in the park due to scale and should be used in conjunction with local-level observations and weather forecasts. The Fire Danger Index is updated daily and provides a 3-day forecast of the Fire Weather Index and Lowveld Fire Danger Index. Fuel Condition data are updated every 10 days and provide a Vegetation Condition Index and a Vegetation Productivity Index, as well as indicating the number of days since the last burn for each area.

Detection and monitoring of fires: MODIS Active Fire detection data are updated regularly on a daily basis (10am and 3pm). MSG Active Fire detection data are updated every 15 minutes.

Assessment of fires: the MODIS Burned Area Product that maps and documents the extent of fires throughout the year is updated every month. MODIS Imagery is updated every 6 hours and provides visual fire monitoring and assessment.

The Park Warden and Fire Management Team should access the available information from the Internet at the: AFIS website - <http://afis.meraka.org.za> and the Global Fire Information Management System (GFIMS) website - <http://www.fao.org/nr/gfims/en>.

The GFIMS provides a fire email alert service whereby registered users of the MODIS-derived Active Fire detection product receive an email whenever a fire is detected in a specified area of interest.



The Controlled Fire Management Report

The Fire Management Team prepares a *Controlled Fire Management Report* based on the monitoring information; the report will address objectives and document the outcomes of implementation. Assessments of programme effectiveness and identification of inadequacies will provide fire management recommendations for the following year.

5. Capacity and Resource requirements

To build the capacity of Mangetti National Park management to safely and efficiently implement its Controlled Fire Management Programme requires a long-term process aimed at developing the skills and knowledge acquired only through firsthand experience. Most of this training takes place in the course of ongoing fire management implementation and requires strategic and regular trainer intervention.

Training and Skills Transfer

On-the-job training is required throughout the duration of the initial fire season to build the capacity of the Fire Management Team to implement the Controlled Fire Management Programme. Ideally, annual follow-up training should be provided in subsequent fire seasons.

A fire management specialist will be required to deliver three training sessions in Mangetti National Park. Training of the Fire Management Team, other park staff, neighbours, and other key stakeholders will coincide with the planning, implementing and monitoring of the programme in its first year.

Session 1: Controlled Fire Management Planning

A Controlled Fire Management Planning Session of two weeks’ duration is required in April/May and will cover all theoretical aspects of the programme; develop the Controlled Fire Management Plan; and commence implementation training.

Part 1, Controlled Fire Management Programme Theory: covering all aspects of the Controlled Fire Management Programme, including:

- Fire issues	- Planning
- Existing fire management	- Implementation
- The Controlled Fire Management Programme	- Monitoring
- The Fire Management Team - roles and responsibilities	- Reporting

Part 2, Fire Management Plan Development: a finalised draft of the Controlled Fire Management Plan is developed for Mangetti NP through the training process whereby data collection, liaison activities, FMZ analyses, and controlled fire management requirement assessments will be carried out.

Part 3, Implementation Activity Introduction: practical introductory training of implementation techniques to achieve the FMZs’ controlled fire management requirements.

Session 2: Controlled Fire Management Implementation

A Controlled Fire Management Implementation Session of four weeks’ duration is required in July to train the Fire Management Team in operational/practical aspects of controlled burning coordination; controlled protection burning; controlled mosaic burning; and uncontrolled wildfire management.

July is a critical time for the Fire Management Team to acquire these skills, in order to ensure sufficient protection and controlled mosaic burning is implemented prior to the wildfire season.

Session 3: Controlled Fire Management Monitoring and Reporting

A Controlled Fire Management Monitoring and Reporting Session of one week’s duration is required in November to train the Fire Management Team in collating park-level and remote monitoring information to assess programme effectiveness to date and develop recommendations.

Equipment

Implementing an effective controlled fire management programme dispenses with the need for costly fire suppression equipment. The following equipment is recommended for the satisfactory implementation of the Controlled Fire Management Programme for Mangetti National Park (Table 3 to Table 5, below).

Vehicles and Drivers

Vehicle and driver availabilities underpin the effectiveness of successful implementation of the Controlled Fire Management Programme throughout this large and remote park.

Suitable 4x4 bakkie vehicles (with competent drivers) are to be prioritised for use in controlled burning operations during the months of May, June and July. A 4x4 bakkie fitted with a twin impellor skid unit constitutes the most suitable mobile fire management equipment for the rough terrain of this remote park.

Personal Protective Equipment (PPE)

The Fire Management Team will require:

Table 3: Personal protective equipment (PPE) required at Mangetti NP

Description	Fire retardant Conti suit
Specifications	100% cotton
Total = 20	
Description	Safety boots (pair)
Specifications	Leather, non-steel-capped
Total = 20	
Description	Broad-brimmed hat
Specifications	100% cotton
Total = 20	
Description	Gloves (pair)
Specifications	Heavy duty leather
Total = 20	
Description	UV protection safety glasses
Total = 20	
Description	First aid burn kit
Total = 2	

PPE can be sourced from safety equipment distributors in Namibia.

Fire Management Equipment

The following fire management equipment is required for controlled burning implementation:

Table 4: Mobile vehicle fire management units required at Mangetti NP

Description	Twin impeller skid unit	
Model	Honda GX160/Davey 2-stage pump	
Specifications	600 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length - 60 m; weight +/- 125 kg	
No. of units, Mangetti National Park: North FMZ		
No. of units, Mangetti National Park: East FMZ		
No. of units, Mangetti National Park: West FMZ		
Total = 4		
Description	Dry chemical powder fire extinguishers	
Specifications	9 kg size conforming to minimum fire ratings specified in SANS 1910:2009 for A, B or C class fires or a combination of ABC class fires	
No. of units, Mangetti National Park: North FMZ		
No. of units, Mangetti National Park: East FMZ		
No. of units, Mangetti National Park: West FMZ		
Total = 10		
Personnel fire management equipment		
Description	Aluminium drip torch	
Specifications	5 L lightweight high-strength aluminium alloy tank with full-length handle, smooth domed-top to prevent fuel accumulation, and check valve in cover to protect against flashbacks	
No. of units, Mangetti National Park: North FMZ		
No. of units, Mangetti National Park: East FMZ		
No. of units, Mangetti National Park: West FMZ		
Total = 10		

Table 5: Personnel fire management equipment required at Mangetti NP

Description	Knapsack spray units 20 L high-density polythene tank with double action plunger valve pump	
Specifications	20 L high-density polythene tank with double action plunger valve pump	
No. of units, Mangetti National Park: North FMZ		
No. of units, Mangetti National Park: East FMZ		
No. of units, Mangetti National Park: West FMZ		
Total = 10		
Description	Fire beaters	
Specifications	Steel or aluminium handled with rubber head	
No. of units, Mangetti National Park: North FMZ	X 10	
No. of units, Mangetti National Park: East FMZ	X 10	
No. of units, Mangetti National Park: West FMZ	X 10	
Total = 30		
Description	Jerry cans	
Specifications	25 L steel	
No. of units, Mangetti National Park: North FMZ		
No. of units, Mangetti National Park: East FMZ		
No. of units, Mangetti National Park: West FMZ		
Total = 10		

Specialised fire management equipment can be sourced from distributors in Namibia and/or South Africa.

Communication

The existing park VHF radio communication network should be strengthened with five handheld VHF radios.

6. Collaboration with Residents and Neighbours of Mangetti National Park

Coordination and collaboration with park residents and neighbours sharing a boundary with the park constitute an important component of the Controlled Fire Management Programme for Mangetti National Park.

The DoF, through the Forest Act of 2001, regulates fire management in the neighbouring private farms, communal area conservancies and communal lands. To implement controlled burning these landholders/land managers require authorisation from the DoF.

Structured, controlled fire management collaboration and coordination between the park Fire Management Team and neighbouring private farms and communal areas enables alignment of fire management objectives and the sharing of resources and workloads.

6.1 Communal Areas and Conservancies

Structured community-based fire management (CBFiM) programmes should be established in communal areas that share a boundary with Mangetti National Park.

Integrated CBFiM programmes established in conservancies neighbouring the north-east parks between 2007 and 2009 provide an operational framework: conservancies register a fire management committee to prepare a fire management plan and apply for a Permit to Burn from the DoF. The fire management committee implements a controlled fire management programme to provide tangible livelihood benefits to the community through reducing uncontrolled fires, improving land use, and enhancing environmental management. The traditional authority oversees the functions of the committee and arbitrates on fire-related disputes within the community.

The CBFiM programmes will require technical and financial support from government and international and local development non-governmental organisations.

6.2 Private Farms

Controlled fire management collaboration and coordination with the numerous privately owned farms adjoining Mangetti NP will require progressive efforts to change deeply entrenched fire exclusion approaches.

In the short term, it is particularly important to ensure that the park's controlled fire management activities do not extend into farms that maintain fire exclusion management. The Park Fire Management Team should notify neighbours and facilitate the sharing of resources, equipment and workloads, particularly when implementing protection burns along shared boundaries.

Development of private-farm-controlled fire management programmes is to be encouraged and supported by park management through joint training, sharing of technical skills and knowledge, and coordinated implementation activities. Controlled fire management programmes and permits are administered through the DoF.



Awareness and education

Increasing the awareness and understanding of the Controlled Fire Management Programme for Mangetti National Park in park staff, residents, neighbours, and visitors is required to facilitate safe and efficient implementation. Pre-season multi-stakeholder meetings and/or agreements establish the foundation for this awareness.

Awareness and education programmes are recommended to foster the commitment and support of key role players and clearly describe their roles and responsibilities within the programme. A multi-media public awareness campaign (electronic and print media such as newspapers, posters and brochures) is required to inform park visitors, civil society, government, and non-government institutions of the programme's objectives and achievements.

Appendix 6

CONTROLLED FIRE MANAGEMENT PROGRAMME,
MUDUMU NATIONAL PARK



1. Introduction

Mudumu National Park covers approximately 737 km² of Zambezi Region, east of the Kwando River. It is centred on the Mudumu Mulapo, a seasonally dry, open channel that drains the mopane woodlands to the east. Riverine woodlands and floodplains are associated along the river while mopane and Kalahari woodlands dominate the savanna ecosystem. The semi-arid tropical climate provides a variable and highly seasonal (October to April) average annual rainfall of approximately 600mm.

Communal lands with a rural population dependent on subsistence farming and natural resources as important sources of livelihoods surround the unfenced park. Collaborative management initiatives to coordinate wildlife and natural resource management with neighbouring communities are well established in the form of communal area conservancies and forests.

2. Fire History, Management and Issues

Fire is a natural and regular occurrence in savanna ecosystems and fire plays a fundamental role in the vegetation, wildlife, and land use management of Mudumu National Park.

Fires occur almost exclusively in the dry season, with early dry season (May to July) fires characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Late dry season (August to November) fires, on the other hand, are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions.

Management of fires in Mudumu NP has been dominated by exclusion, protection, and suppression policies, dating back several decades. This has resulted in extensive uncontrolled wildfires in the late dry season occurring throughout most (70 to 80 per cent) of the park's area every year. The vast majority of these fires are caused by the uncoordinated use of fire as a land management tool in neighbouring communal areas, including during slash-and-burn agriculture, natural product harvesting, and livestock grazing.

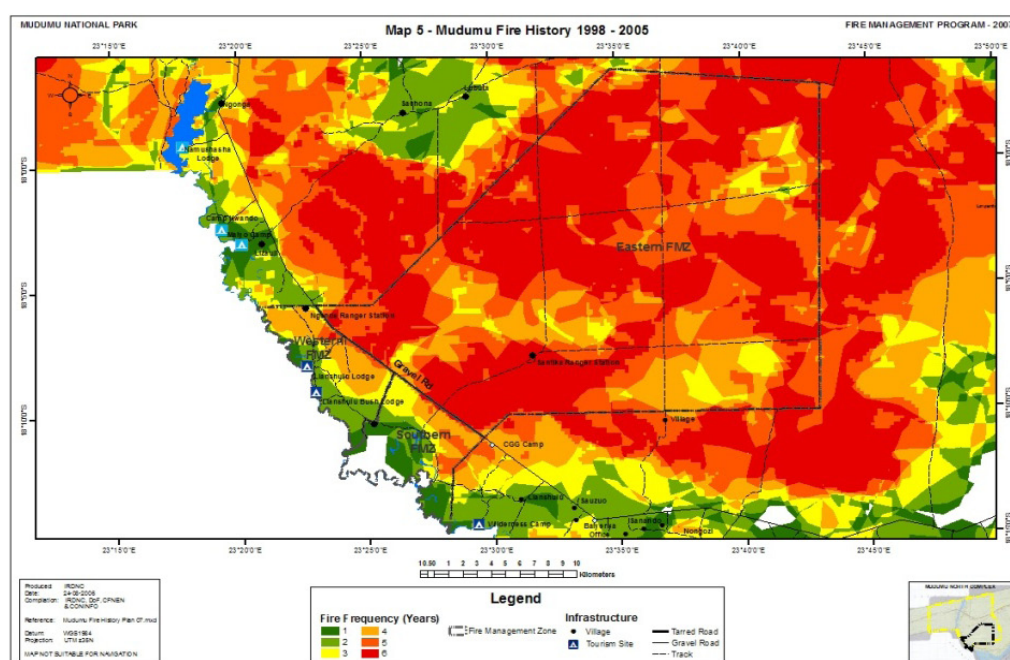


Figure 1: Mudumu National Park fire history, 1998 to 2005

Between 1998 and 2005, approximately 70 to 80 per cent of the park was affected annually (Figure 1).

Fire frequency has been lower in areas adjacent to the Kwando River but frequent fires are experienced in the Mudumu Mulapo and surrounding mopane woodlands.

Over the past two decades, the annual fire regime has contributed to the degradation of the park's ecosystem and limited management objectives by affecting vegetation structure and species composition; reducing ecosystem function and productivity; threatening high value and vulnerable ecosystems; and reducing habitat and biological diversity.

In 2007, Mudumu National Park embarked on the controlled burning programme across the Mudumu North Complex that is central to what is now known as the Zambezi Regional Integrated Fire Management Programme.

The **fire management programme** aims to use controlled burning to reduce the area burnt annually and shift the seasonality of burning to the early dry season. Collaborative implementation with neighbouring conservancies is structured to share resources and workloads to jointly manage uncontrolled wildfires, enhance land use, and improve the environment.

Planning involves designating preparation, controlled burning and assessment activities through the park's annual work plan process.

Implementation is of an ad hoc nature and relies on the skills and knowledge of one key staff member; it is however, generally consistent and well organised. Controlled burning has been successfully established in approximately 40 per cent of the park in the early dry season of each year (see Table 1 and Figure 2). This is contributing significantly to rehabilitating and maintaining the park's ecosystems and habitats, protecting infrastructure, and enhancing economic benefits from consumptive and non-consumptive tourism.

Uncontrolled fires continue to occur from both external sources and internal sources and a large proportion of the park continues to burn annually. However, the seasonality of more than half of these burns has been shifted to the early dry season, significantly improving habitat management.

Implementing the programme remains a challenge due to limited access along the firebreak network.

Table 1: Mudumu National Park fires, 2007 to 2011

	FIRE SEASON									
	2007*		2008*		2009^		2010^		2011^	
	Area (km ²)	% of park	Area (km ²)	% of park	Area (km ²)	% of park	Area (km ²)	% of park	Area (km ²)	% of park
Controlled burn	285	39	301	41	41	6	198	27	142	19
Uncontrolled fire	360	47	271	37	255	35	249	34	381	52
Total	645	86	572	78	296	41	447	61	523	71

* Fires mapped manually from monthly MODIS 250 m Satellite imagery.

^ Automated fire mapping - MODIS Burned Area Product 500 m detects large intense fires only.

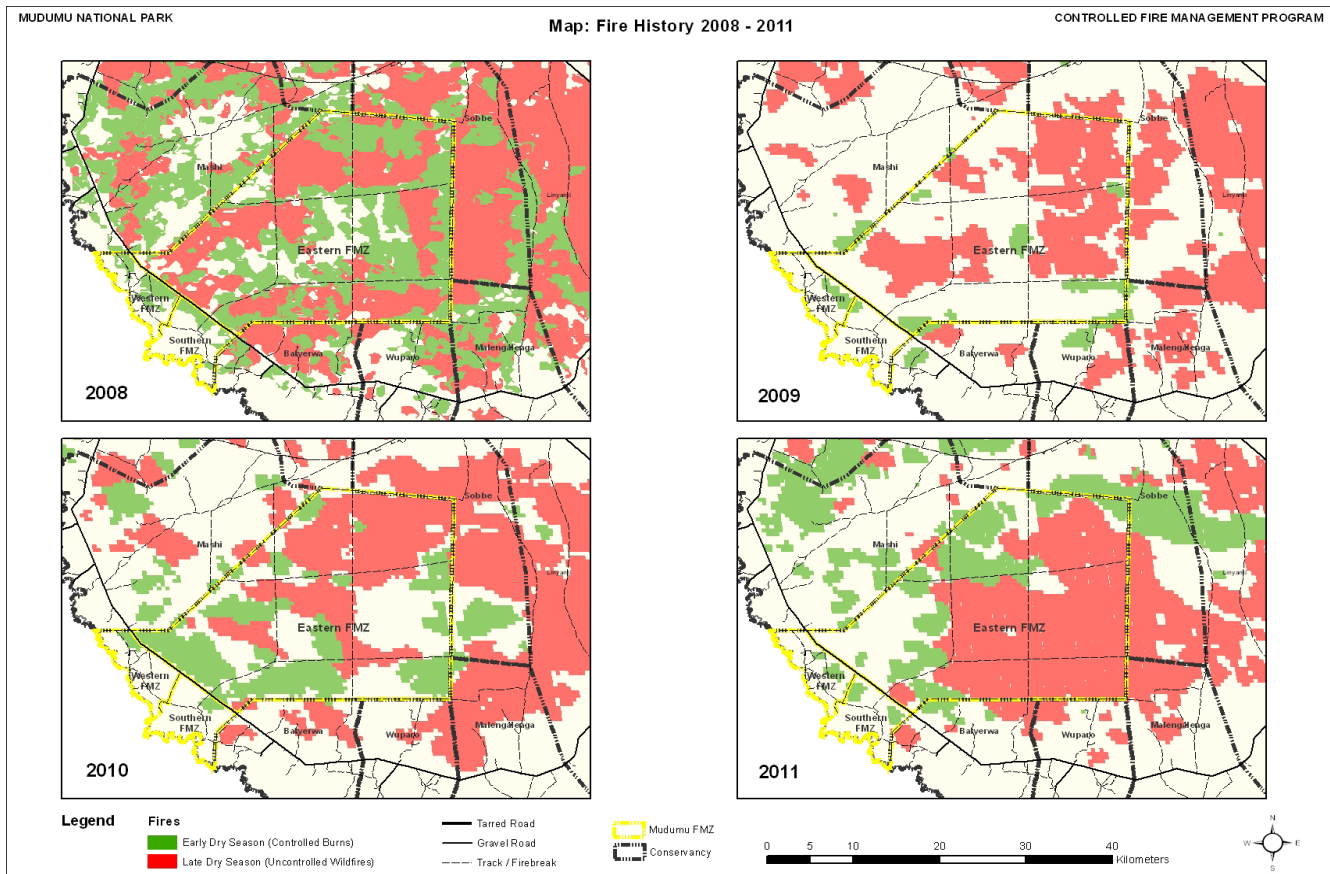


Figure 2: Mudumu National Park fire occurrences, 2008 to 2011

Monitoring is limited to ground observations of fire intensity, extent, and the effect on habitats, but minimal records are kept of implementation or uncontrolled fire occurrence. Remotely sensed fire maps are inconsistently provided to park management for an overview of fire distribution.

3. Resources and Capacity

Skills and knowledge in fire management planning, implementation and monitoring are limited to one key staff member. Other staff members have acquired fire management experience over the years, but lack the leadership skills required to coordinate a fire programme.

Vehicles and drivers: availabilities limit implementation of the programme.

Fire management equipment: the existing three drip torches, three water knapsack spray units, and ten fire beaters represent the absolute minimum required for controlled fire management operations.

Personal protective equipment: the existing staff uniforms are the minimum required for controlled fire management operations.

Communication: the existing park VHF radio communication network is adequate.

Neighbours: Mudumu NP's neighbouring conservancies/community forests are an integral component of the Zambezi Regional Integrated Fire Management Programme initiated in 2007. □□Community-based fire management (CBFiM) programmes in each conservancy/community forest were established with fire

management committees to prepare a fire management plan and apply for a Permit to Burn from the DoF. The fire management committees implemented controlled fire management programmes that provided tangible livelihood benefits to communities through reducing uncontrolled fires and improving land use and environmental management. The traditional authority oversaw the functions of the committee and arbitrated on fire-related disputes within the community.

Since withdrawal of support funding in 2010, the fire management initiatives have ceased to be implemented with any meaningful effect, however. Communities have reverted to the uncoordinated use of fire and uncontrolled wildfires continue to impact negatively on livelihoods and natural resource management.

The DoF provides fire management support by means of discouraging burning through education and awareness campaigns, creating firebreak networks, and community training in wildfire suppression. SWOT analysis: this analysis evaluates and summarises the implementation of the existing integrated fire management programme at Mudumu National Park.

Table 2: SWOT analysis of the existing integrated fire management programme at Mudumu NP

	Beneficial	Detrimental
Internal	STRENGTHS - Five years' controlled burning programme experience - A key staff member with substantial fire skills and knowledge driving the programme - Established as a core park management activity - Consistent implementation with substantial controlled burns established every year - Other experienced staff	WEAKNESSES - Strategic planning lacks a structured process - Programme depends on one staff member - Implementation ad hoc and falls short of some fire management objectives - Vehicle and driver availabilities limit implementation - Road/firebreak maintenance challenges restrict access - Local and remote monitoring insufficient
External	OPPORTUNITIES - Well-established collaboration with neighbouring community and conservancy structures - Multi-stakeholder collaborative management forums well established (MNC and MSC). - Previous CBFiM programmes in neighbouring conservancies/community forests can easily be revitalized - Strengthen partnership with DoF to facilitate controlled burning in neighbouring areas - High-profile area for development aid funding support	THREATS Uncontrolled wildfires from neighbouring areas Poachers igniting fires inside park Management fires impacting neighbours Weak partnership with DoF hinders facilitation of controlled burning in neighbouring areas

4. The Controlled Fire Management Programme

The *Controlled Fire Management Programme for Mudumu National Park* represented by this document is based on the strategic implementation of controlled burning to manipulate the park's ecosystems to enhance park management objectives.

The programme objective builds on the existing patchwork mosaic burning fire management approach developed in 2007 to safely and efficiently implement controlled fire management activities that:

- minimise wildfire hazard;
- enhance park use; and
- improve ecosystems and habitats.

The programme implements an overarching strategic early dry season burning regime to establish a mosaic of interlinked burnt patches and corridors representing approximately 50 per cent of the park. This creates an early dry season firebreak network that minimises the occurrence and extent of late dry season fires. The specific location, timing, and method of controlled burning are all guided by park use and habitat requirements.

The Fire Management Team carries out a structured annual planning, implementation and monitoring process based on three fire management zones (FMZs). This enables adaptable and robust fire management that addresses the diversity of habitats and park use while achieving overall park fire management objectives.

This effectively dispenses with the need for the costly and hazardous conventional fire management practices of establishing firebreak networks and undertaking wildfire suppression.

4.1 The Fire Management Team

The Fire Management Team is the functional group of five key MET staff members based at Ngenda in the NP, whose role is to carry out the structured planning, implementation, and monitoring of the programme within the FMZs every year.

The Fire Management Team Leader works in close collaboration with the Park Warden to coordinate and supervise the Controlled Fire Management Programme.

It is important that the Fire Management Team members are maintained in their roles for as long as possible in order to build fire management skills and knowledge and ensure development and improvement of the programme.

4.2 Fire Management Zones

Mudumu NP has been divided into three FMZs based on habitat, infrastructure, park use, and fire history/issues.

- **Eastern FMZ:** to the eastern side of the main road, encompassing the Mudumu Mulapo and surrounding woodlands.
- **Western FMZ:** the Kwando floodplain and adjoining woodlands between the northern boundary, main road, Kwando River and access road to Nakatwa Ranger Station.
- **Southern FMZ:** the Kwando floodplain and adjoining woodlands south of the access road to Nakatwa Ranger Station between the southern boundary, main road and Kwando River.

Controlled fire management is tailored to the specific requirements of each FMZ through a structured planning, implementation and monitoring process.

4.3 Planning

Controlled fire management planning is carried out at the end of the summer/wet season (March/April) every year. A planning process involving information collection and analysis at the FMZ level is required to determine the key planning factors for developing the controlled fire management requirements for the following fire season.

Fire Management Zone Analysis

A combination of satellite data, field surveys, and liaison with park staff, concessionaires and neighbours is used to collect and analyse key information. A series of questions highlights the information required, as follows:

Fire history and issues - When did the fires occur in the last three years? What was the source of fire (controlled or uncontrolled)? Where did the fires come from? How much rain fell last year? How much fuel load exists? What and where are the fire hazards?

Park use - Where is the infrastructure (boreholes, lodges, fences, etc.) located? What are the locations and condition of roads, tracks and firebreaks? What areas do the park visitors frequent? What important events took place in the park and what are management's objectives?

Ecosystem and habitat maintenance - What and where are the different habitats? What are the wildlife habitat requirements? How have the habitats been affected by fire? Is there bush encroachment? What are the important habitat functions and services for the park?

Controlled Fire Management Requirements

Specific controlled fire management requirements for each FMZ are to be developed to minimise wildfire hazard, enhance park use, and improve habitats.

Wildfire hazard reduction - Strategic controlled burns are to be employed to reduce fuel loads along boundaries, roads and infrastructure. Patchwork mosaic burning is to be used to fragment fuel loads throughout the park area.

Enhance park use - Strategic controlled burns are to be employed to protect life and property; facilitate efficient management; enhance visitor aesthetics and experience; and enhance economic benefits.

Improving ecosystems and habitats - Strategic controlled burns are to be employed to maintain and enhance habitats; reduce fire intensity; improve ecosystem productivity and function; and establish a less repetitious annual burning regime to promote diversity of habitats and increase biodiversity.



The Controlled Fire Management Plan

The *Controlled Fire Management Plan for Mudumu National Park* documents the analysis and the controlled fire management requirements for each FMZ for the fire season to come. Development of the plan focuses on the operational aspects of implementation by employing a 'what, when, where and how' approach, as follows:

- What implementation activity is most suitable?
- When should the activity be employed?
- Where is the implementation activity needed?
- How should the activity be implemented?

Operating procedures for implementation and monitoring activities are detailed in the plan.

Standard Operating Procedures

Development of standard operating procedures (SOPs) for implementation and monitoring activities takes place during the planning process and the SOPs form an integral part of the Controlled Fire Management Plan for Mudumu National Park.

SOPs will document and detail the work processes to be followed by the Fire Management Team and other park staff to facilitate consistent compliance with fire management implementation.

4.4 Implementation

Controlled fire management is implemented in the early dry season (May to July) when fires are characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Light winds, cool temperatures, and fuel (grass, litter etc.) that is only partly dried out all limit the intensity and extent of these fires.

Controlled Burning Coordination

The Fire Management Team Leader supervises and coordinates all controlled burning operations of the Fire Management Team in each of the three FMZs. *The Controlled Fire Management Plan* guides the specific location, timing, and method of controlled burning.

Many factors influence fire behaviour and controlled burning operations. Flexibility is required to adapt to prevailing conditions and preparation considers the following:

Coordination liaison and awareness: it is important to coordinate, collaborate, and increase awareness of controlled burning objectives and activities with park staff and residents, neighbours and visitors.

Access: knowledge of locations and condition of roads, tracks and firebreaks to provide access and a safe environment to implement controlled burning.

Start early in the season: early controlled burning is the key to an effective controlled fire management programme. Areas should be targeted as soon as they are dry enough to burn.

Start late in the day: in the afternoon, winds stabilise and the extent of the fire is under control before nightfall, thus reducing the possible uncontrolled spread of the burned area.

Conditions: weather conditions; fuel type and load; landscape features; and fire behaviour.

Control: active suppression of controlled burns is not desirable. Prior to ignition, the fire controller must be prepared for the control of burn in all possible eventualities.

Controlled Protection Burning

Strategic controlled burns within each FMZ are required to establish buffer zones of burnt areas around infrastructure, resources, and boundaries in order to protect life and property; prevent controlled fires escaping; and minimise the uncontrolled wildfire hazard.

Removing fire fuel load: resources and infrastructure are secured by removing fuel loads through controlled burning to establish buffer zones. Fires later in the year will not be able to reach resources or infrastructure with no fuel to burn around them.

Burn against the wind: protection burns, where possible, are implemented against the wind to create a fire of low intensity and low rate of spread that is easier to control.

Boundary protection: protection burns are established, where required, along park or internal boundaries to ensure controlled mosaic burns or uncontrolled fires do not traverse these boundaries.

Controlled Mosaic Burning

Strategic controlled burns are required to establish a mosaic of burnt and un-burnt patches throughout each FMZ. The protection burns enable controlled mosaic burning to be implemented safely and efficiently.

The specific location, timing, and method of controlled mosaic burning are guided by the FMZ controlled fire management requirements developed in the Controlled Fire Management Plan: controlled burns of varying extent, intensity, patch scale, and timing are implemented throughout the early dry season.

Combined with the protection burns, a mosaic of interlinked burnt patches and corridors representing approximately 50 per cent of the FMZs creates an effective early dry season firebreak network that minimises the occurrence and extent of late dry season fires.

Controlled burning is implemented by igniting fires from vehicles along roads and tracks or by walking across country.

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The management of uncontrolled wildfire relies on the patchwork of interlinked burnt patches and corridors established during the controlled protection and mosaic burning, which creates an early dry season firebreak network.

Uncontrolled wildfire management is important in the late dry season (August to November) when fires are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions and fully cured fuel.

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Detection and communication: the detection of wildfires is the responsibility of all park staff members



and they are required to communicate the location of the fire within the FMZ in question to the Fire Management Team. Early detection and swift communication are essential.

Assessment: the Fire Management Team needs to assess the uncontrolled wildfire and determine if it is a hazard to life and property, a threat to park use, or jeopardises ecosystems and habitats. The fire controller must have reliable knowledge and understanding of the area, particularly previously burnt areas, for safe and efficient management.

Response: minimising the threat or hazard of a wildfire involves focusing resources and tactics to protect life and property, contain the wildfire within a controlled area, and suppress and extinguish the wildfire (as a last resort).

Collaboration: if containing the wildfire is beyond the capabilities of the Fire Management Team and park staff, they must notify neighbours and request assistance from them.

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To build the capacity of Mudumu National Park management to safely and efficiently implement its Controlled Fire Management Programme requires a long-term process aimed at developing the skills and knowledge acquired only through first hand experience. Most of this training takes place in the course of ongoing fire management implementation and requires strategic and regular trainer intervention.

5.1 Training and Skills Transfer

On-the-job training is required throughout the duration of the initial fire season to build the capacity of the Fire Management Team to implement the Controlled Fire Management Programme. Ideally, annual follow-up training should be provided in subsequent fire seasons.

A fire management specialist will be required to deliver three training sessions in Mudumu National Park. Training of fire management teams, other park staff, neighbours, and other key stakeholders will coincide with the planning, implementing and monitoring of the programme in its first year.

Session 1: Controlled Fire Management Planning

A Controlled Fire Management Planning Session of two weeks' duration is required in April/May and will cover all theoretical aspects of the programme; develop the Controlled Fire Management Plan; and commence implementation training.



Part 1, Controlled Fire Management Programme Theory: covering all aspects of the Controlled Fire Management Programme, including:

- Fire issues	- Planning
- Existing fire management	- Implementation
- The Controlled Fire Management Programme	- Monitoring
- The Fire Management Team - roles and responsibilities	- Reporting

Part 2, Fire Management Plan Development: a finalised draft of the Controlled Fire Management Plan is developed for Mudumu NP through the training process whereby data collection, liaison activities, FMZ analyses, and controlled fire management requirement assessments will be carried out.

Part 3, Implementation Activity Introduction: practical introductory training of implementation techniques to achieve the FMZs’ controlled fire management requirements.

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A Controlled Fire Management Implementation Session of four weeks’ duration is required in July to train the Fire Management Team in operational/practical aspects of controlled burning coordination; controlled protection burning; controlled mosaic burning; and uncontrolled wildfire management.

July is a critical time for the Fire Management Team to acquire these skills, in order to ensure sufficient protection and controlled mosaic burning is implemented prior to the wildfire season.

Session 3: Controlled Fire Management Monitoring and Reporting

A Controlled Fire Management Monitoring and Reporting Session of one week’s duration is required in November to train the Fire Management Team in collating park-level and remote monitoring information to assess programme effectiveness to date and develop recommendations.

5.2 Equipment

Implementing an effective controlled fire management programme dispenses with the need for costly fire suppression equipment. The following equipment is recommended for the satisfactory implementation of the Controlled Fire Management Programme for Mudumu National Park (Table 3 to Table 5, below).

Vehicles and Drivers

Vehicle and driver availabilities underpin the effectiveness of successful implementation of the Controlled Fire Management Programme throughout this large and remote park.

Suitable 4x4 bakkie vehicles (with competent drivers) are to be prioritised for use in controlled burning operations during the months of May, June and July. A 4x4 bakkie fitted with a twin impellor skid unit constitutes the most suitable mobile fire management equipment for the rough terrain of this remote park.

Personal Protective Equipment (PPE)

The Fire Management Team will require:

Table 3: Personal protective equipment (PPE) required at Mudumu NP

Description	Fire retardant Conti suit
Specifications	100% cotton
Total = 40	
Description	Safety boots (pair)
Specifications	Leather, non-steel-capped
Total = 40	
Description	Broad-brimmed hat
Specifications	100% cotton
Total = 40	
Description	Gloves (pair)
Specifications	Heavy duty leather
Total = 40	
Description	UV protection safety glasses
Total = 40	
Description	First aid burn kit
Total = 5	

PPE can be sourced from safety equipment distributors in Namibia.

Fire Management Equipment

The following fire management equipment is required for controlled burning implementation:

Table 4: Mobile vehicle fire management units required at Mudumu NP

Description	Twin impeller skid unit		
Model	Honda GX160/Davey 2-stage pump		
Specifications	600 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length - 60 m; weight +/- 125 kg.		
No. of units, Mudumu National Park: Eastern FMZ			
No. of units, Mudumu National Park: Western FMZ			
No. of units, Mudumu National Park: Southern FMZ			
Total = 5			

Description	Dry chemical powder fire extinguishers	
Specifications	9 kg size conforming to minimum fire ratings specified in SANS 1910:2009 for A, B or C class fires or a combination of ABC class fires	
No. of units, Mudumu National Park: Eastern FMZ		X 2
No. of units, Mudumu National Park: Western FMZ		X 2
No. of units, Mudumu National Park: Southern FMZ		X 2
Total = 6		

Table 5: Personnel fire management equipment required at Mudumu NP

Description	Aluminium drip torch	
Specifications	5 L lightweight high-strength aluminium alloy tank with full-length handle, smooth domed-top to prevent fuel accumulation, and check valve in cover to protect against flashbacks	
No. of units, Mudumu National Park: Eastern FMZ		
No. of units, Mudumu National Park: Western FMZ		
No. of units, Mudumu National Park: Southern FMZ		
Total = 10		
Description	Knapsack spray units 20 L high-density polythene tank with double action plunger valve pump	
Specifications	20 L high-density polythene tank with double action plunger valve pump	
No. of units, Mudumu National Park: Western FMZ		
No. of units, Mudumu National Park: Western FMZ		
No. of units, Mudumu National Park: Western FMZ		
Total = 20		
Description	Fire beaters	
Specifications	Steel or aluminium handled with rubber head	
No. of units, Mudumu National Park: Eastern FMZ		
No. of units, Mudumu National Park: Western FMZ		
No. of units, Mudumu National Park: Southern FMZ		
Total = 40		
Description	Jerry cans	
Specifications	25 L steel	
No. of units, Mudumu National Park: Eastern FMZ		
No. of units, Mudumu National Park: Western FMZ		
No. of units, Mudumu National Park: Southern FMZ		
Total = 10		

Specialised fire management equipment can be sourced from distributors in Namibia and/or South Africa.

Communication

The existing park VHF radio communication network should be strengthened with five handheld VHF radios.

6. Collaboration with Residents and Neighbours of Mudumu National Park

Coordination and collaboration with park residents and neighbours sharing a boundary with the park constitute an important component of the Controlled Fire Management Programme for Mudumu National Park.

6.1 Communal Areas and Conservancies

Structured, controlled fire management collaboration and coordination between the park's Fire Management Team and neighbouring conservancies and community forests enables alignment of fire management objectives as well as the sharing of resources and workloads.

The DoF, through the Forest Act of 2001, regulates fire management in the neighbouring communal area conservancies. To implement controlled burning these institutions require authorisation from the DoF.

The community-based fire management (CBFiM) programmes established in the neighbouring conservancies/community forests can provide the operational framework for future collaborations and should be revitalized. The park Fire Management Team implements direct liaison coordination, joint training and implementation with the CBFiM institutional structures.

The CBFiM programmes will require technical and financial support from government and international and local development non-governmental organisations.

6.2 International Collaboration

International collaboration with Botswana and Angola should build on existing regional initiatives, including the Southern African Development Committee (SADC) Regional Fire Management Programme and the SADC Protocol on Forestry (2002).

Integration with the Kavango Zambezi Trans-frontier Conservation Area (KAZA TFCA) should be facilitated through the Namibian Integrated Development Plan as well as other existing structures.

6.3 Awareness and Education

Increasing the awareness and understanding of the Controlled Fire Management Programme for Mudumu National Park in park staff, residents, neighbours, and visitors is required to facilitate safe and efficient implementation. Pre-season multi-stakeholder meetings and/or agreements establish the foundation for this awareness.

Awareness and education programmes are recommended to foster the commitment and support of key role players and clearly describe their roles and responsibilities within the programme. A multi-media public awareness campaign (electronic and print media such as newspapers, posters and brochures) is required to inform park visitors, civil society, government, and non-government institutions of the programme's objectives and achievements.



Appendix 7

CONTROLLED FIRE MANAGEMENT PROGRAMME,
WATERBERG PLATEAU PARK

1. Introduction

Waterberg Plateau Park covers approximately 405 km² of Otjozondjupa Region, west of the Kalahari plains. It is centred on and around the Waterberg Plateau escarpment and contains a diverse variety of habitats ranging from broadleaf savanna to forest.

This fenced park is an important conservation area for the protection and breeding of rare and endangered animal species. The semi-arid climate provides a variable and highly seasonal (October to April) average annual rainfall of approximately 450mm.

Private farms surround the park and collaborative conservation with the neighbouring farms and communal area conservancies to the east has been facilitated through the establishment of the Greater Waterberg Landscape conservation area (GWLCA).

2. Fire History, Management and Issues

Fire is a natural and regular occurrence in savanna ecosystems and fire plays a fundamental role in the vegetation, wildlife, and land use management of Waterberg Plateau Park.

Fires occur almost exclusively in the dry season, with early dry season (May to July) fires characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Late dry season (August to November) fires, on the other hand, are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions.

Management of fires in Waterberg Plateau NP has consisted of fire exclusion policies dating back to the 1960s. In the event of a fire, this was allowed to burn the block it started in.

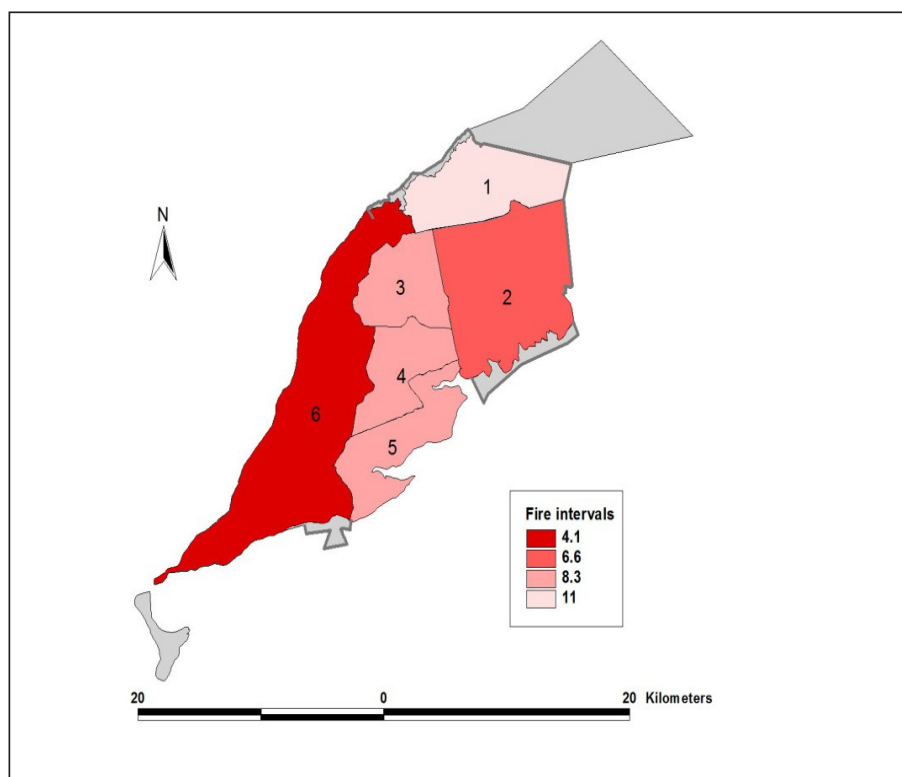


Figure 1: Fire return interval in each fire management zone from 1977 to 2009

In the 1980s, a controlled fire management programme commenced based on rotational block burning techniques to reduce fuel loads and uncontrolled fire hazard. Since then, fire occurrence in Waterberg Plateau Park has been dominated by a late dry season fire regime of infrequent high-intensity fires originating from neighbours, intentional management burns, and lightning. Between 1977 and 2009 (a 32-year period) a fire return interval ranging from 4 to 11 years was been experienced in the six fire management zones (see Figure 1).

The low frequency of fire occurrence in the park has enabled accumulation of very high fuel loads and significant thickening of woody cover. This has affected the park management objectives through the degradation of the park's ecosystems, affecting vegetation structure and species composition; reducing ecosystem function and productivity; threatening high value and vulnerable ecosystems; and reducing habitat and biological diversity.

The 2010 *Waterberg Plateau Park Management Plan* details a revised controlled burning programme based on finer-scale block burning techniques to establish a more diverse distribution of late dry season fires throughout the park.

Planning involves a complex decision-making process based on rainfall, fuel loads, and game carrying capacity to determine which eighteen fire blocks to burn in September and October each year, up to a maximum of 50 per cent of the park's area.

Implementation has been hindered by the mismatch between the scope of activities designed in the planning process and the capacity of management to carry out the practical tasks necessary; a gap in controlled fire management experience, skills and knowledge exists among park staff that constrains their ability to safely and efficiently implement the programme as it stands. Suppression of uncontrolled fires has continued as the mainstay of fire management in the park over the past several years as a consequence.

In the past four years, the incidence of significant fire has been limited to a single uncontrolled late dry season fire entering from a neighbouring farm that burnt 61 km² of FMZs 1 and 2 in 2010 (15 per cent of the park's area) (see Figure 2). The escarpment acts as a natural firebreak and the north-eastern park boundary represents the area of ingress for fire from neighbours. A lightning fire occurred in 2011 in FMZ 2, however its spread was limited due to suppression by management.

Monitoring involves collating a complex array of measurement for vegetation (tree and grass) structure, species composition, and fuel loads. Again this has been hindered by impracticalities in terms of the capacity of the existing park management to effectively carry out the activities involved.

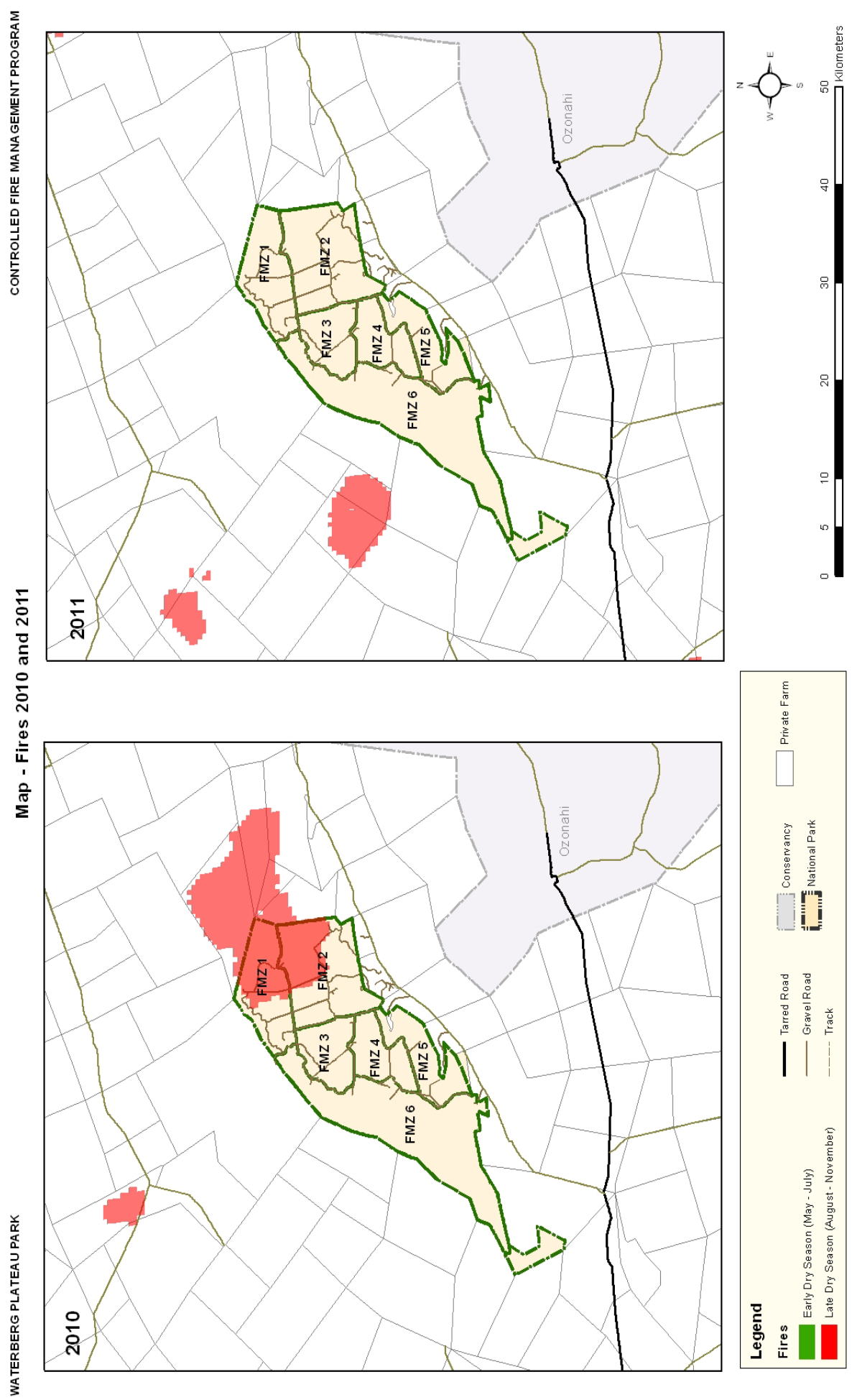


Figure 2: Waterberg Plateau Park fire occurrences, 2010 and 2011

3. Resources and Capacity

Skills and knowledge of fire management planning, implementation, and monitoring are limited among staff.

Vehicles and drivers: availabilities limit implementation of the programme.

Fire management equipment: 4 bakkie 600 litre skid units are adequate to implement safe and efficient controlled fire management operations.

Personal protective equipment: the existing staff uniforms are the minimum required for controlled fire management operations.

Communication: the existing park VHF radio communication network is adequate.

Neighbours: the occurrence and management of fires in the areas neighbouring Waterberg Plateau Park, particularly those sharing a boundary, is an integral aspect of park fire management.

There are numerous (approximately 20) privately owned farms surrounding Waterberg Plateau Park. The majority of farms are involved in livestock or game production with the remainder operating ecotourism ventures associated with the park. Management of fire on these farms consists of exclusion and suppression approaches: land tenure of relatively small blocks and intensive management capacity (firebreaks, fencing, and intensive stocking) has limited the extent and occurrence of fires in the past.

Wildfire suppression along bordering areas takes place through collaborative management with park management; structured collaboration forums are coordinated through the Greater Waterberg Landscape conservation area management body.

Four communal area conservancies (African Wild Dog, Okamatapati, Otjituuo, and Ozonahi) are incorporated into the Greater Waterberg Landscape conservation area (GWLCA), although none shares a boundary with Waterberg Plateau Park.

Use of fire as a land management tool is not widespread in the nearby communal lands although uncontrolled wildfires do occur – particularly in the eastern areas, with up to 20 per cent of land affected. Management consists mainly of prevention and suppression approaches; the DoF provides limited support in the form of firebreak maintenance and wildfire suppression.

SWOT analysis: this analysis evaluates and summarises the existing implementation of fire management at Waterberg Plateau Park.

Table 1: SWOT analysis of existing fire management practices at Waterberg Plateau Park

	Beneficial	Detrimental
Internal	STRENGTHS - Recent policy change from entrenched fire exclusion and suppression approaches - Well-established and dedicated management structure - Fire management skills and experience in upper management - Good access roads throughout - Escarpment forms natural firebreak	WEAKNESSES - Limited controlled fire management experience, skills and knowledge of staff - Very high accumulated fuel loads - Risk of excessive management fires impacting on intensive park use, i.e., sufficient wildlife grazing/browsing availability - Vehicle and driver availabilities limit implementation - Insufficient equipment to safely and efficiently implement controlled fires - Local habitat monitoring insufficient
External	OPPORTUNITIES - Existing fire management collaboration with neighbours to build upon - Established collaboration structures with neighbouring conservancies - Multi-stakeholder collaborative management forums well established (GWLCA) - Strengthen partnership with DoF to facilitate controlled burning in neighbouring areas - High-profile area for development aid funding support	THREATS - Uncontrolled wildfires from neighbouring areas - Management fires impacting neighbouring properties, particularly private farms - Deeply entrenched exclusion and suppression management in private farms - Weak partnership with DoF hinders facilitation of controlled burning in neighbouring areas - Management influenced by public perception and political motivations

4. The Controlled Fire Management Programme

The Controlled Fire Management Programme for Waterberg Plateau Park represented by this document is based on the strategic implementation of controlled burning to manipulate the park ecosystem to enhance park management objectives.

The programme builds on the existing rotational block burning fire management approach to safely and efficiently implement controlled fire management activities that:

- minimise wildfire hazard;
- enhance park use; and
- improve ecosystems and habitats.

The programme implements an overarching strategic early dry season burning regime to establish a mosaic of interlinked burned patches and corridors, representing approximately 35 per cent of the park area. This creates an early dry season firebreak network that minimises the occurrence and extent of late dry season fires subsequently.

The specific location, timing, and method of controlled burning is guided by park land-use and habitat requirements: in this small, fenced and intensively used park, a fine-scale patchwork mosaic of fires distributed throughout the early dry season is required to ensure sufficient grazing for wildlife.

The Fire Management Team carries out a structured annual planning, implementation, and monitoring process based on fire management zones (FMZs). This enables adaptable and robust fire management activities that address the diversity of habitats and park use while achieving overall park fire management objectives.

This effectively dispenses with the need for the costly and hazardous conventional fire management practices of establishing firebreak networks and undertaking wildfire suppression.

4.1 The Fire Management Team

The Fire Management Team is the functional group of five key MET staff members based at Waterberg Plateau Park whose role is to carry out the structured planning, implementation, and monitoring of the programme within the FMZs every year. The Team Leader coordinates and supervises the Controlled Fire Management Programme.

It is important that the Fire Management Team members are maintained in their roles for as long as possible to build fire management skills and knowledge and ensure development and improvement of the programme.

4.2 Fire Management Zones

Waterberg Plateau Park has historically been divided into six fire management zones based on habitat, infrastructure, park use, and fire history/issues and these will be maintained in the programme (see Figure 2).

Controlled fire management is tailored to the specific requirements of each FMZ through a structured planning, implementation, and monitoring process.

FMZs are not considered as 'burning blocks' and block-burning techniques should generally be avoided. Very high fuel loads exist and use of the extensive road network to contain management fires will be important until fuel loads are reduced to manageable levels over the first few years of the programme.

4.3 Planning

Controlled fire management planning is carried out at the end of the summer/wet season (March/April) every year. A planning process involving information collection and analysis at the FMZ level is required to determine the key planning factors for developing the controlled fire management requirements for the following fire season.

Fire Management Zone Analysis

A combination of satellite data, field surveys, and liaison with park staff, concessionaires and neighbours is used to collect and analyse key information. A series of questions highlights the information required, as follows:

Fire history and issues - When did the fires occur in the last three years? What was the source of fire

(controlled or uncontrolled)? Where did the fires come from? How much rain fell last year? How much fuel load exists? What and where are the fire hazards?

Park use - Where is the infrastructure (boreholes, lodges, fences, etc.) located? What are the locations and condition of roads, tracks and firebreaks? What areas do the park visitors frequent? What important events took place in the park and what are management's objectives?

Ecosystem and habitat maintenance - What and where are the different habitats? What are the wildlife habitat requirements? How have the habitats been affected by fire? Is there bush encroachment? What are the important habitat functions and services for the park?

Controlled Fire Management Requirements

Specific controlled fire management requirements for each FMZ are to be developed to minimise wildfire hazard, enhance park use, and improve habitats.

Wildfire hazard reduction - Strategic controlled burns are to be employed to reduce fuel loads along boundaries, roads and infrastructure. Patchwork mosaic burning is to be used to fragment fuel loads throughout the park area.

Enhance park use – Strategic controlled burns are to be employed to protect life and property; facilitate efficient management; enhance visitor aesthetics and experience; and enhance economic benefits.

Improving ecosystems and habitats - Strategic controlled burns are to be employed to maintain and enhance habitats; reduce fire intensity; improve ecosystem productivity and function; and establish a less repetitious annual burning regime to promote diversity of habitats and increase biodiversity.

The Controlled Fire Management Plan

The Controlled Fire Management Plan for Waterberg Plateau Park documents the analysis and the controlled fire management requirements for each FMZ for the fire season to come. Development of the plan focuses on the operational aspects of implementation by employing a 'what, when, where and how' approach, as follows:

- What implementation activity is most suitable?
- When should the activity be employed?
- Where is the implementation activity needed?
- How should the activity be implemented?

Operating procedures for implementation and monitoring activities are detailed in the plan.

Standard Operating Procedures

Development of standard operating procedures (SOPs) for implementation and monitoring activities takes place during the planning process and the SOPs form an integral part of the *Controlled Fire Management Plan for Waterberg Plateau Park*.

SOPs will document and detail the work processes to be followed by the Fire Management Team and other park staff to facilitate consistent compliance with fire management implementation.



4.4 Implementation

Controlled fire management is implemented in the early dry season (May to July) when fires are characterised by low intensity, a high degree of patchiness, and a tendency to extinguish spontaneously overnight. Light winds, cool temperatures, and fuel (grass, litter etc.) that is only partly dried out all limit the intensity and extent of these fires.

Controlled Burning Coordination

The Fire Management Team Leader supervises and coordinates all controlled burning operations of the Fire Management Team in each FMZ. The Controlled Fire Management Plan guides the specific location, timing, and method of controlled burning.

Many factors influence fire behaviour and controlled burning operations. Flexibility is required to adapt to prevailing conditions and preparation considers the following:

Coordination liaison and awareness: it is important to coordinate, collaborate, and increase awareness of controlled burning objectives and activities with park staff and residents, neighbours and visitors.

Access: knowledge of locations and condition of roads, tracks and firebreaks to provide access and a safe environment to implement controlled burning.

Start early in the season: early controlled burning is the key to an effective controlled fire management programme. Areas should be targeted as soon as they are dry enough to burn.

Start late in the day: in the afternoon, winds stabilise and the extent of the fire is under control before nightfall, thus reducing the possible uncontrolled spread of the burned area.

Conditions: weather conditions; fuel type and load; landscape features; and fire behaviour.

Control: active suppression of controlled burns is not desirable. Prior to ignition, the fire controller must be prepared for the control of burn in all possible eventualities.

Controlled Protection Burning

Strategic controlled burns within each FMZ are required to establish buffer zones of burnt areas around infrastructure, resources, and boundaries in order to protect life and property; prevent controlled fires escaping; and minimise the uncontrolled wildfire hazard.

Removing fire fuel load: resources and infrastructure are secured by removing fuel loads through controlled burning to establish buffer zones. Fires later in the year will not be able to reach resources or infrastructure with no fuel to burn around them.

Burn against the wind: protection burns, where possible, are implemented against the wind to create a fire of low intensity and low rate of spread that is easier to control.

Boundary protection: protection burns are established, where required, along park or internal boundaries to ensure controlled mosaic burns or uncontrolled fires do not traverse these boundaries.

Controlled Mosaic Burning

Strategic controlled burns are required to establish a mosaic of burnt and un-burnt patches throughout each FMZ. The protection burns enable controlled mosaic burning to be implemented safely and efficiently.

The specific location, timing, and method of controlled mosaic burning are guided by the FMZ controlled fire management requirements developed in the Controlled Fire Management Plan: controlled burns of varying extent, intensity, patch scale, and timing are implemented throughout the early dry season.

Combined with the protection burns, a mosaic of interlinked burnt patches and corridors representing approximately 35 per cent of the FMZs creates an effective early dry season firebreak network that minimises the occurrence and extent of late dry season fires.

Controlled burning is implemented by igniting fires from vehicles along roads and tracks or by walking across country.

Uncontrolled Wildfire Management

The management of uncontrolled wildfire relies on the patchwork of interlinked burnt patches and corridors established during the controlled protection and mosaic burning, which creates an early dry season firebreak network.

Uncontrolled wildfire management is important in the late dry season (August to November) when fires are characterised by high intensity, low levels of patchiness, and a tendency to spread due to the hot, dry and windy conditions and fully cured fuel.

Prevention: coordination, liaison and awareness are critical to inform park staff, residents, tour operators, visitors and neighbours about the hazards of wildfires and the use of prevention measures against wildfires.

Detection and communication: the detection of wildfires is the responsibility of all park staff members and they are required to communicate the location to the fire management team in the FMZ in question. Early detection and swift communication are essential.

Assessment: the fire management team needs to assess the uncontrolled wildfire and determine if it is a hazard to life and property, a threat to park use, or jeopardises ecosystems and habitats. The fire controller must have reliable knowledge and understanding of the area, particularly previously burnt areas, for safe and efficient management.

Response: minimising the threat or hazard of a wildfire involves focusing resources and tactics to protect life and property, contain the wildfire within a controlled area, and suppress and extinguish the wildfire (as a last resort).

Collaboration: if containing the wildfire is beyond the capabilities of the fire management team and park staff, they must notify neighbours and request assistance from them.



4.5 Monitoring

Monitoring of the Controlled Fire Management Programme activities is required throughout the year and involves a combination of park-level observations and remotely sensed information.

Park Level

At the park level, the Fire Management Team is responsible for monitoring within all the FMZs.

Controlled burning monitoring: follow-up surveillance and monitoring is conducted to ensure that controlled protection and mosaic burning achieve the intended objective(s).

Controlled fire management register: a register of location, timing, and nature of controlled burns and wildfires in each FMZ must be maintained.

Vegetation monitoring: monitoring sites (in a range of habitats) in each FMZ are to be marked and surveyed annually towards the end of the wet season (i.e., in April). Photographs, data on basic vegetation structure, and fire information are to be collected.

Remote Monitoring

The Etosha Ecological Institute is responsible for the remote monitoring of fires through the AMESD Advanced Fire Information System. Relevant information and maps are to be developed and communicated, with training in GIS and remote sensing skills and knowledge undertaken. Use of AFIS products will incorporate:

Prediction of fires is of limited use in the park due to scale and should be used in conjunction with local-level observations and weather forecasts. The Fire Danger Index is updated daily and provides a 3-day forecast of the Fire Weather Index and Lowveld Fire Danger Index. Fuel Condition data are updated every 10 days and provide a Vegetation Condition Index and a Vegetation Productivity Index, as well as indicating the number of days since the last burn for each area.

Detection and monitoring of fires: MODIS Active Fire detection data are updated regularly on a daily basis (10am and 3pm). MSG Active Fire detection data are updated every 15 minutes.

Assessment of fires: the MODIS Burned Area Product that maps and documents the extent of fires throughout the year is updated every month. MODIS Imagery is updated every 6 hours and provides visual fire monitoring and assessment.

The wardens and Fire Management Team should access the available information from the Internet at the: AFIS website - <http://afis.meraka.org.za> and the Global Fire Information Management System (GFIMS) website - <http://www.fao.org/nr/gfims/en>.

The GFIMS provides a fire email alert service whereby registered users of the MODIS-derived Active Fire detection product receive an email whenever a fire is detected in a specified area of interest.

The Controlled Fire Management Report

The Fire Management Team prepares a Controlled Fire Management Report based on the monitoring information; the report will address objectives and document the outcomes of implementation. Assessments of programme effectiveness and identification of inadequacies will provide fire management recommendations for the following year.

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Vehicle and driver availabilities underpin the effectiveness of successful implementation of the Controlled Fire Management Programme throughout this large and remote park.

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Personal Protective Equipment (PPE)

The Fire Management Team will require:

Table 2: Personal protective equipment (PPE) required at Waterberg Plateau Park

Description	Fire retardant Conti suit
Specifications	100% cotton
Total = 40	
Description	Safety boots (pair)
Specifications	Leather, non-steel-capped
Total = 40	
Description	Broad-brimmed hat
Specifications	100% cotton
Total = 40	
Description	Gloves (pair)
Specifications	Heavy duty leather
Total = 40	
Description	UV protection safety glasses
Total = 40	
Description	First aid burn kit
Total = 20	

PPE can be sourced from safety equipment distributors in Namibia.

Fire Management Equipment

The following fire management equipment is required for controlled burning implementation:

Table 3: Mobile vehicle fire management units required at Waterberg Plateau Park

Description	Twin impeller skid unit
Model	Honda GX160/Davey 2-stage pump
Specifications	600 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length - 60 m; weight +/- 125 kg.
Total = 6	
Description	Dry chemical powder fire extinguishers
Specifications	9 kg size conforming to minimum fire ratings specified in SANS 1910:2009 for A, B or C class fires or a combination of ABC class fires
Total = 10	
Description	Off-road trailer unit
Model	2 x Honda GX160/Davey 2-stage pump
Specifications	5.5 HP engine capacity, 40 mm suction diameter, pump flow rate 400 LMP/10 bar, hose length – 60 m
Total = 2	
Description	Support 2500 L truck skid unit
Model	Honda GX 160/Davey 2-stage pump
Specifications	2500 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length – 60 m.
Total = 2	
Description	Off-road 4x4 2-tonne fire truck
Model	Samil 20 or Unimong truck 2 x Honda GX160/Davey 2-stage pump
Specifications	1500 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10 bar; hose length – 60 m
Total = 1	
Description	Off-road 4x4 5-tonne fire truck
Model	Samil 50 2 x Honda GX160/Davey 2-stage pump
Specifications	4000 L tank; 5.5 HP engine capacity; 40 mm suction diameter; pump flow rate 400 LMP/10bar; hose length – 60 m
Total = 1	

Table 4: Personnel fire management equipment required at Waterberg Plateau Park

Description	Aluminium drip torch
Specifications	5 L lightweight high-strength aluminium alloy tank with full-length handle, smooth domed-top to prevent fuel accumulation, and check valve in cover to protect against flashbacks
Total = 10	
Description	Knapsack spray units 20 L high-density polythene tank with double action plunger valve pump
Specifications	20 L high-density polythene tank with double action plunger valve pump
Total = 40	
Description	Fire beaters
Specifications	Steel or aluminium handled with rubber head
Total = 40	
Description	Jerry cans
Specifications	25 L steel
Total = 10	

Specialised fire management equipment can be sourced from distributors in Namibia and/or South Africa.

Communication

The existing park VHF radio communication network should be strengthened with five handheld VHF radios.

6. Collaboration with Residents and Neighbours of Waterberg Plateau Park

Coordination and collaboration with park residents and neighbours sharing a boundary with the park constitute an important component of the Controlled Fire Management Programme for Waterberg Plateau Park.

The DoF, through the Forest Act of 2001, regulates fire management in the neighbouring private farms, communal area conservancies and communal lands. To implement controlled burning these landholders/land managers require authorisation from the DoF.

Structured controlled fire management collaboration and coordination between the park's Fire Management Team and neighbouring conservancies, private farms and communal areas enables alignment of fire management objectives and the sharing of resources and workloads.

The Greater Waterberg Landscape conservation area provides an ideal platform to establish collaborative fire management.

6.1 Private Farms

Controlled fire management collaboration and coordination with the numerous privately owned farms surrounding Waterberg Plateau Park will require progressive efforts to build upon the existing partnerships. The park's Fire Management Team must notify neighbours of its activities and facilitate the sharing of resources, equipment and workloads – particularly when implementing protection burns along shared boundaries.

In the short term, it is particularly important to ensure that the park's controlled fire management activities do not extend into farms that maintain fire exclusion management.

Development of private-farm-controlled fire management programmes is to be encouraged and supported by park management through joint training, sharing of technical skills and knowledge, and coordinated implementation activities. Controlled fire management programmes and permits are administered through the DoF.

6.2 Communal Area Conservancies

Integrated CBFiM programmes established in conservancies neighbouring the north-east parks between 2007 and 2009 provide an operational framework for coordinated activities: conservancies register a fire management committee to prepare a fire management plan and apply for a Permit to Burn from the DoF. The fire management committee implements a controlled fire management programme to provide tangible livelihood benefits to the community through reducing uncontrolled fires, improving land use, and enhancing environmental management. The traditional authority oversees the functions of the committee and arbitrates on fire-related disputes within the community.

A structured community-based fire management (CBFiM) programme should be established in Ozonahi Conservancy, which is close to Waterberg Plateau Park. This CBFiM programme will require technical and financial support from government and international and local development non-government.

6.3 Awareness and Education

Increasing the awareness and understanding of the Controlled Fire Management Programme for Waterberg Plateau Park in park staff, residents, neighbours, and visitors is required to facilitate safe and efficient implementation. Pre-season multi-stakeholder meetings and/or agreements establish the foundation for this awareness.

Awareness and education programmes are recommended to foster the commitment and support of key role players and clearly describe their roles and responsibilities within the programme. A multi-media public awareness campaign (electronic and print media such as newspapers, posters and brochures) is required to inform park visitors, civil society, government, and non-government institutions of the programme's objectives and achievements.



APPENDIX 8

SMALL PROTECTED AREAS' FIRE MANAGEMENT

Daan Viljoen Game Park
Gross-Barmen Hot Springs
Hardap Game Park
Naute Game Park
Popa Game Park
South West Nature Park
Von Bach Game Park

DAAN VILJOEN GAME PARK

Daan Viljoen Game Park covers approximately 40 km² of the Khomas Region. It is located in part of the highland shrubland savanna ecosystem that surrounds Windhoek. The semi-arid climate provides a highly variable and seasonal (October to April) average annual rainfall of approximately 400 mm. The park's proximity to major tourist routes and Windhoek makes it an ideal destination for day visits and tourist stopovers. The city of Windhoek borders the park to east, while private farms surround the park along the other boundaries.

Management of fire in Daan Viljoen GP is dominated by fire exclusion, protection, and suppression policies. In recent years, good rainfall has increased the uncontrolled wildfire hazard from neighbouring farms. Management of game numbers to reduce fuel loads is a strategy used to minimise this fire hazard at Daan Viljoen GP. Minimal to no fire management resources and capacity exist in the park.

Controlled fire management is not recommended in this small reserve. Existing strategies to manage uncontrolled wildfires should be strengthened with basic equipment, training, collaboration with neighbours, and roads/firebreak maintenance.

GROSS-BARMEN HOT SPRINGS

Gross-Barmen Hot Springs is a small (1 km²) popular day resort for both Namibians and tourists. It is sited at the hot spring in the highland shrubland ecosystem near Okahandja. It is too small to effectively manage as a game reserve and fire is not a major issue or management priority in this intensively used and modified habitat.

HARDAP GAME PARK

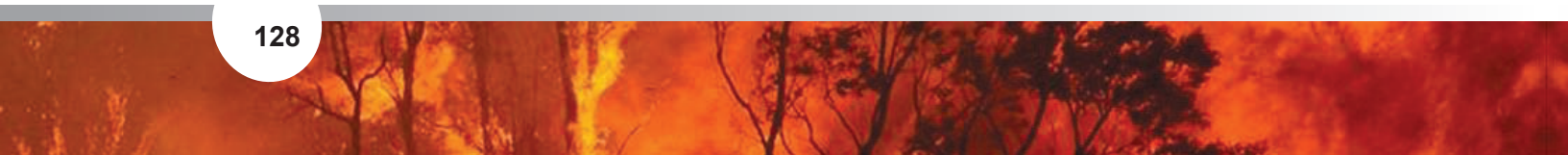
Hardap Game Park covers approximately 252 km² of the Hardap Region. It is centred around the Hardap Dam on the Fish River, in an open savanna and mountainous ecosystem. The arid climate provides a highly variable and seasonal (October to April) average annual rainfall of approximately 200 mm. Home to a small population of black rhino, it is also an important bird habitat. Private farms surround the park.

Management of fire in this low rainfall area is dominated by fire exclusion, protection, and suppression policies. Fires are not common although waste management fires from neighbouring farms entered the park in 2009 and 2012. Management of game numbers to reduce fuel loads is a strategy used to minimise this fire hazard at Hardap GP. Minimal to no fire management resources and capacity exist in the park.

Controlled fire management is not recommended in the low rainfall park. Existing strategies to manage uncontrolled wildfires should be strengthened with basic equipment, training, collaboration with neighbours, and road/firebreak maintenance.

NAUTE GAME PARK

Naute Game Park covers approximately 225 km² of the /Karas Region. It is centred around the Naute Dam on the Lowen River, surrounded by grassy plains and shrubland ecosystems. The arid climate provides a highly variable and seasonal (October to April) average annual rainfall of approximately 200 mm. Private farms surround the park.



Management of fire in this low rainfall area is dominated by fire exclusion, protection, and suppression policies. Management of game numbers to reduce fuel loads is a strategy used to minimise this fire hazard at Naute GP. Minimal to no fire management resources and capacity exist in the park.

Controlled fire management is not recommended in the low rainfall park. Existing strategies to manage uncontrolled wildfires should be strengthened with basic equipment, training, collaboration with neighbours, and road/firebreak maintenance.

POPA GAME PARK

Popa Game Park is a small (0.25 km²) rest camp on the banks of the Kavango River in Kavango-East Region. It is too small to effectively manage as a game reserve and fire is not a major issue or management priority in the intensively used and modified habitat.

SOUTH WEST NATURE PARK

South West Nature Park (the National Botanic Garden of Namibia) covers approximately 0.12 km² in the centre of Windhoek. Fire is not a major issue or management priority in the intensively used and modified park.

VON BACH GAME PARK

Von Bach Game Park covers approximately 43 km² of the Otjozondjupa Region. It is centred on the Von Bach Dam in the Swakop River catchment in a tree/shrub savanna ecosystem. The semi-arid climate provides a highly variable and seasonal (October to April) average annual rainfall of approximately 400 mm.

Management of fire is dominated by fire exclusion, protection and suppression policies. In recent years, good rainfall has increased the uncontrolled wildfire hazard from neighbouring farms. Management of game numbers to reduce fuel loads is a strategy used to minimise this fire hazard at Von Bach GP. Minimal to no fire management resources and capacity exist in the park.

Controlled fire management is not recommended in the low rainfall park. Existing strategies to manage uncontrolled wildfires should be strengthened with basic equipment, training, collaboration with neighbours, and road/firebreak maintenance.

APPENDIX 9

DESERT BIOME PROTECTED AREAS' FIRE MANAGEMENT

/Ai-/Ais Hot Springs and Huns Mountain park
Cape Cross Seal Reserve
Dorob National Park
Namib-Naukluft Park
Skeleton Coast Park
Tsau //Khaeb National Park



/AI-/AIS HOT SPRINGS AND HUNS MOUNTAIN PARK

/Ai-/Ais Hot Springs and Huns Mountain Park covers approximately 4,611 km² of the /Karas Region. It is centred on the Fish River Canyon in the Desert Karoo biome. The arid climate provides a highly variable average annual rainfall of approximately 90 mm. It is bordered by South Africa along the Orange River to the south, Sperrgebiet National Park to the west, and private farms to the north and east. Fire rarely occurs in this very low rainfall area and is not a major issue or management priority.

CAPE CROSS SEAL RESERVE

Cape Cross Seal Reserve covers approximately 60 km² of rocky and sandy Atlantic coastline in the Erongo Region. It is centred on the Cape fur seal colony along the Desert biome coastline. Fire does not occur here and is therefore not a management issue in the reserve.

DOROB NATIONAL PARK

Dorob National Park covers approximately 8,147 km² along the west coast of the Erongo Region. It is characterised by the Atlantic coastline, with its sandy beaches and dune hummocks, to the west and extensive lichen fields and gravel plains inland. In this Desert biome fire does not occur and is therefore not a management issue in the park.

NAMIB-NAUKLUFT PARK

The Namib-Naukluft Park covers approximately 49,768 km² of the Erongo, Hardap and /Karas regions. The park consists of three distinctive vegetation biomes: Namib Desert, Succulent Karoo, and Nama Karoo. The arid climate provides a highly variable average annual rainfall of approximately 100 mm. Private farms adjoin the park along its eastern boundary.

Fires occur occasionally in the higher rainfall areas of the eastern section, particularly during wet cycles. Management of fire is dominated by fire exclusion, protection, and suppression policies. Minimal to no fire management resources and capacity exist in the park. Existing strategies to manage uncontrolled wildfires could be strengthened with basic equipment, training, and collaboration with neighbours.

SKELETON COAST PARK

The Skeleton Coast Park covers approximately 16,390 km² of the Kunene Region, along the west coast of Namibia. It is characterised by the Atlantic coastline with its sandy beaches and sand dunes; inland are ephemeral riverbeds and canyons and extensive mountain ranges to the east. As average annual rainfall rarely exceeds 10 mm in this Desert biome, fire does not occur and is therefore not a management issue in the park.

TSAU //KHAEB NATIONAL PARK

Tsau //Khaeb (formerly Sperrgebiet) National Park covers approximately 22,000 km² of the /Karas Region, along the west coast of the country. It is characterised by the Atlantic coastline with its sandy beaches and rocky headlands and sand dunes; gravel plains and mountain ranges occur inland. Average annual rainfall rarely exceeds 50 mm in this Desert biome so fire does not occur and is not a management issue in the park.

NOTES

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